

Model Testing Report

Experiment: bank_aml_tm_10k_local

Generated: 2026-04-08 19:10:39

Segmentation Column: value_date

Segments Defined: 4

Segment	Rows	Status
value_date_quarter=2024Q1	2122	completed
value_date_quarter=2024Q2	2448	completed
value_date_quarter=2024Q3	2652	completed
value_date_quarter=2024Q4	2778	completed

Overall - 1) Model Effectiveness (1/3)

Metric	Figure	Explanation
auc_roc	0.8709	Measures ranking power across all thresholds. 0.8709 indicates strong separation. Next: compare AUC-ROC across segments and keep threshold tuning separate from ranking validation.
auc_pr	0.5977	Measures precision-recall performance under class imbalance; the observed event rate is 15.74%. Next: use this together with alert capacity to decide whether the high-score queue is practically useful.
precision	0.6630	Shows the share of flagged cases that are true positives at the current threshold. Next: if this is below review expectations, raise the threshold or improve exclusion features.
recall	0.4225	Shows the share of true positives captured at the current threshold. Next: if missed cases are too high, lower the threshold or add stronger risk features.
f1	0.5161	Balances precision and recall in one score. Next: use it as a threshold-selection aid when both missed cases and false alerts matter.
ks_statistic	0.5994	Measures score separation between positive and negative classes. Next: low KS suggests the model needs better features or a different specification.

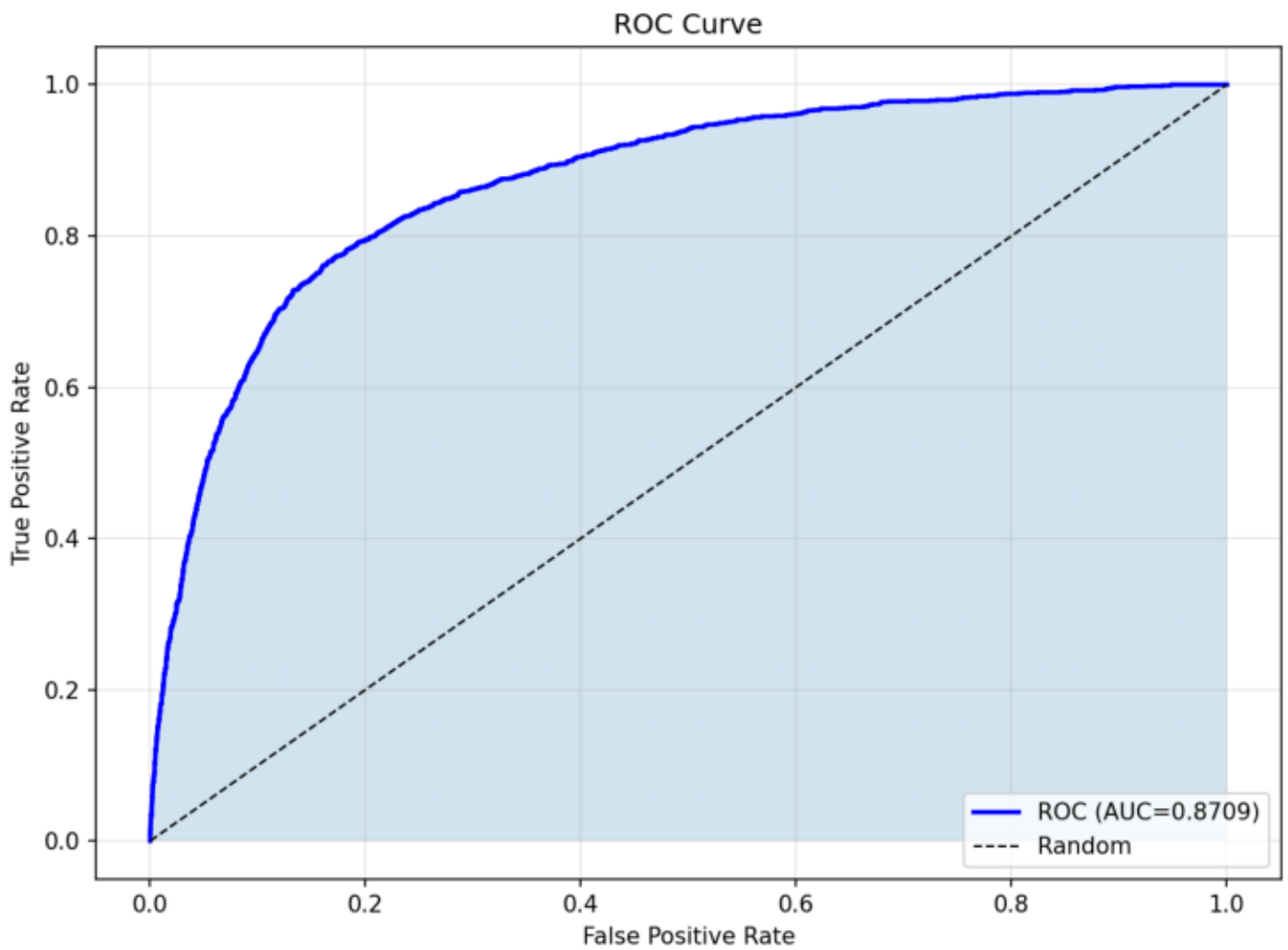
Overall - 1) Model Effectiveness (2/3)

Metric	Figure	Explanation
ks_threshold	0.1800	This is the threshold that maximizes KS separation. Next: treat it as a candidate operating point, then compare it with business capacity and policy limits.
confusion_matrix[TN]	8088	True negatives correctly cleared. Next: preserve this base while tuning the alert threshold.
confusion_matrix[FP]	338	False positives create review cost without SAR conversion. Next: reduce this count with threshold tuning or sharper risk features.
confusion_matrix[FN]	909	False negatives are missed SAR cases. Next: if this count is unacceptable, lower the threshold or add features that capture suspicious behavior earlier.
confusion_matrix[TP]	665	True positives are correctly escalated cases. Next: assess whether this volume is sufficient relative to expected SAR capture goals.
precision_at_k[10]	1.0000	Shows how accurate the top-10 highest-risk queue is. Next: align K with investigator capacity and confirm that top alerts stay high quality.
precision_at_k[50]	0.9000	Shows how accurate the top-50 highest-risk queue is. Next: align K with investigator capacity and confirm that top alerts stay high quality.
precision_at_k[100]	0.8400	Shows how accurate the top-100 highest-risk queue is. Next: align K with investigator capacity and confirm that top alerts stay high quality.

Overall - 1) Model Effectiveness (3/3)

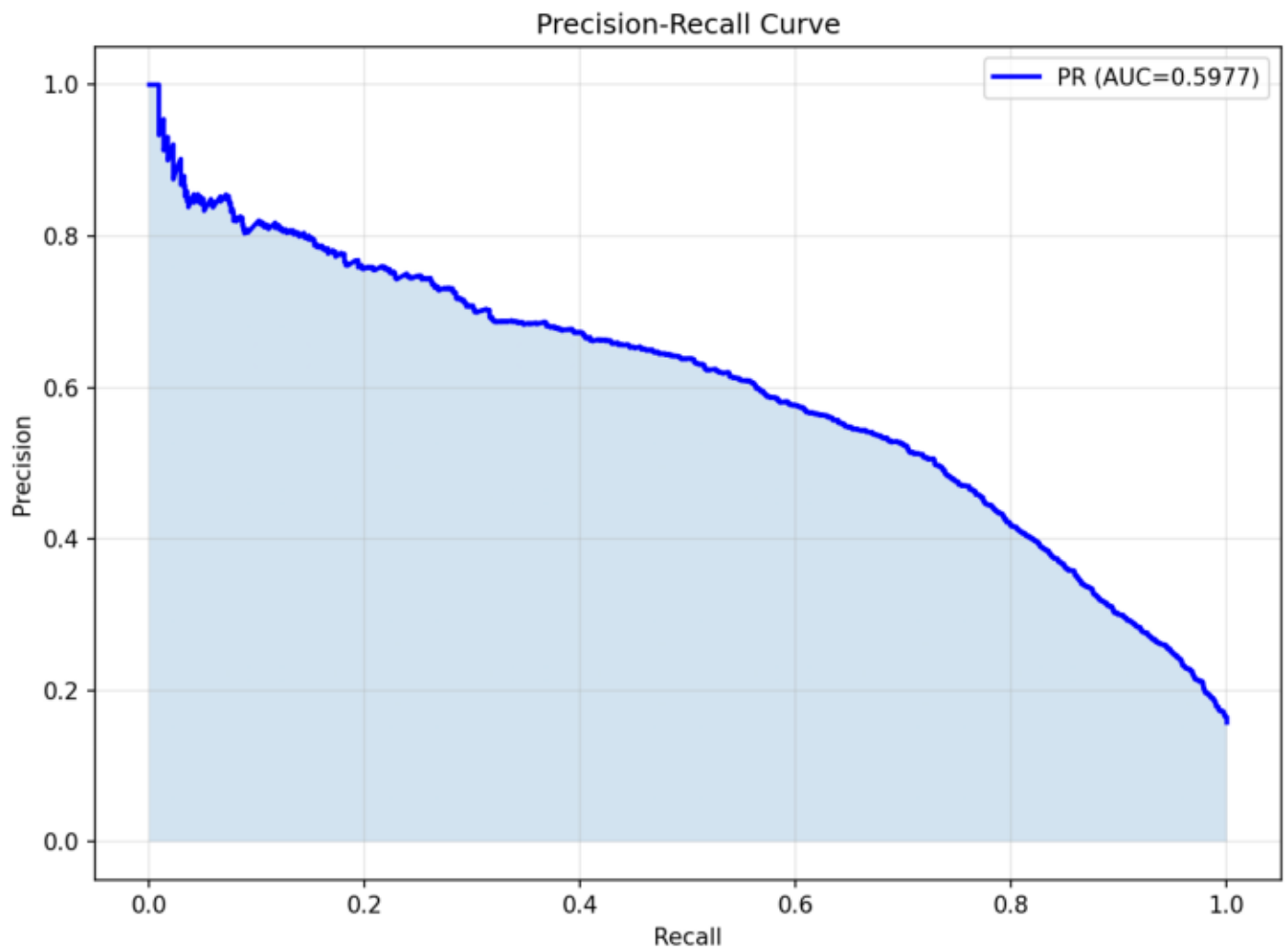
Metric	Figure	Explanation
precision_at_k[200]	0.8150	Shows how accurate the top-200 highest-risk queue is. Next: align K with investigator capacity and confirm that top alerts stay high quality.
precision_at_k[500]	0.7500	Shows how accurate the top-500 highest-risk queue is. Next: align K with investigator capacity and confirm that top alerts stay high quality.
recall_at_k[10]	0.0064	Shows how much of the positive population is captured within the top-10 queue. Next: increase K if coverage is too low, or improve ranking quality if capacity cannot expand.
recall_at_k[50]	0.0286	Shows how much of the positive population is captured within the top-50 queue. Next: increase K if coverage is too low, or improve ranking quality if capacity cannot expand.
recall_at_k[100]	0.0534	Shows how much of the positive population is captured within the top-100 queue. Next: increase K if coverage is too low, or improve ranking quality if capacity cannot expand.
recall_at_k[200]	0.1036	Shows how much of the positive population is captured within the top-200 queue. Next: increase K if coverage is too low, or improve ranking quality if capacity cannot expand.
recall_at_k[500]	0.2382	Shows how much of the positive population is captured within the top-500 queue. Next: increase K if coverage is too low, or improve ranking quality if capacity cannot expand.

Overall - 1) Model Effectiveness



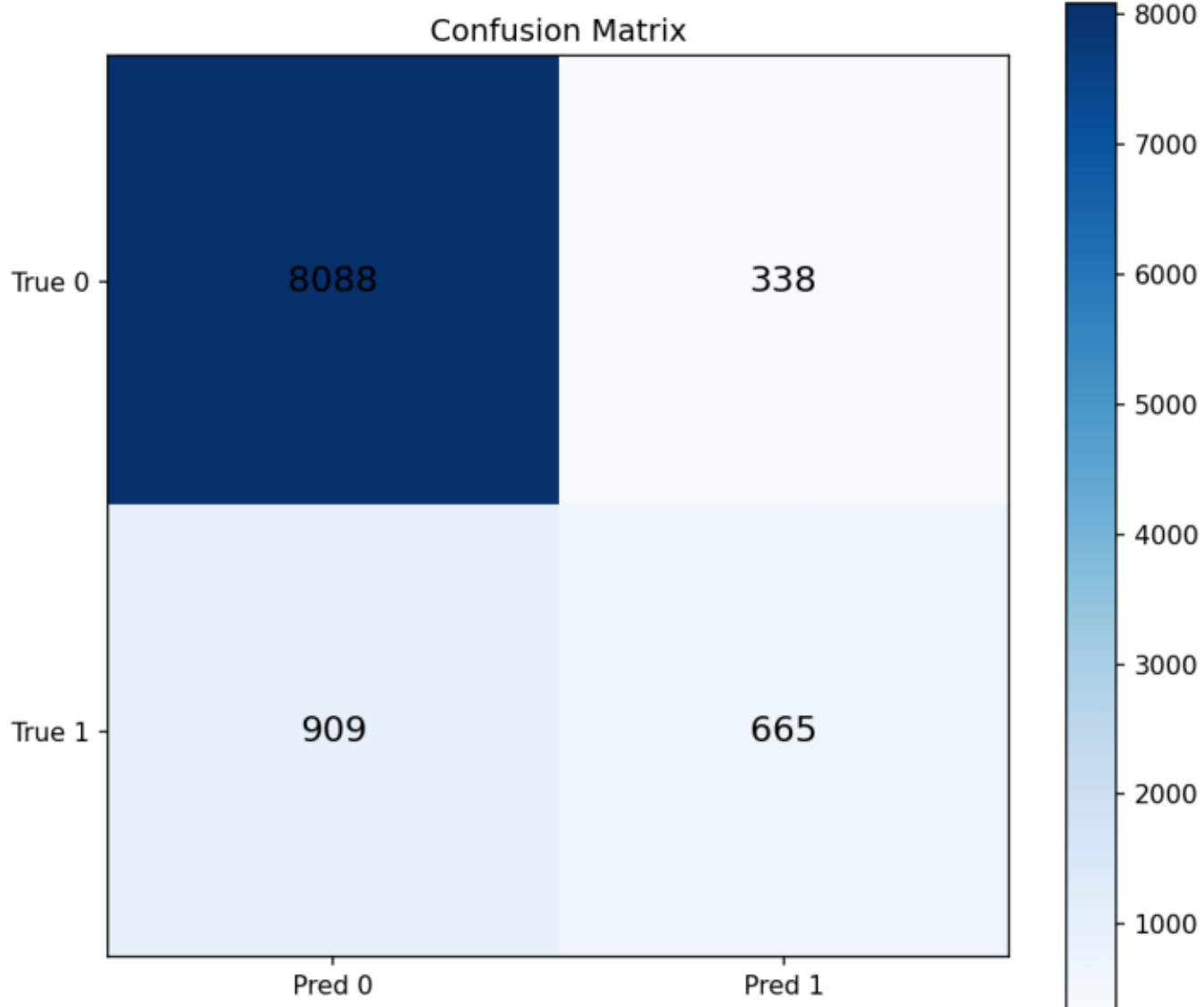
roc_curve: AUC-ROC=0.8709; curve above diagonal indicates strong separation.

Overall - 1) Model Effectiveness



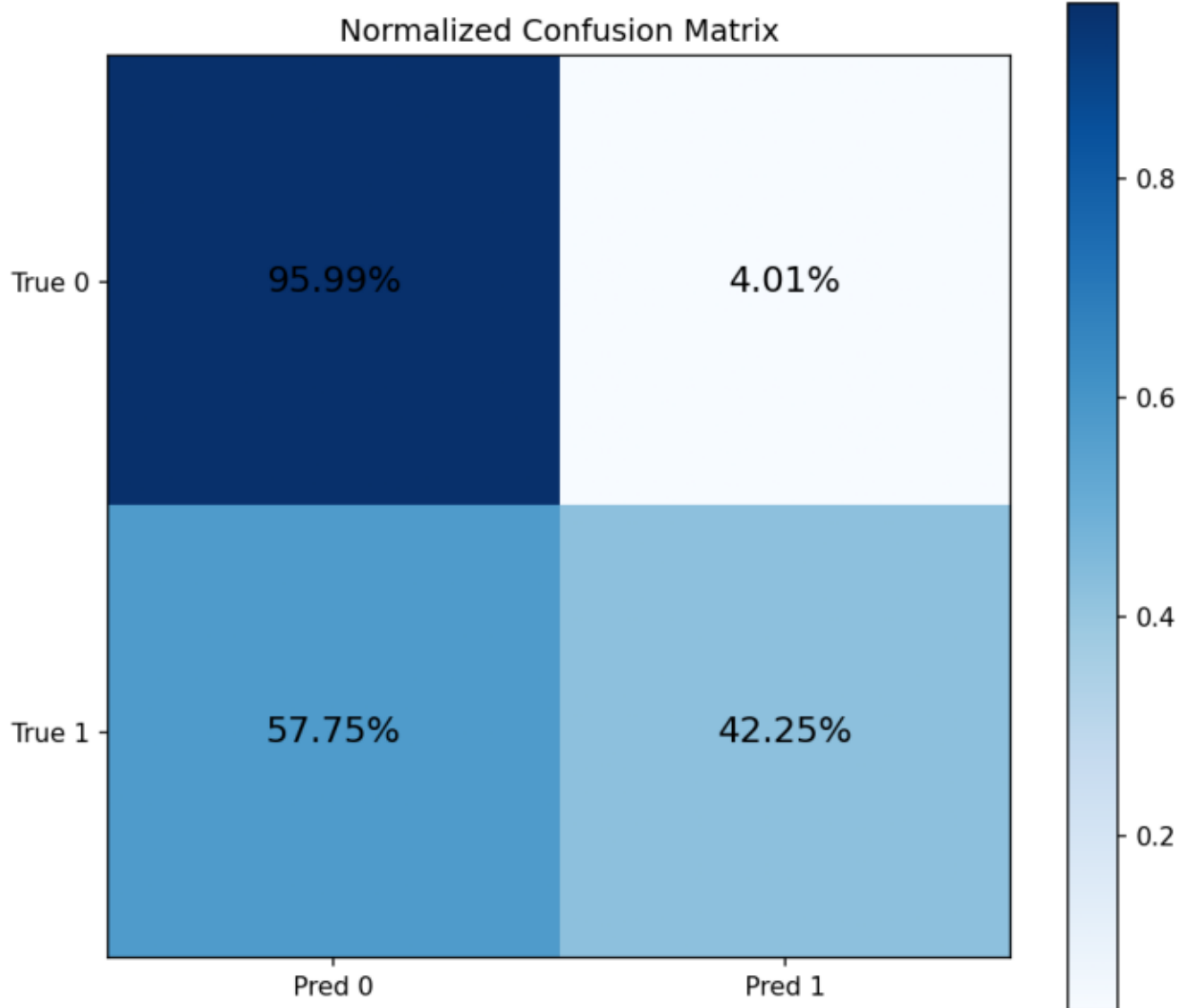
pr_curve: AUC-PR=0.5977 vs baseline 15.74%; higher curve implies better precision at recall.

Overall - 1) Model Effectiveness



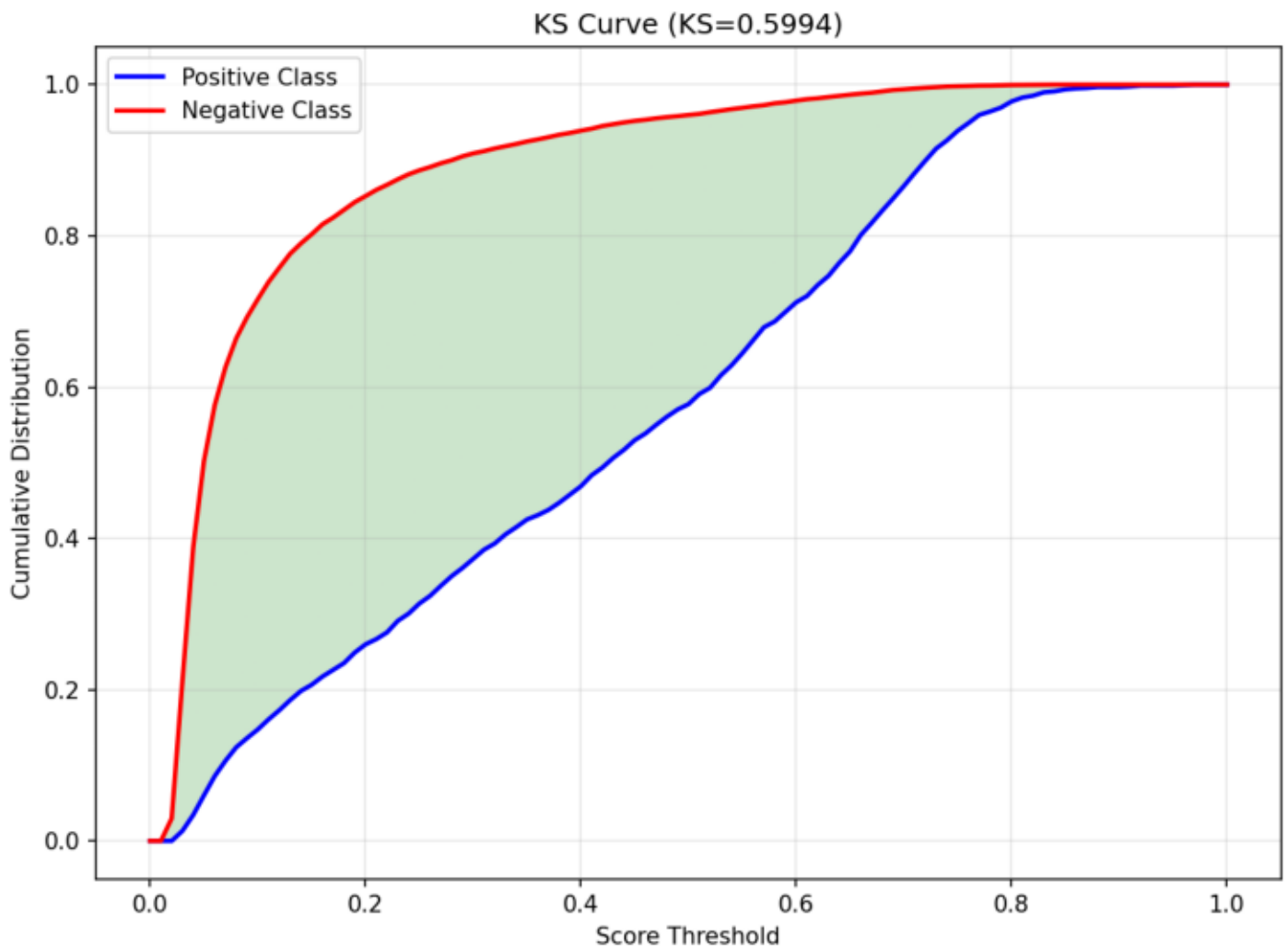
confusion_matrix: At threshold 0.50: TP=665, FP=338, FN=909, TN=8088.

Overall - 1) Model Effectiveness



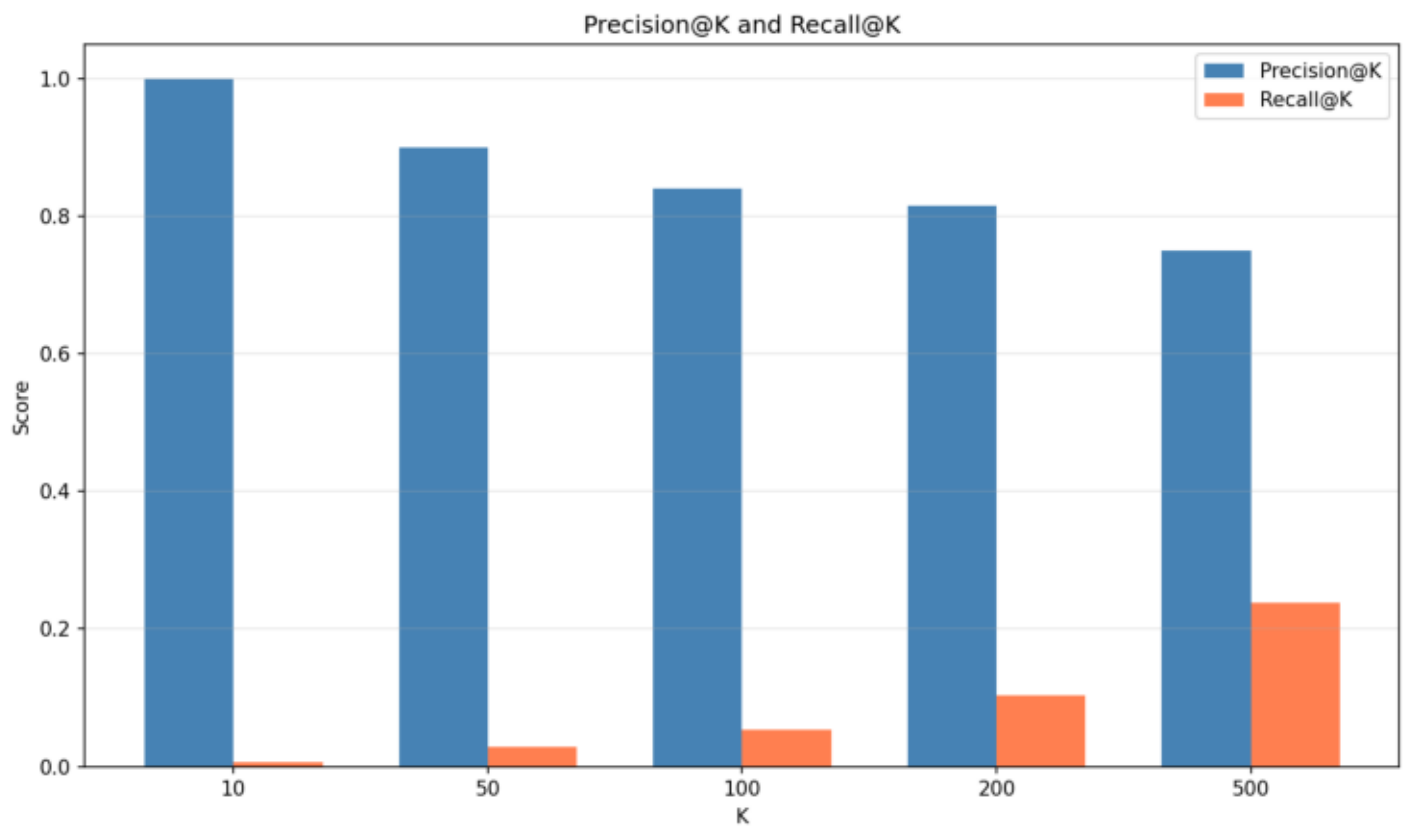
confusion_matrix_norm: Normalized rates: TPR=42.25%, TNR=95.99%.

Overall - 1) Model Effectiveness



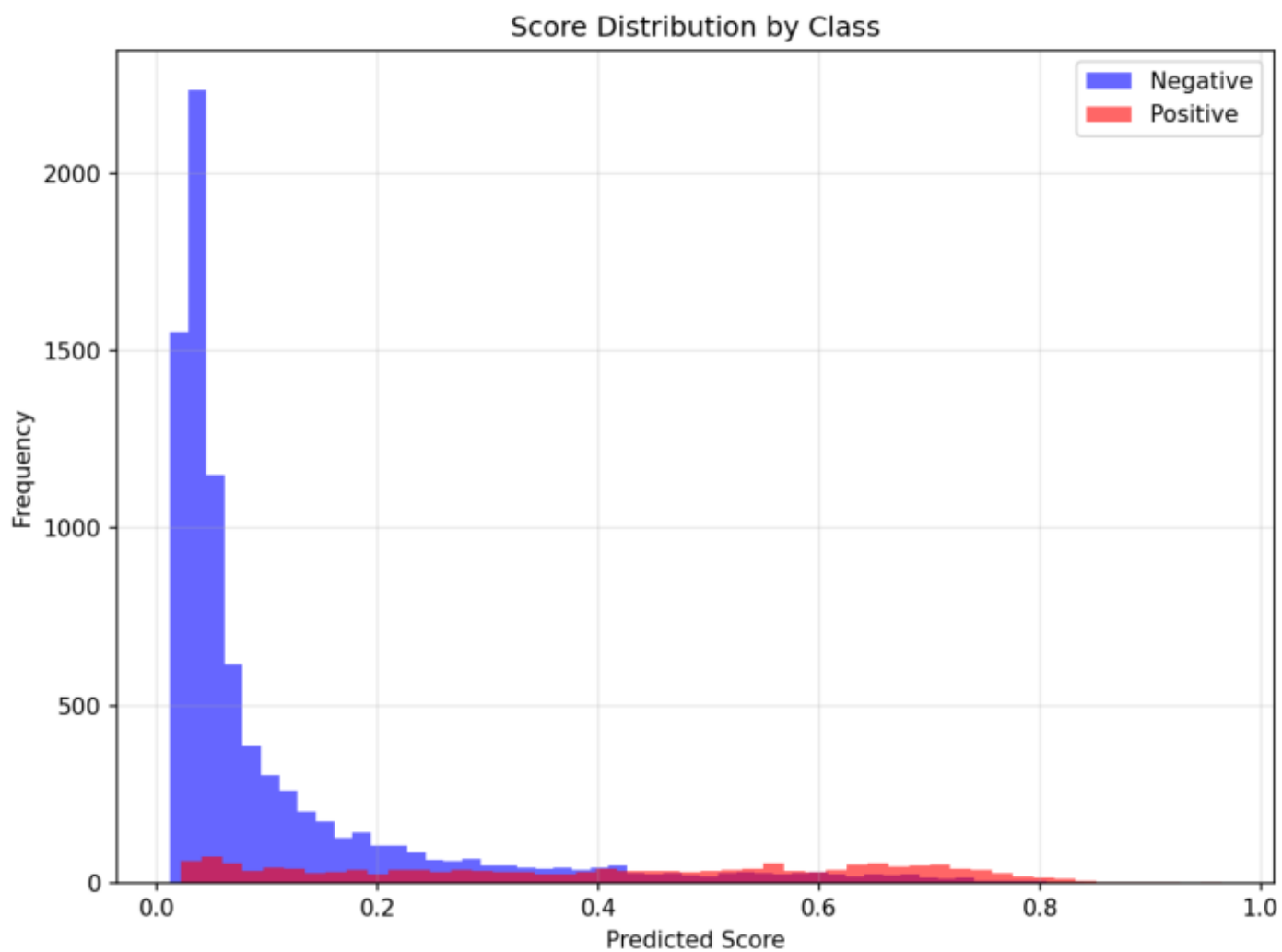
ks_curve: Maximum separation KS=0.5994 at threshold 0.18.

Overall - 1) Model Effectiveness



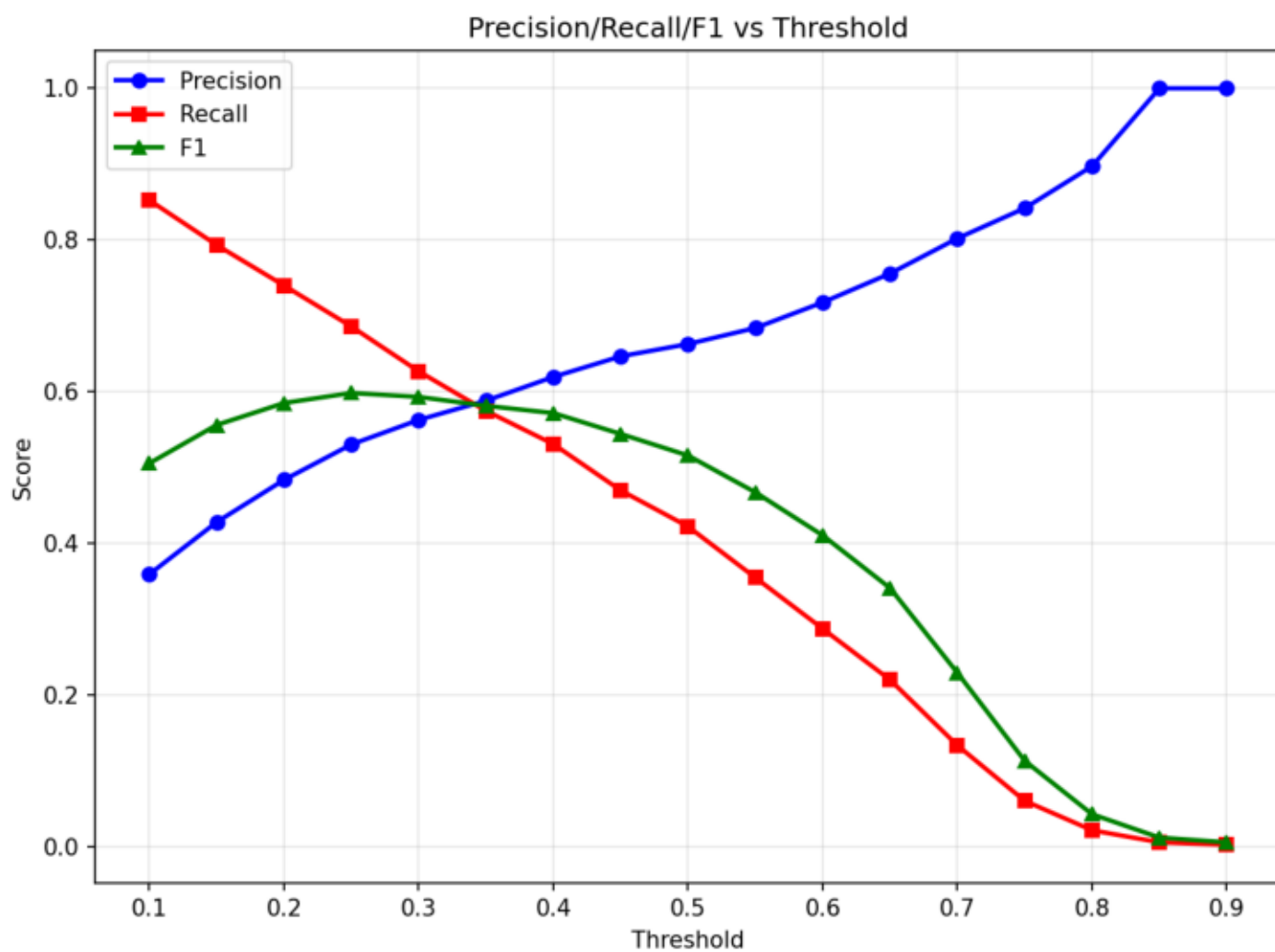
precision_recall_at_k: Bars show precision/recall as you expand the review queue.

Overall - 1) Model Effectiveness



score_distribution: Mean score: positive=0.412, negative=0.108 (larger gap is better).

Overall - 1) Model Effectiveness



threshold_analysis: Best F1 in tested grid is 0.5985 at threshold 0.25.

Overall - 2) Model Efficiency (1/4)

Metric	Figure	Explanation
fpr	0.0401	Measures the share of negatives incorrectly flagged; 0.0401 is low. Next: if review demand is too high, raise the threshold or add features that better suppress false alerts.
tn	8088	Counts negatives correctly cleared without manual review. Next: preserve this clearance rate while checking that recall does not fall below policy expectations.
fp	338	Counts non-events that would still be sent to review. Next: use this with staffing capacity to decide whether threshold or feature refinement is required.
threshold	0.5000	This is the active operating cutoff used to flag alerts. Next: tune it jointly with recall, precision, and review capacity rather than in isolation.
fpr_at_thresholds[thresholds.t_0.05]	0.4973	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.05. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.10]	0.2836	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.10. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.

Overall - 2) Model Efficiency (2/4)

Metric	Figure	Explanation
fpr_at_thresholds[thresholds.t_0.15]	0.1978	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.15. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.20]	0.1474	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.20. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.25]	0.1133	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.25. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.30]	0.0909	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.30. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.35]	0.0751	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.35. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.40]	0.0609	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.40. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.

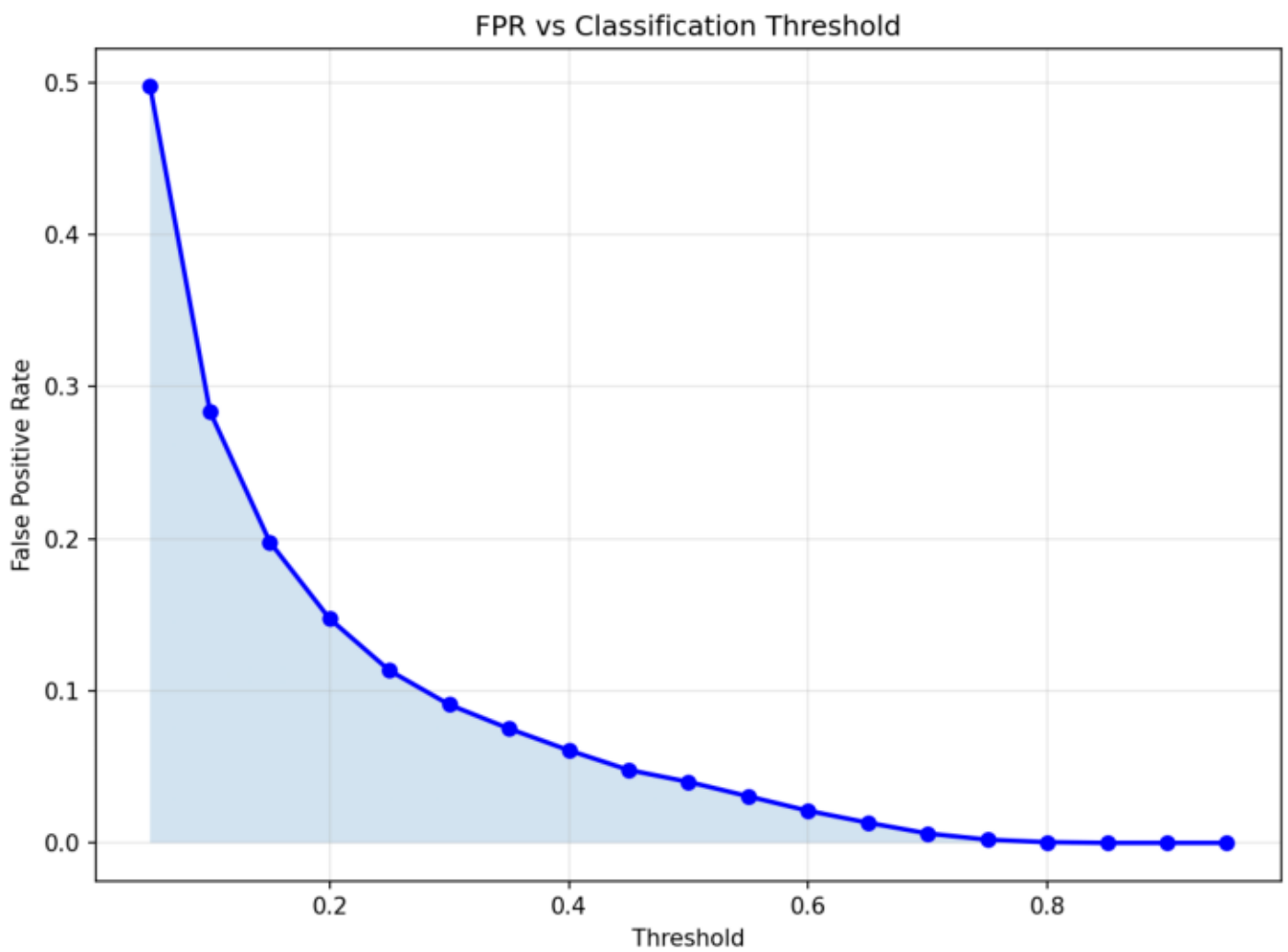
Overall - 2) Model Efficiency (3/4)

Metric	Figure	Explanation
fpr_at_thresholds[thresholds.t_0.45]	0.0479	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.45. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.50]	0.0401	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.50. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.55]	0.0306	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.55. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.60]	0.0211	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.60. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.65]	0.0133	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.65. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.70]	0.0062	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.70. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.

Overall - 2) Model Efficiency (4/4)

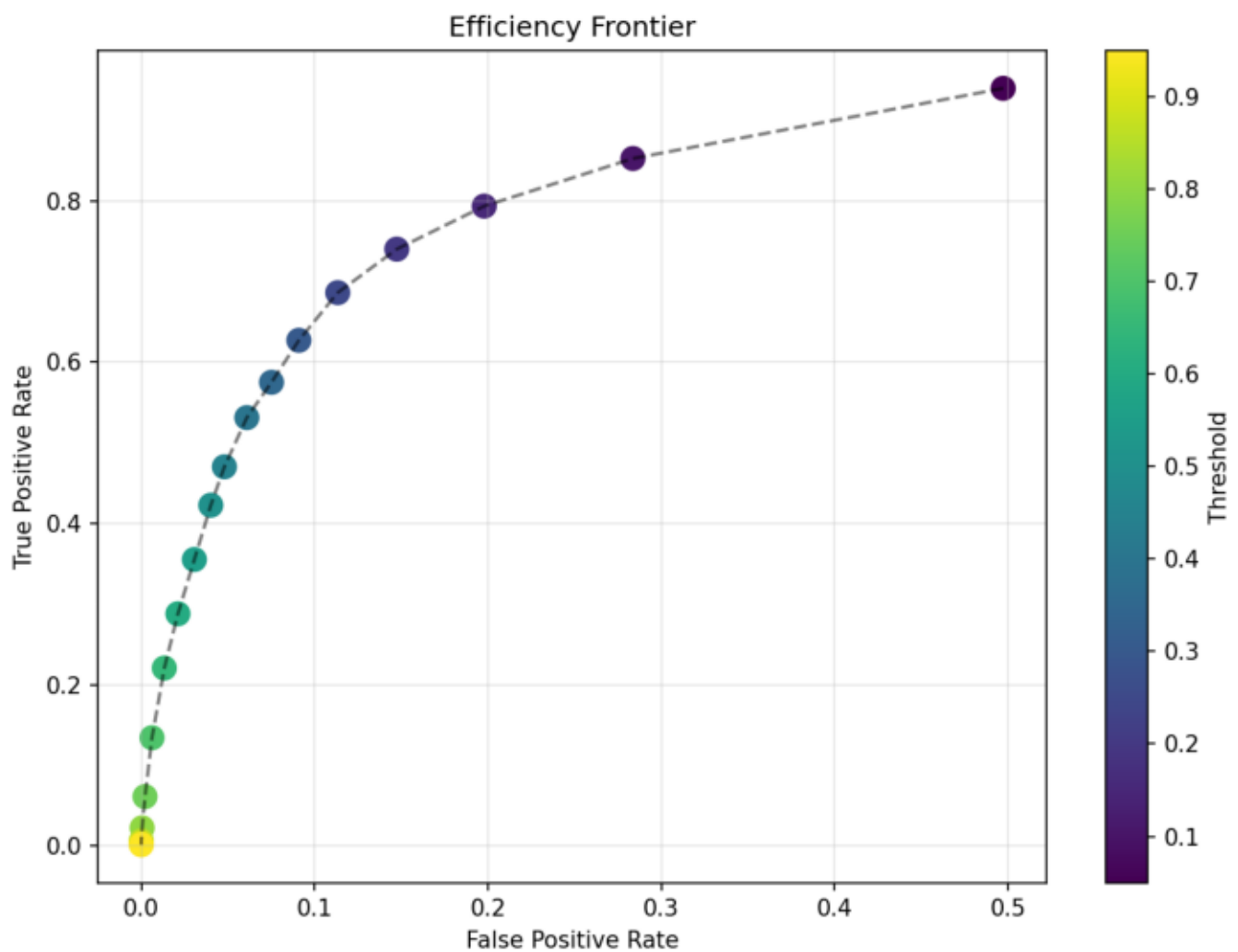
Metric	Figure	Explanation
fpr_at_thresholds[thresholds.t_0.75]	0.0021	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.75. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.80]	0.0005	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.80. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.85]	0.0000	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.85. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.90]	0.0000	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.90. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.95]	0.0000	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.95. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.

Overall - 2) Model Efficiency



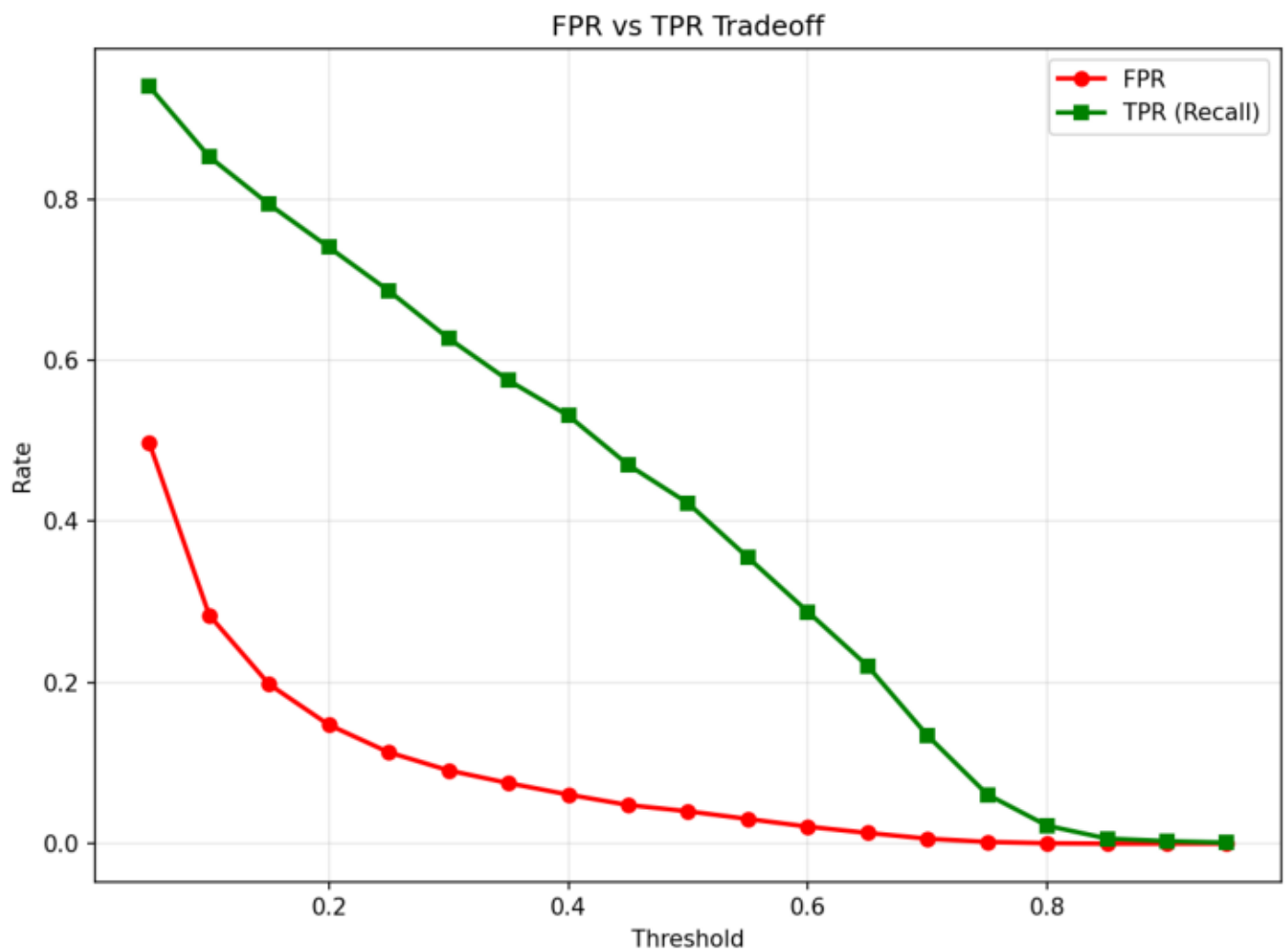
fpr_vs_threshold: FPR decreases as the threshold increases; use it to pick an operating point.

Overall - 2) Model Efficiency



efficiency_frontier: Each point shows the FPR/TPR tradeoff; move toward the top-left for better efficiency.

Overall - 2) Model Efficiency



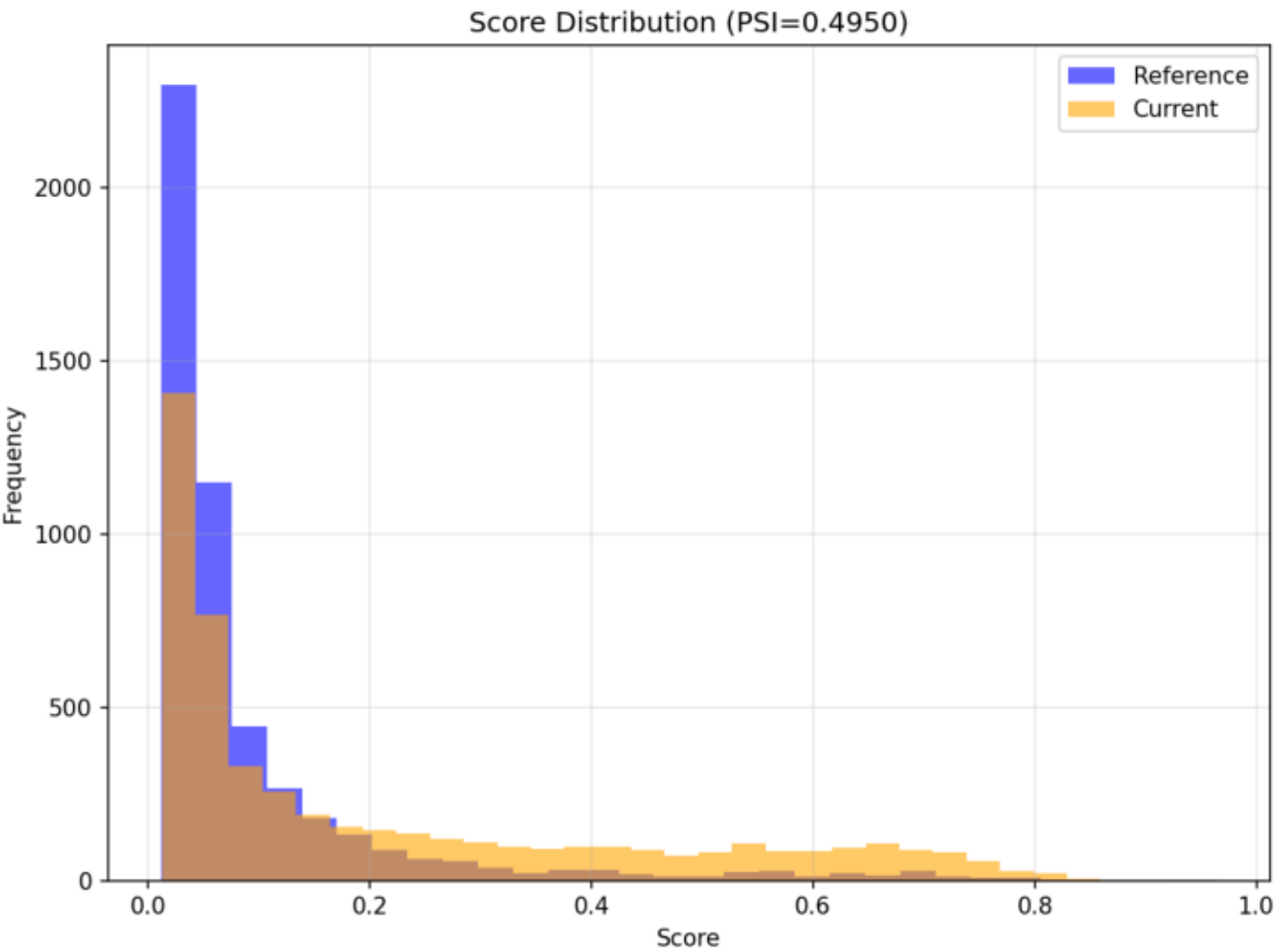
Overall - 3) Model Stability (1/2)

Metric	Figure	Explanation
psi	0.4950	Measures score-distribution shift between reference and current populations; 0.4950 is material. Next: investigate segment mix, data drift, and recalibration needs when PSI is material.
cv_auc_roc_mean	0.8711	Average AUC-ROC across validation folds. Next: compare it with overall AUC-ROC; a large gap can indicate overfitting or unstable sampling.
cv_auc_roc_std	0.0063	Standard deviation of AUC-ROC across folds. Next: if variability is high, simplify the model or increase training stability before deployment.
cv_auc_pr_mean	0.5992	Average AUC-PR across validation folds. Next: confirm that precision-recall performance remains acceptable under resampling.
cv_auc_pr_std	0.0250	Standard deviation of AUC-PR across folds. Next: high dispersion suggests unstable alert quality and warrants deeper validation.
bootstrap_auc_roc_mean	0.8512	Bootstrap mean AUC-ROC estimates expected ranking performance under repeated sampling. Next: compare it with the point estimate to judge optimism in the current test run.
bootstrap_auc_roc_ci_lower	0.8354	Lower confidence bound for bootstrap AUC-ROC. Next: use it as a conservative performance floor in model-risk discussions.
bootstrap_auc_roc_ci_upper	0.8630	Upper confidence bound for bootstrap AUC-ROC. Next: review the interval width with the lower bound to assess uncertainty.

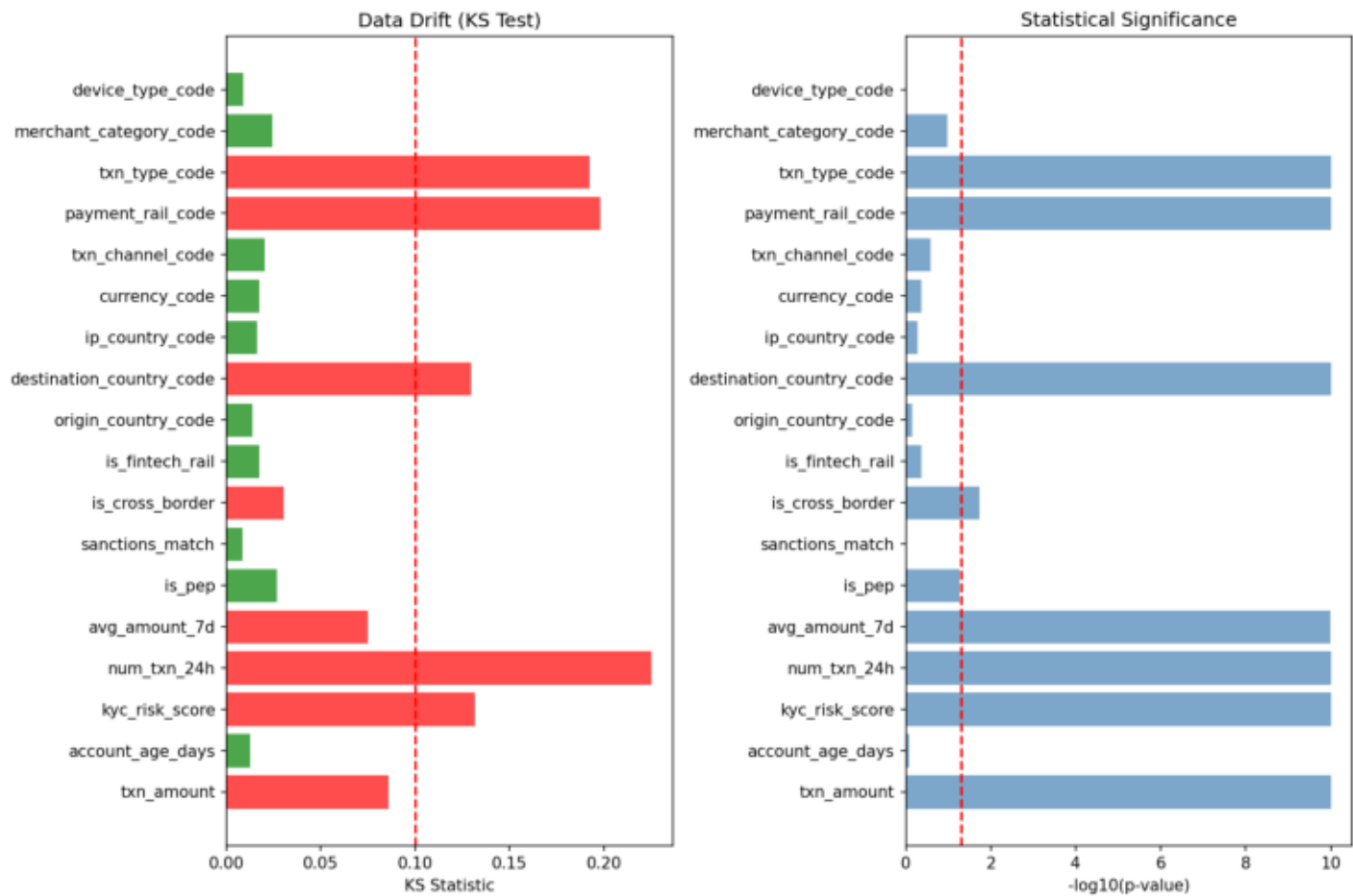
Overall - 3) Model Stability (2/2)

Metric	Figure	Explanation
concept_drift_detected	False	No material concept-drift signal was detected in the evaluated slices. Next: retain ongoing monitoring because stable historical results do not guarantee future stability.
concept_drift_score	0.0251	Quantifies the magnitude of performance change across ordered slices. Next: rising scores should trigger deeper review of segment behavior and model refresh timing.

Overall - 3) Model Stability

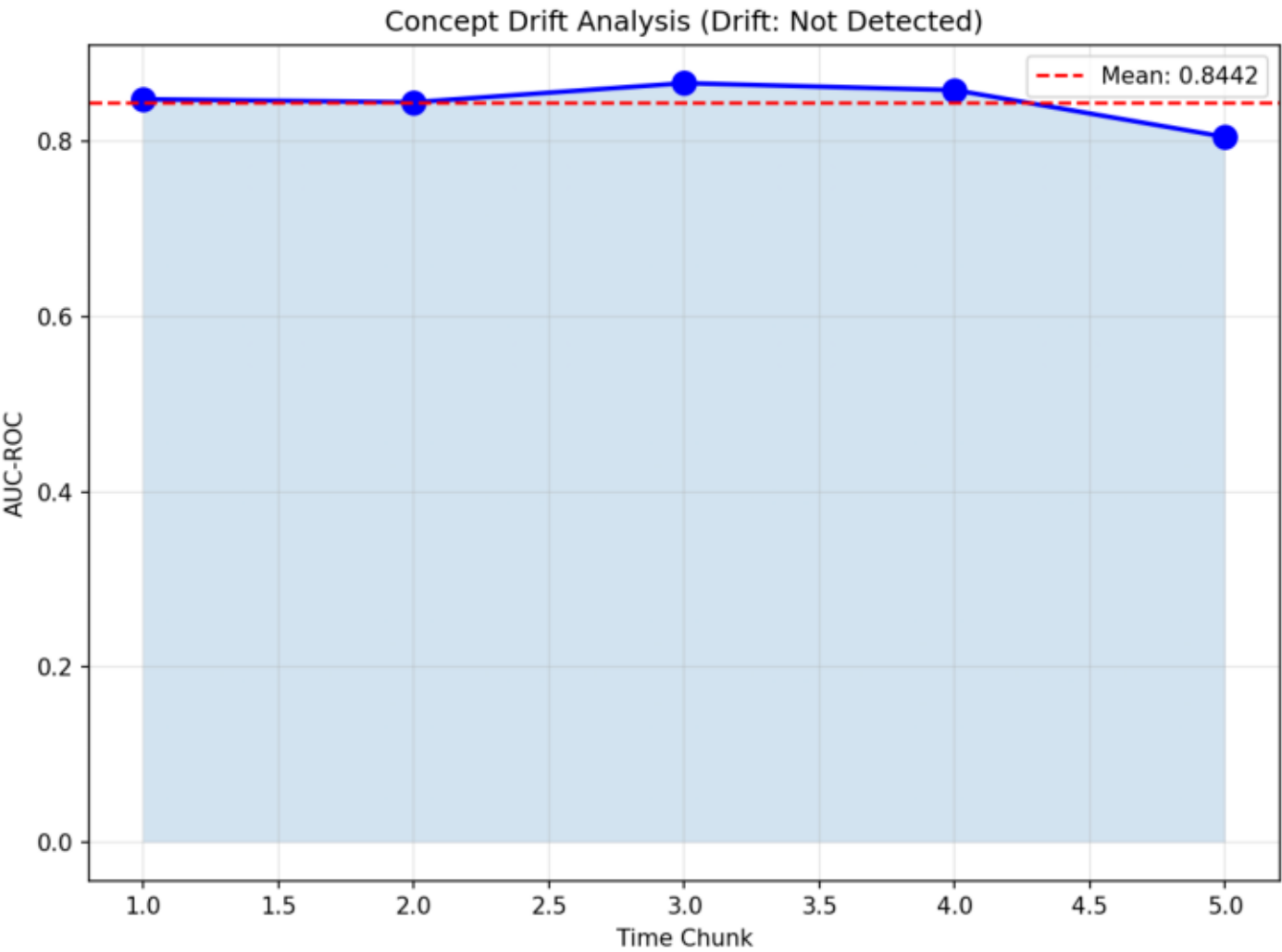


Overall - 3) Model Stability



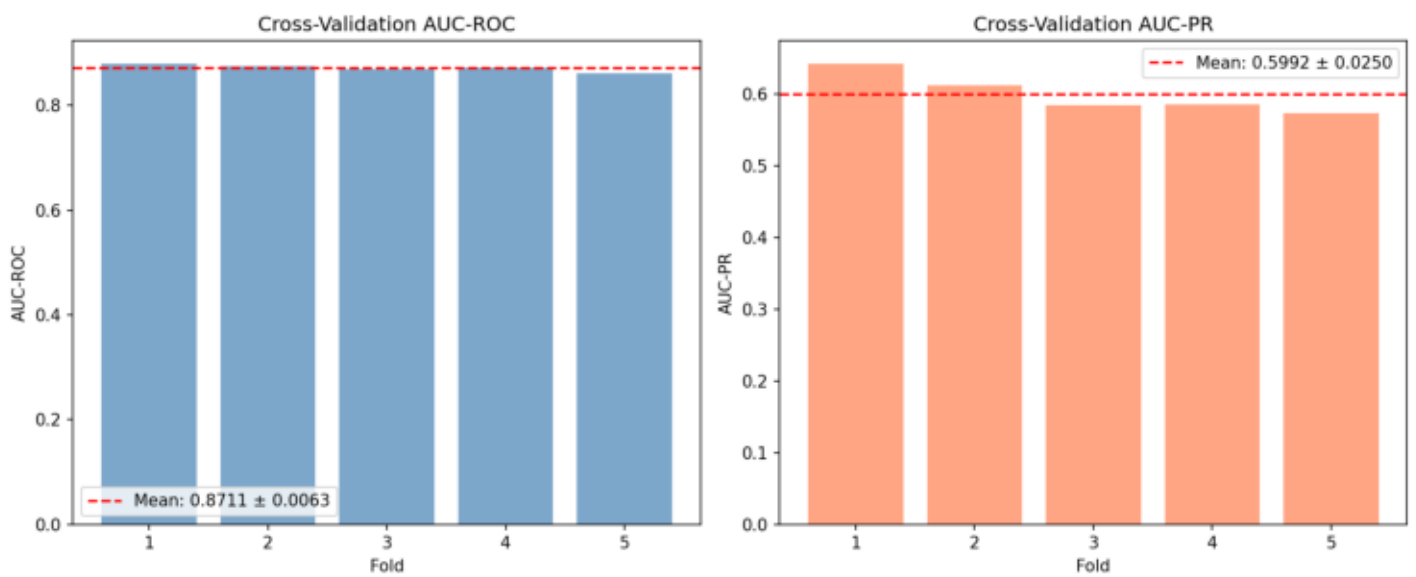
data_drift_heatmap: 8/18 features show drift; red bars highlight flagged features.

Overall - 3) Model Stability



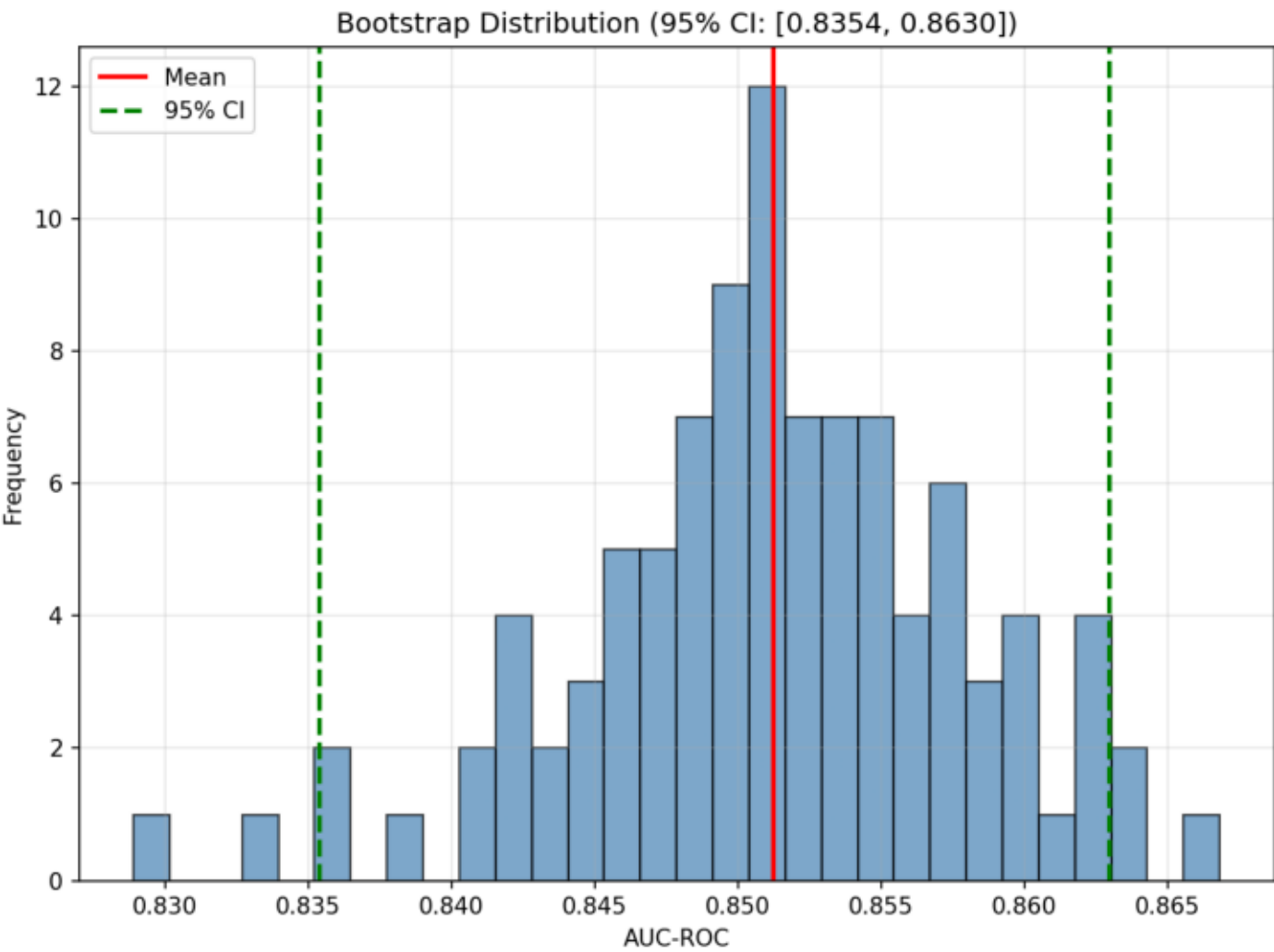
concept_drift: AUC across time chunks shows whether performance is stable over time.

Overall - 3) Model Stability

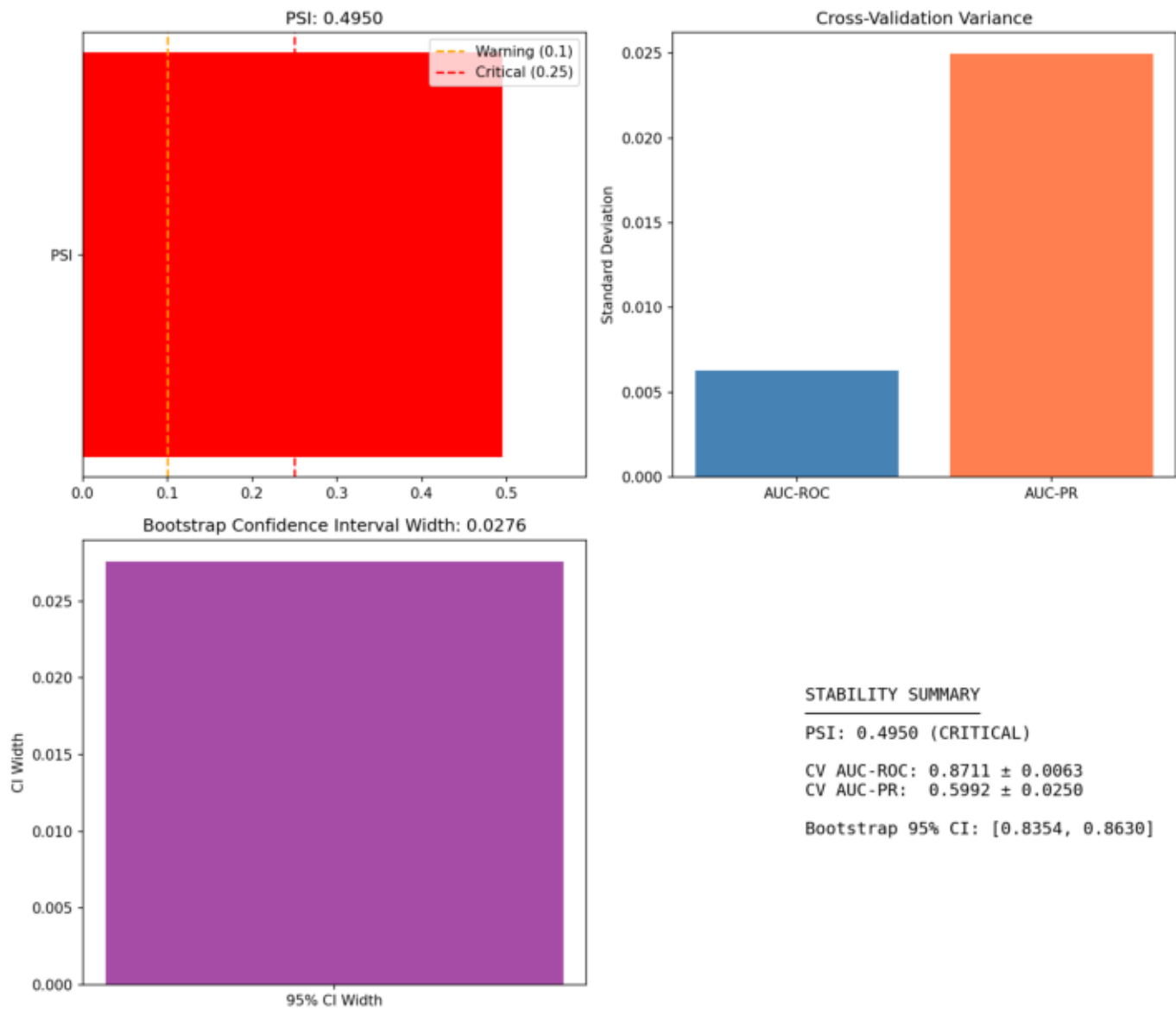


cv_results: Fold scores cluster around ROC 0.8711 and PR 0.5992.

Overall - 3) Model Stability



Overall - 3) Model Stability



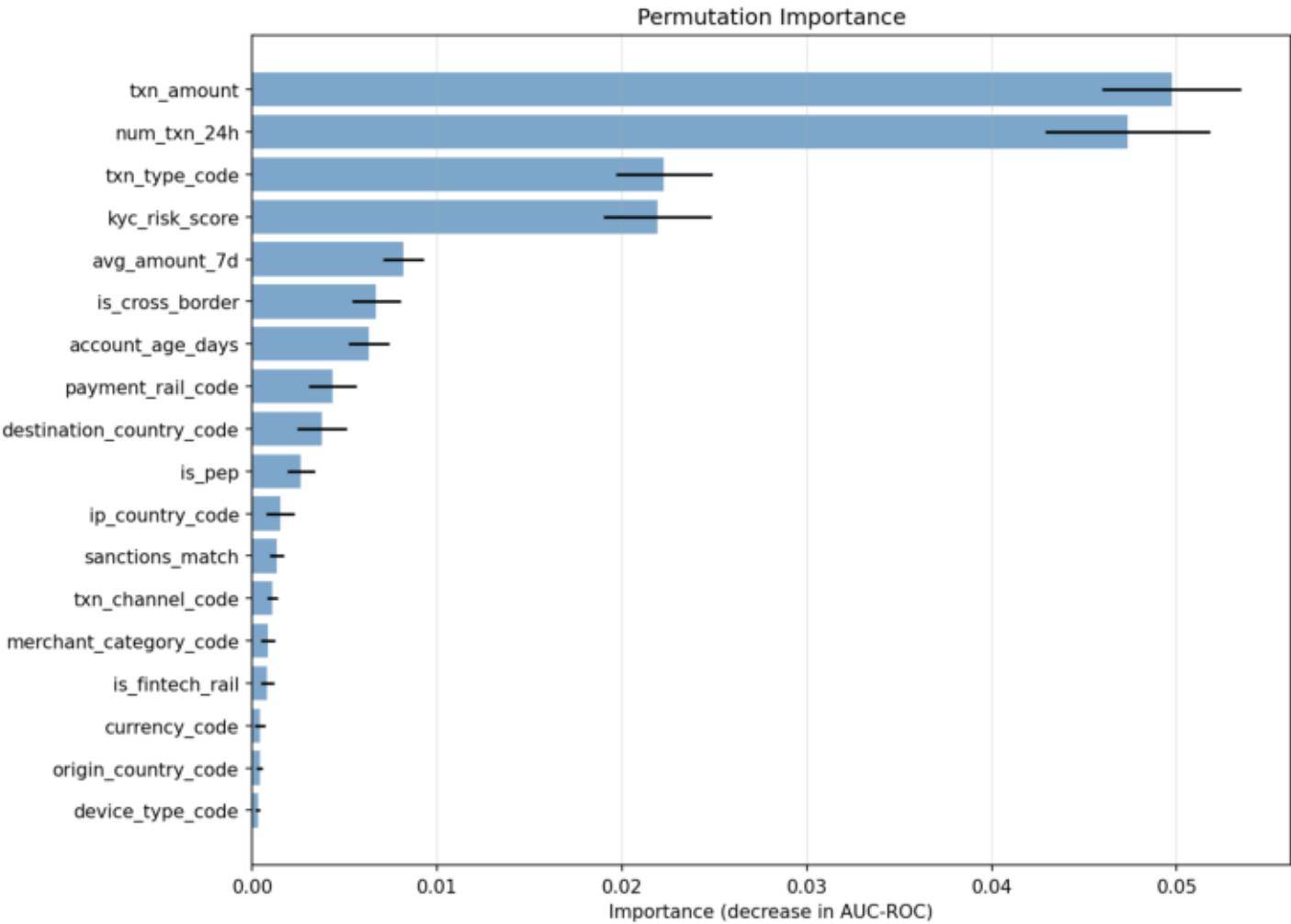
Overall - 4) Model Interpretability (1/2)

Metric	Figure	Explanation
model_type	tree	Identifies the estimator family so explanation methods can be interpreted in the right context. Next: confirm the chosen explanation methods are appropriate for this model class.
methods_used	permutation, pdp, ice, lime	Lists the interpretability methods produced in this run. Next: compare signals across methods before using them in documentation or model challenge.
shap_status	unavailable	Summarizes interpretability evidence for this run. Next: use these outputs to document model drivers and challenge unexpected dependencies.
shap_reason	No module named 'shap'	Summarizes interpretability evidence for this run. Next: use these outputs to document model drivers and challenge unexpected dependencies.
perm_top_features	txn_amount, num_txn_24h, txn_type_code, kyc_risk_score, avg_amount_7d ...	Shows features with the largest performance drop when permuted, which highlights global importance. Next: validate that the top drivers align with business logic, feature governance, and policy expectations.
lime_instances	3	Counts the locally explained observations reviewed with LIME. Next: examine representative true-positive and false-positive cases to understand local decision logic.

Overall - 4) Model Interpretability (2/2)

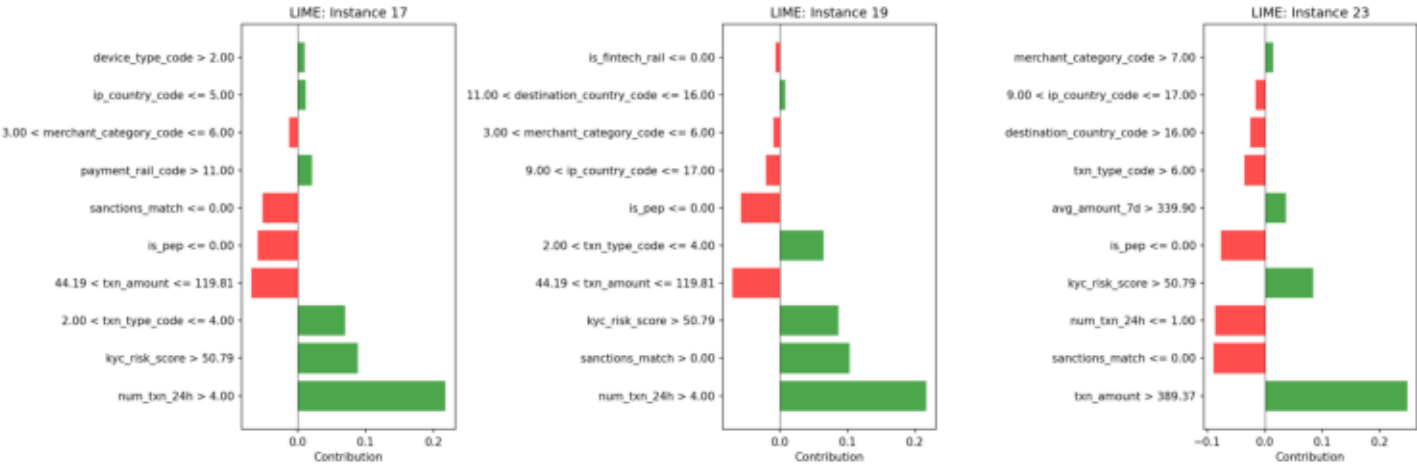
Metric	Figure	Explanation
pdp_features	txn_amount, avg_amount_7d, account_age_days, kyc_risk_score, num_txn_24h ...	Lists features with partial-dependence plots. Next: review whether average model response is monotonic, stable, and operationally sensible.
ice_features	txn_amount, avg_amount_7d, account_age_days, kyc_risk_score	Lists features with individual conditional expectation plots. Next: inspect heterogeneity and interactions that may be hidden in average-only views.

Overall - 4) Model Interpretability



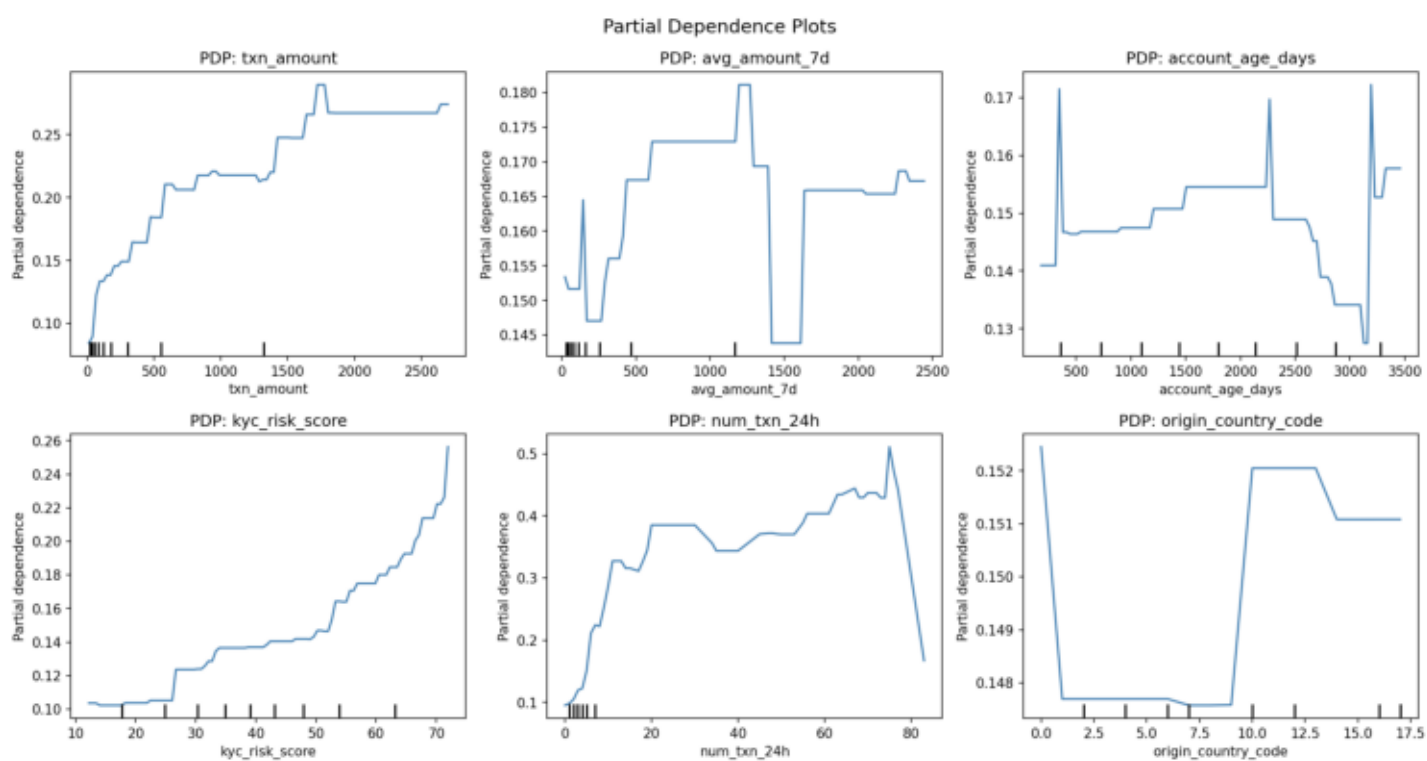
permutation_importance: Bars show how much AUC drops when each feature is permuted.

Overall - 4) Model Interpretability



lime_explanation: LIME bars show local feature contributions for selected instances.

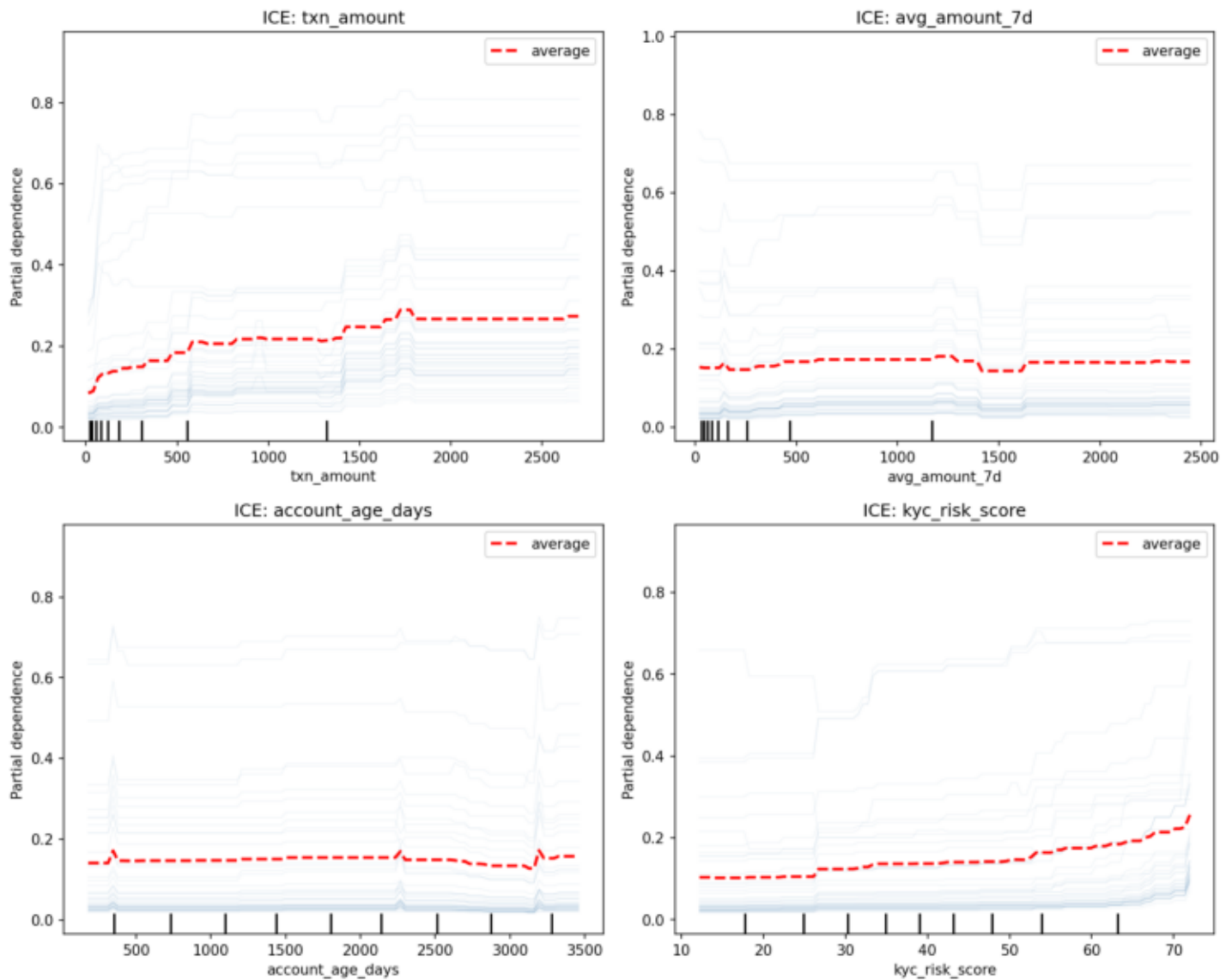
Overall - 4) Model Interpretability



pdp: PDP curves show average model response as a feature varies.

Overall - 4) Model Interpretability

Individual Conditional Expectation (ICE) Plots



ice: ICE curves show per-instance responses as a feature varies.

value_date_quarter=2024Q1 - 1) Model Effectiveness (1/3)

Metric	Figure	Explanation
auc_roc	0.8592	Measures ranking power across all thresholds. 0.8592 indicates strong separation. Next: compare AUC-ROC across segments and keep threshold tuning separate from ranking validation.
auc_pr	0.5323	Measures precision-recall performance under class imbalance; the observed event rate is 6.97%. Next: use this together with alert capacity to decide whether the high-score queue is practically useful.
precision	0.8305	Shows the share of flagged cases that are true positives at the current threshold. Next: if this is below review expectations, raise the threshold or improve exclusion features.
recall	0.3311	Shows the share of true positives captured at the current threshold. Next: if missed cases are too high, lower the threshold or add stronger risk features.
f1	0.4734	Balances precision and recall in one score. Next: use it as a threshold-selection aid when both missed cases and false alerts matter.
ks_statistic	0.5499	Measures score separation between positive and negative classes. Next: low KS suggests the model needs better features or a different specification.

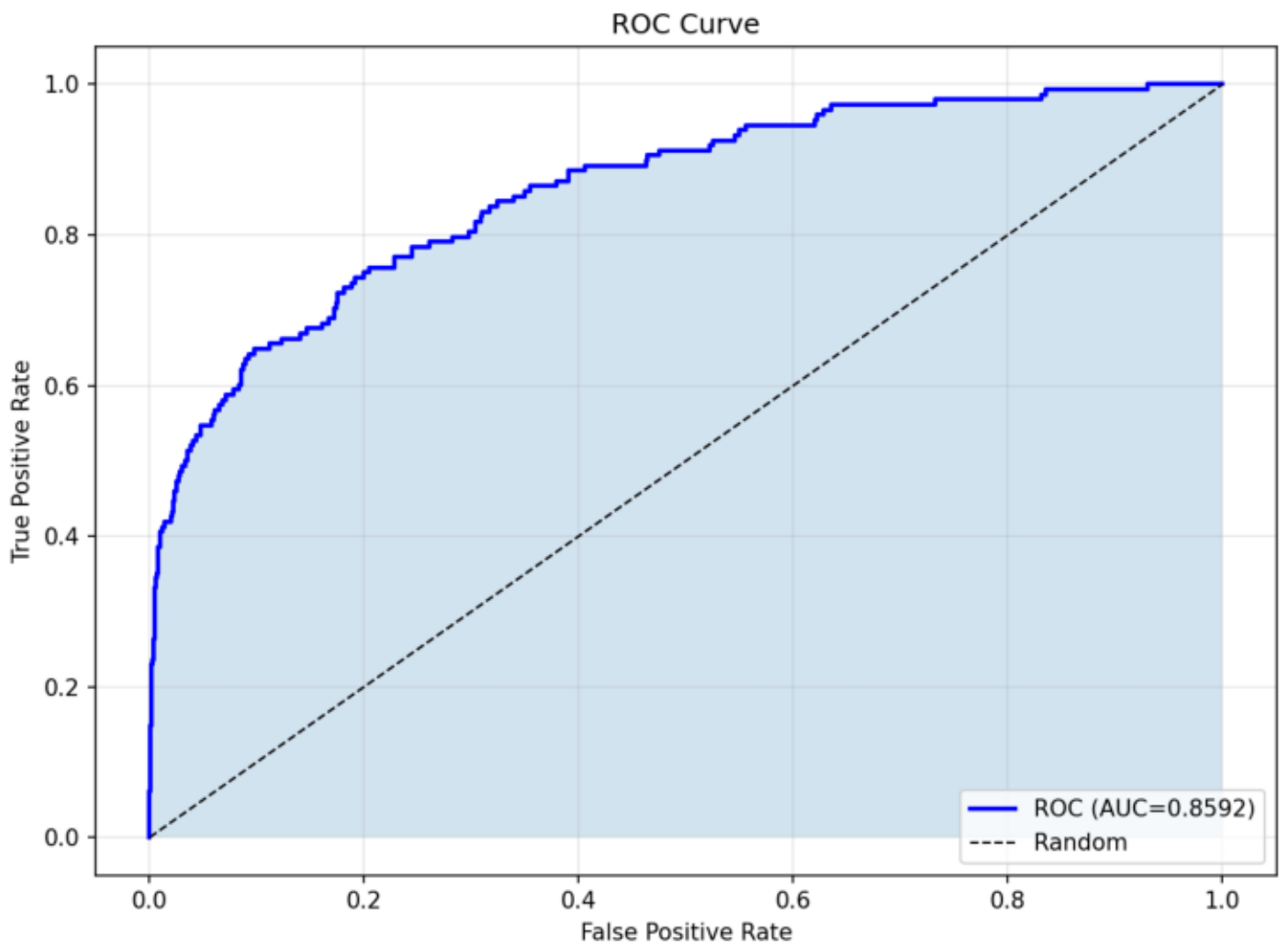
value_date_quarter=2024Q1 - 1) Model Effectiveness (2/3)

Metric	Figure	Explanation
ks_threshold	0.1400	This is the threshold that maximizes KS separation. Next: treat it as a candidate operating point, then compare it with business capacity and policy limits.
confusion_matrix[TN]	1964	True negatives correctly cleared. Next: preserve this base while tuning the alert threshold.
confusion_matrix[FP]	10	False positives create review cost without SAR conversion. Next: reduce this count with threshold tuning or sharper risk features.
confusion_matrix[FN]	99	False negatives are missed SAR cases. Next: if this count is unacceptable, lower the threshold or add features that capture suspicious behavior earlier.
confusion_matrix[TP]	49	True positives are correctly escalated cases. Next: assess whether this volume is sufficient relative to expected SAR capture goals.
precision_at_k[10]	0.9000	Shows how accurate the top-10 highest-risk queue is. Next: align K with investigator capacity and confirm that top alerts stay high quality.
precision_at_k[50]	0.8200	Shows how accurate the top-50 highest-risk queue is. Next: align K with investigator capacity and confirm that top alerts stay high quality.
precision_at_k[100]	0.6200	Shows how accurate the top-100 highest-risk queue is. Next: align K with investigator capacity and confirm that top alerts stay high quality.

value_date_quarter=2024Q1 - 1) Model Effectiveness (3/3)

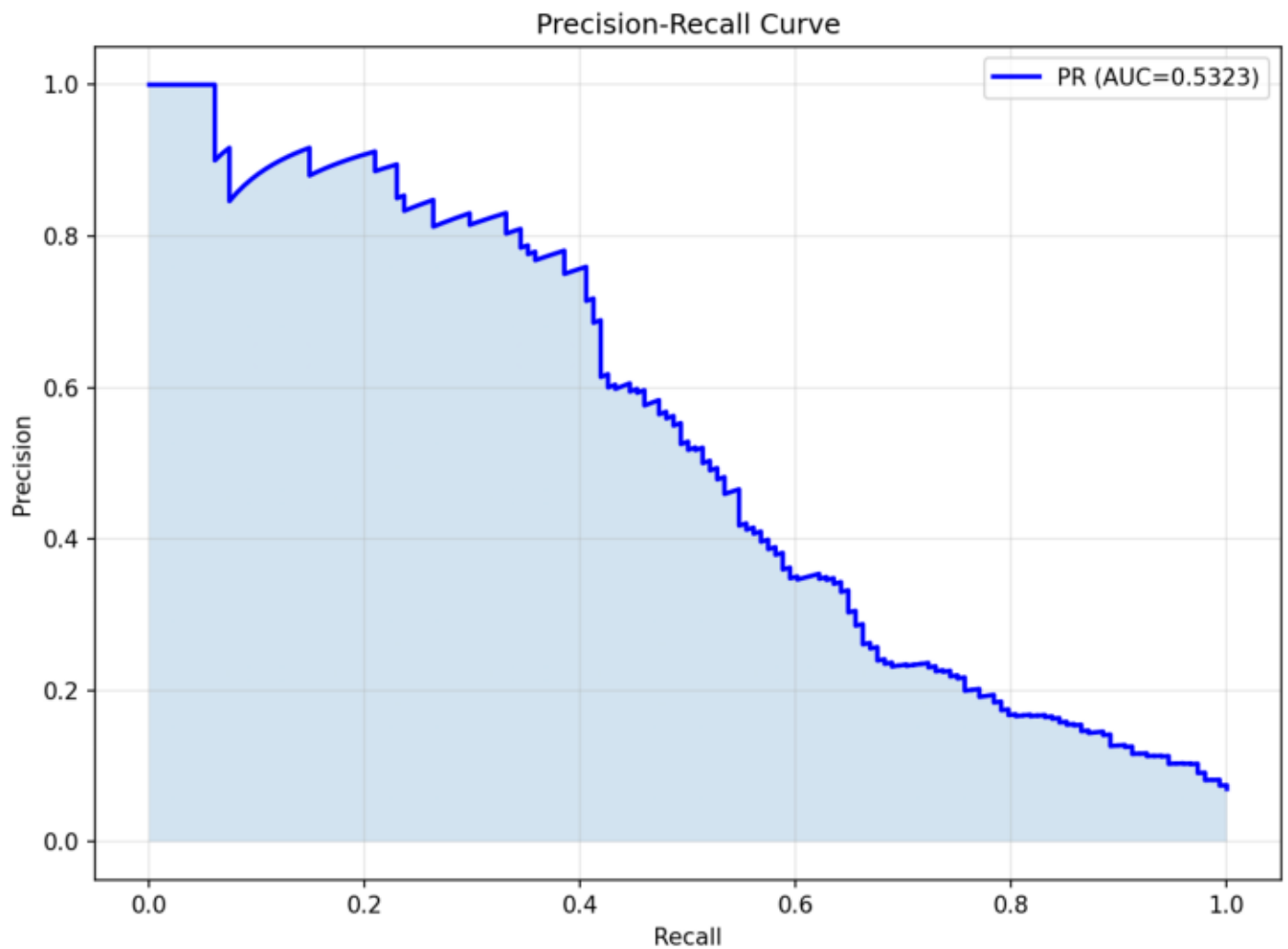
Metric	Figure	Explanation
precision_at_k[200]	0.4150	Shows how accurate the top-200 highest-risk queue is. Next: align K with investigator capacity and confirm that top alerts stay high quality.
precision_at_k[500]	0.2200	Shows how accurate the top-500 highest-risk queue is. Next: align K with investigator capacity and confirm that top alerts stay high quality.
recall_at_k[10]	0.0608	Shows how much of the positive population is captured within the top-10 queue. Next: increase K if coverage is too low, or improve ranking quality if capacity cannot expand.
recall_at_k[50]	0.2770	Shows how much of the positive population is captured within the top-50 queue. Next: increase K if coverage is too low, or improve ranking quality if capacity cannot expand.
recall_at_k[100]	0.4189	Shows how much of the positive population is captured within the top-100 queue. Next: increase K if coverage is too low, or improve ranking quality if capacity cannot expand.
recall_at_k[200]	0.5608	Shows how much of the positive population is captured within the top-200 queue. Next: increase K if coverage is too low, or improve ranking quality if capacity cannot expand.
recall_at_k[500]	0.7432	Shows how much of the positive population is captured within the top-500 queue. Next: increase K if coverage is too low, or improve ranking quality if capacity cannot expand.

value_date_quarter=2024Q1 - 1) Model Effectiveness



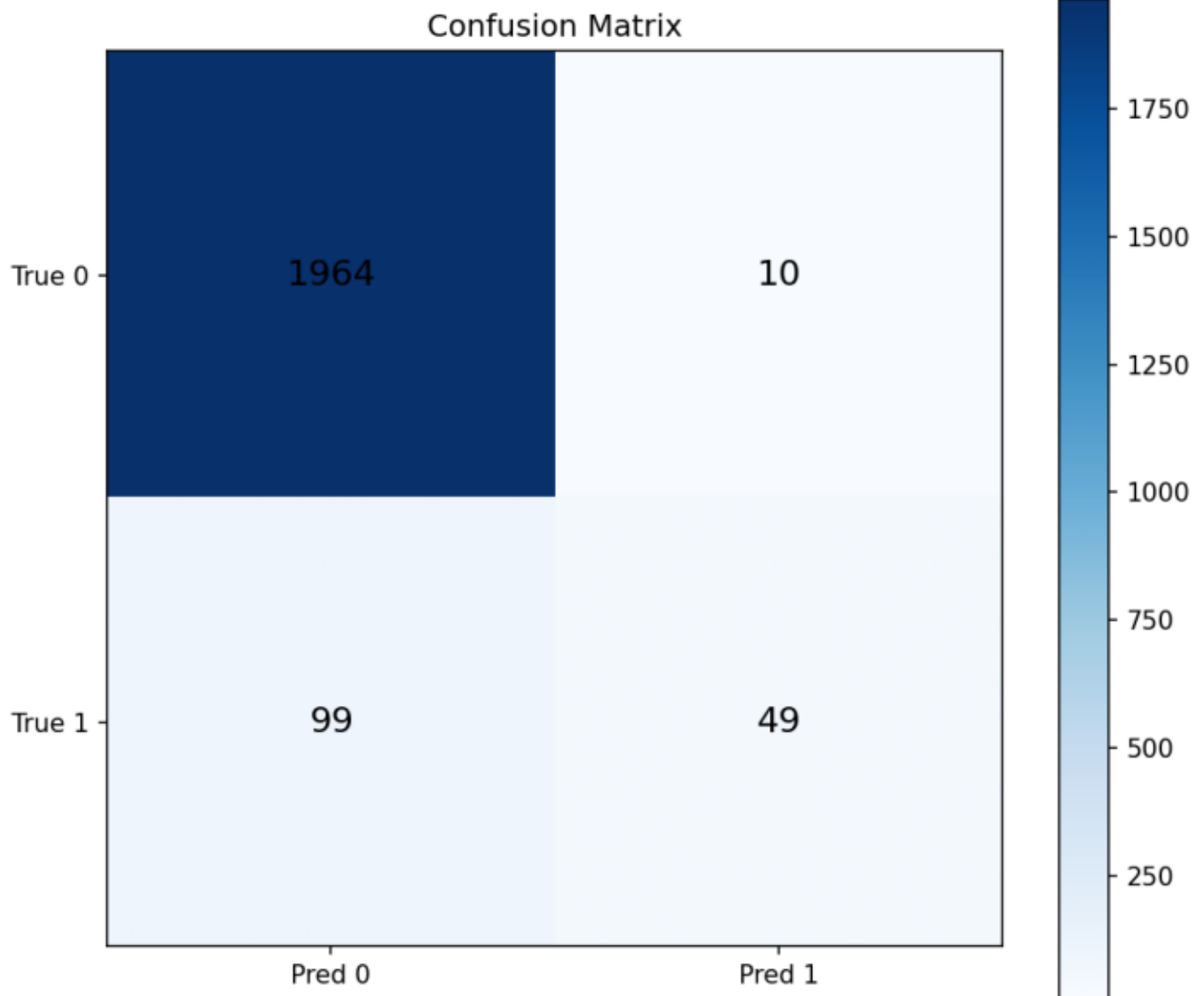
roc_curve: AUC-ROC=0.8592; curve above diagonal indicates strong separation.

value_date_quarter=2024Q1 - 1) Model Effectiveness



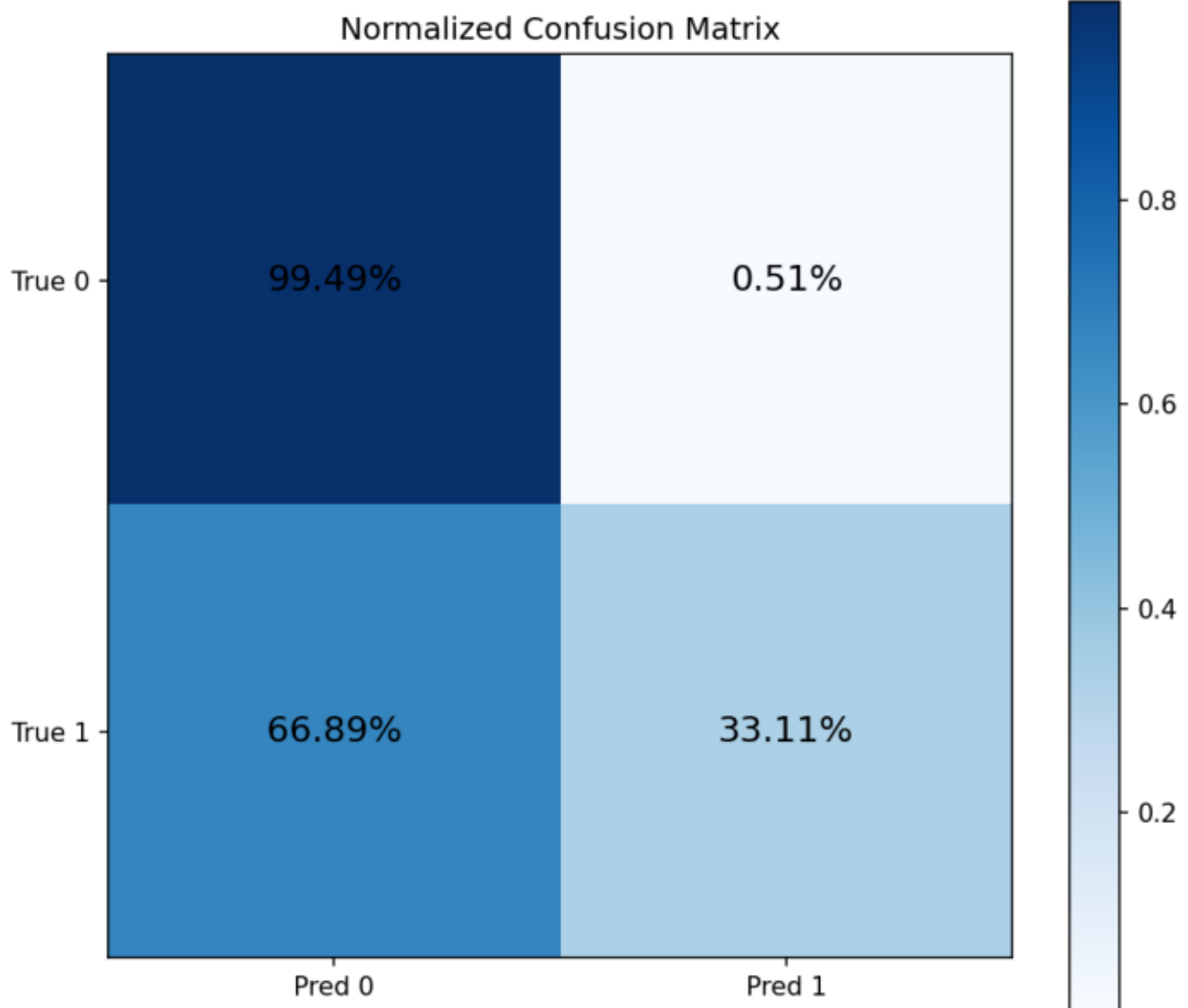
pr_curve: AUC-PR=0.5323 vs baseline 6.97%; higher curve implies better precision at recall.

value_date_quarter=2024Q1 - 1) Model Effectiveness



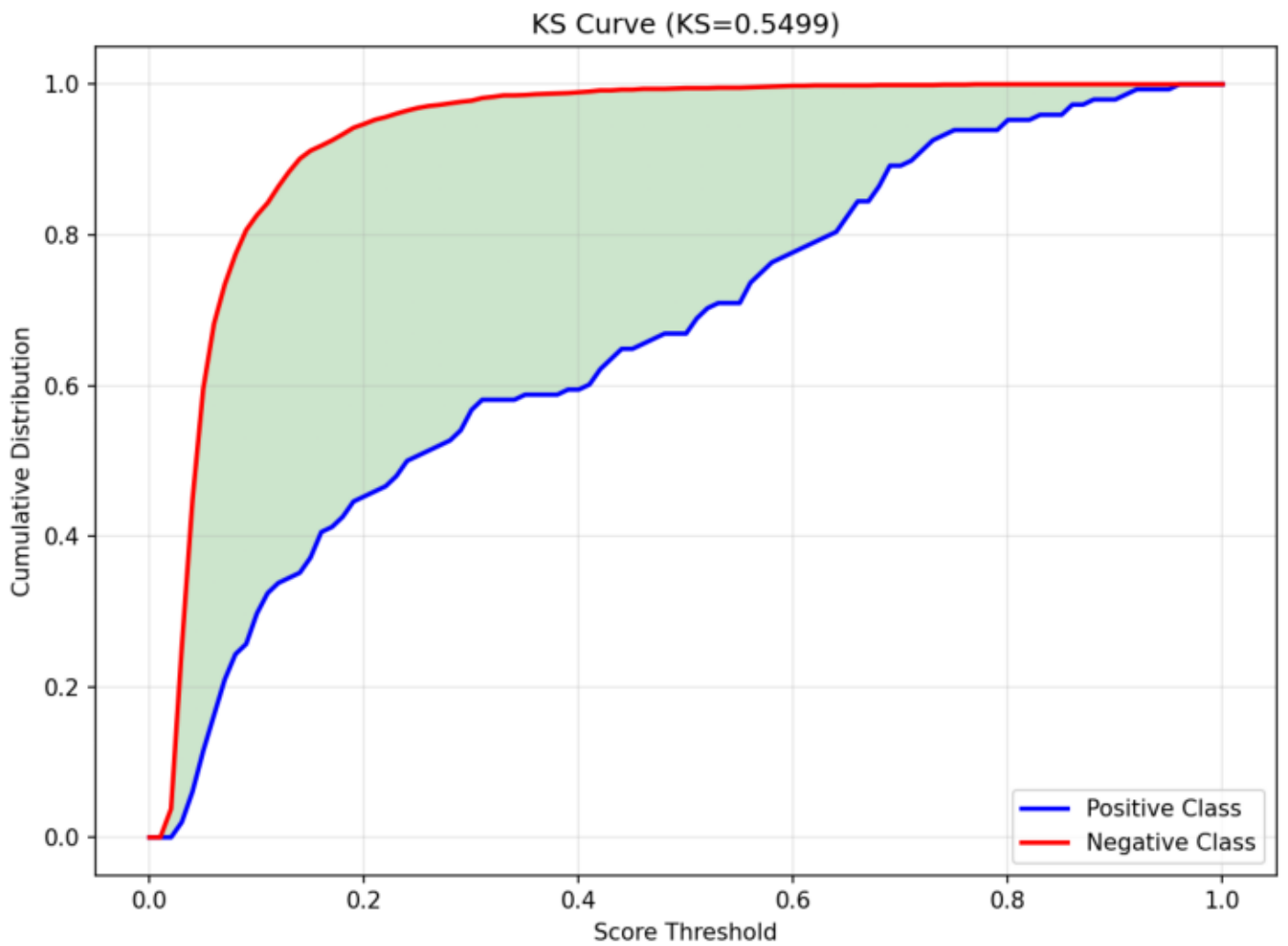
confusion_matrix: At threshold 0.50: TP=49, FP=10, FN=99, TN=1964.

value_date_quarter=2024Q1 - 1) Model Effectiveness



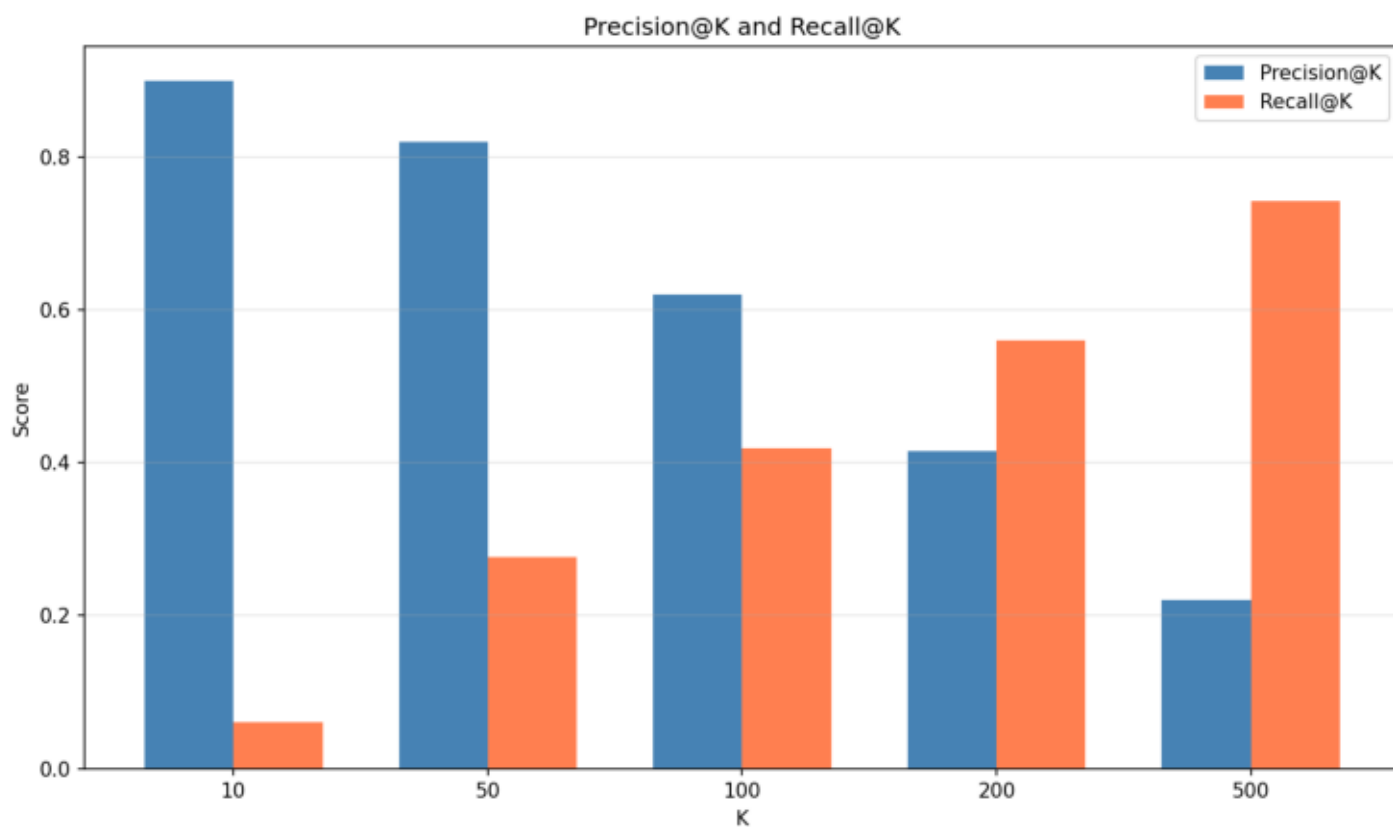
confusion_matrix_norm: Normalized rates: TPR=33.11%, TNR=99.49%.

value_date_quarter=2024Q1 - 1) Model Effectiveness



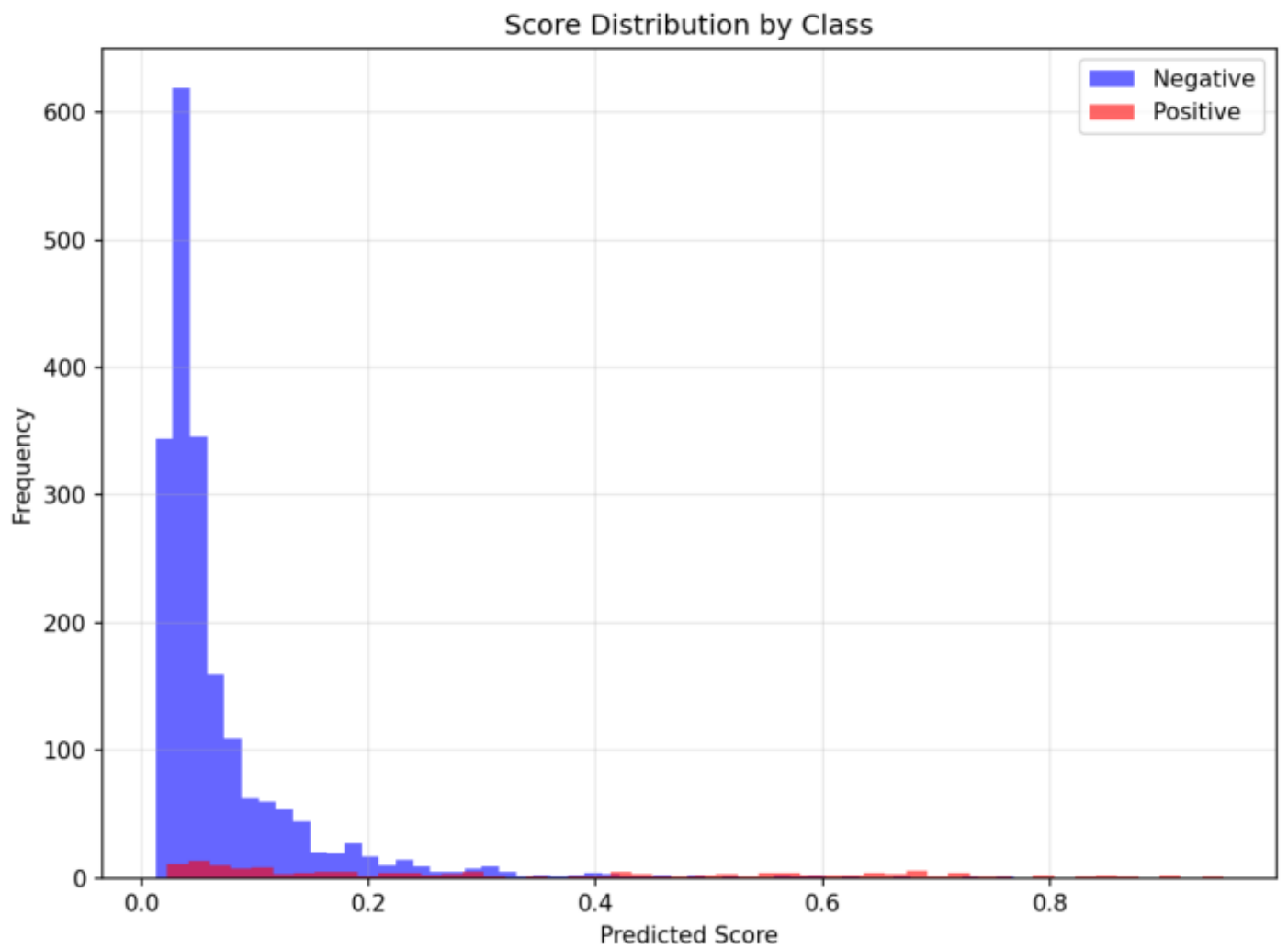
ks_curve: Maximum separation KS=0.5499 at threshold 0.14.

value_date_quarter=2024Q1 - 1) Model Effectiveness



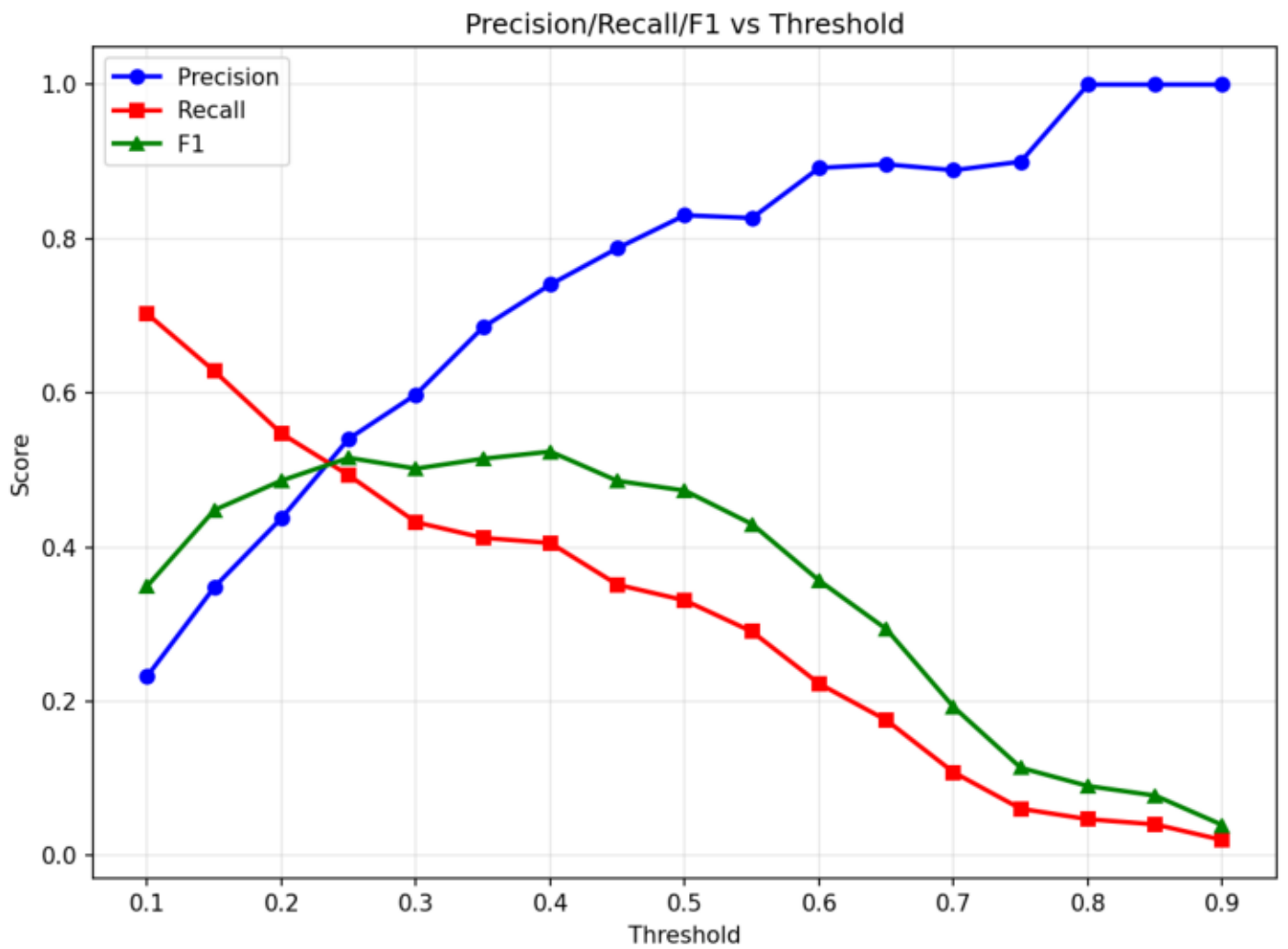
precision_recall_at_k: Bars show precision/recall as you expand the review queue.

value_date_quarter=2024Q1 - 1) Model Effectiveness



score_distribution: Mean score: positive=0.332, negative=0.069 (larger gap is better).

value_date_quarter=2024Q1 - 1) Model Effectiveness



threshold_analysis: Best F1 in tested grid is 0.5240 at threshold 0.40.

value_date_quarter=2024Q1 - 2) Model Efficiency (1/4)

Metric	Figure	Explanation
fpr	0.0051	Measures the share of negatives incorrectly flagged; 0.0051 is low. Next: if review demand is too high, raise the threshold or add features that better suppress false alerts.
tn	1964	Counts negatives correctly cleared without manual review. Next: preserve this clearance rate while checking that recall does not fall below policy expectations.
fp	10	Counts non-events that would still be sent to review. Next: use this with staffing capacity to decide whether threshold or feature refinement is required.
threshold	0.5000	This is the active operating cutoff used to flag alerts. Next: tune it jointly with recall, precision, and review capacity rather than in isolation.
fpr_at_thresholds[thresholds.t_0.05]	0.4043	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.05. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.10]	0.1738	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.10. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.

value_date_quarter=2024Q1 - 2) Model Efficiency (2/4)

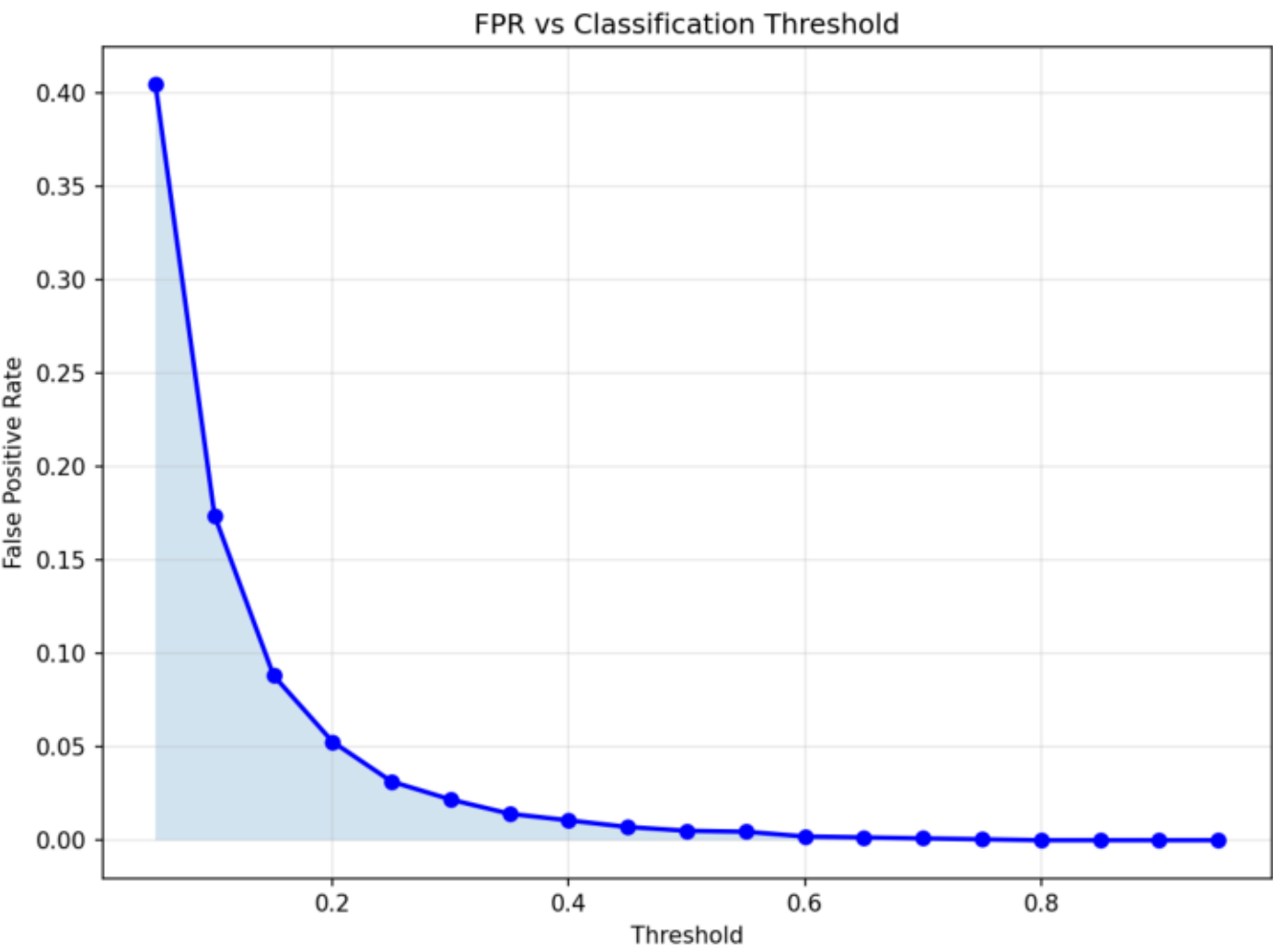
Metric	Figure	Explanation
fpr_at_thresholds[thresholds.t_0.15]	0.0881	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.15. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.20]	0.0527	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.20. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.25]	0.0314	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.25. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.30]	0.0218	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.30. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.35]	0.0142	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.35. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.40]	0.0106	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.40. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.

value_date_quarter=2024Q1 - 2) Model Efficiency (3/4)

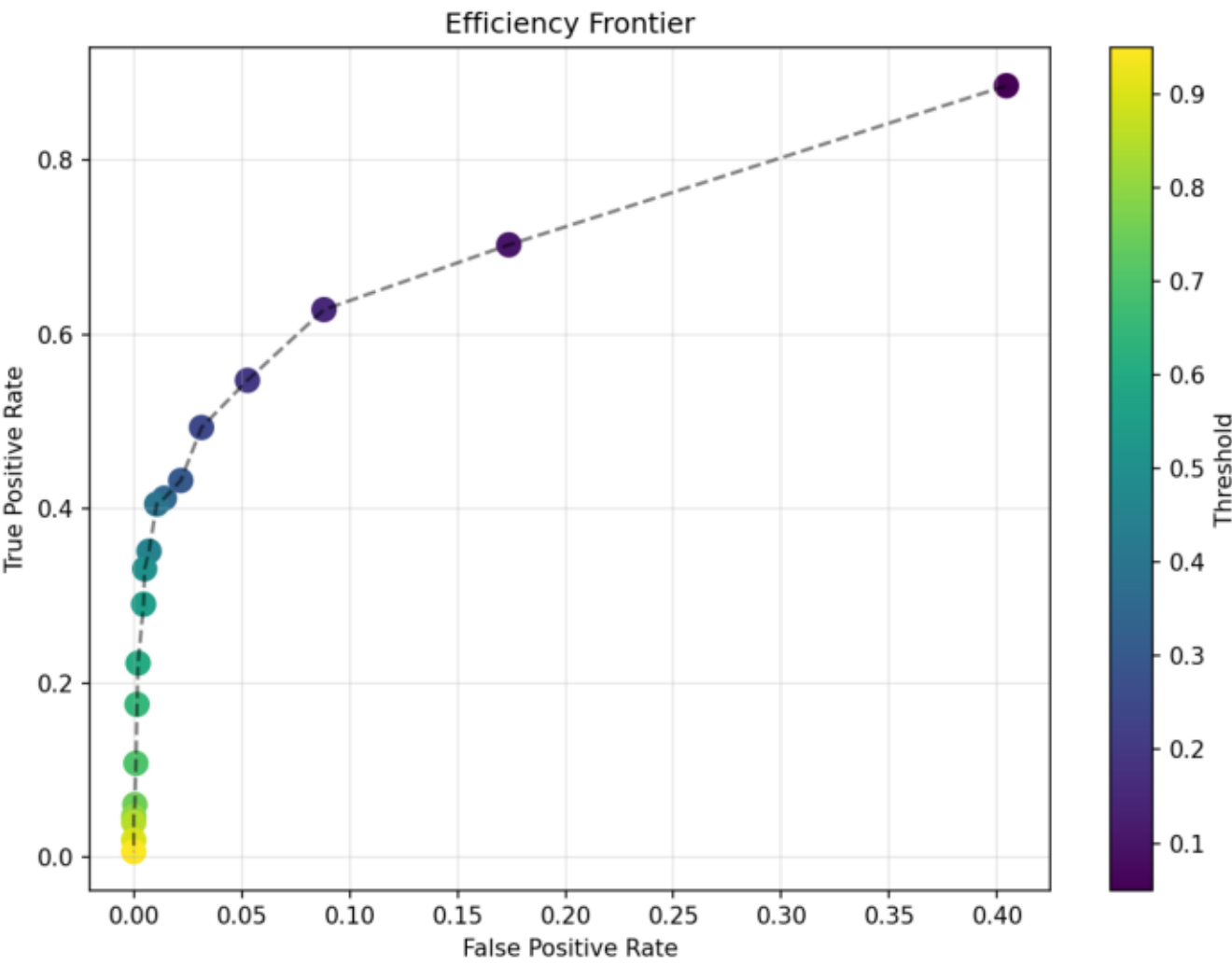
Metric	Figure	Explanation
fpr_at_thresholds[thresholds.t_0.45]	0.0071	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.45. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.50]	0.0051	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.50. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.55]	0.0046	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.55. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.60]	0.0020	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.60. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.65]	0.0015	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.65. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.70]	0.0010	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.70. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.

value_date_quarter=2024Q1 - 2) Model Efficiency (4/4)

Metric	Figure	Explanation
fpr_at_thresholds[thresholds.t_0.75]	0.0005	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.75. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.80]	0.0000	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.80. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.85]	0.0000	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.85. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.90]	0.0000	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.90. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.95]	0.0000	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.95. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.

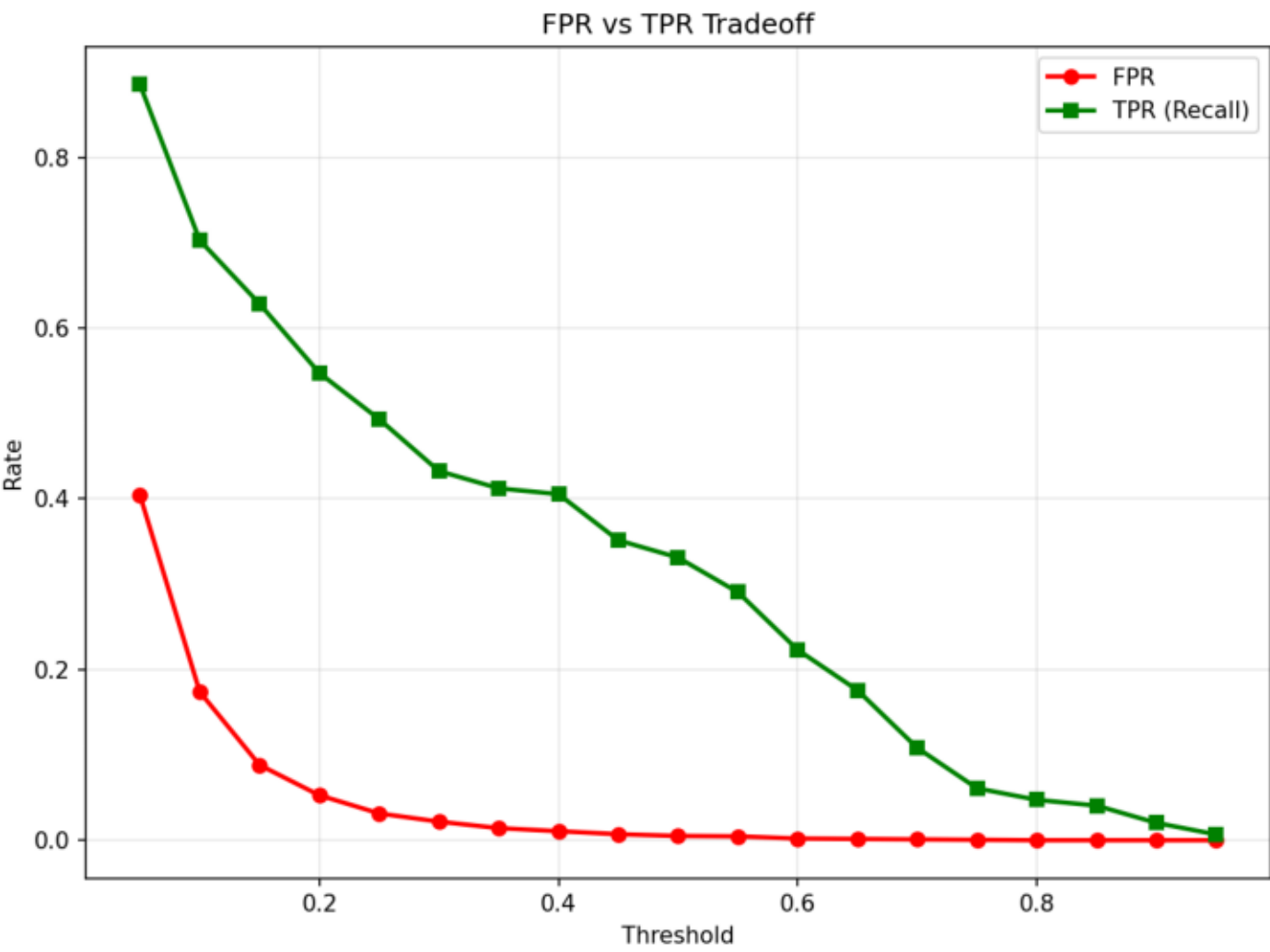


value_date_quarter=2024Q1 - 2) Model Efficiency



efficiency_frontier: Each point shows the FPR/TPR tradeoff; move toward the top-left for better efficiency.

value_date_quarter=2024Q1 - 2) Model Efficiency



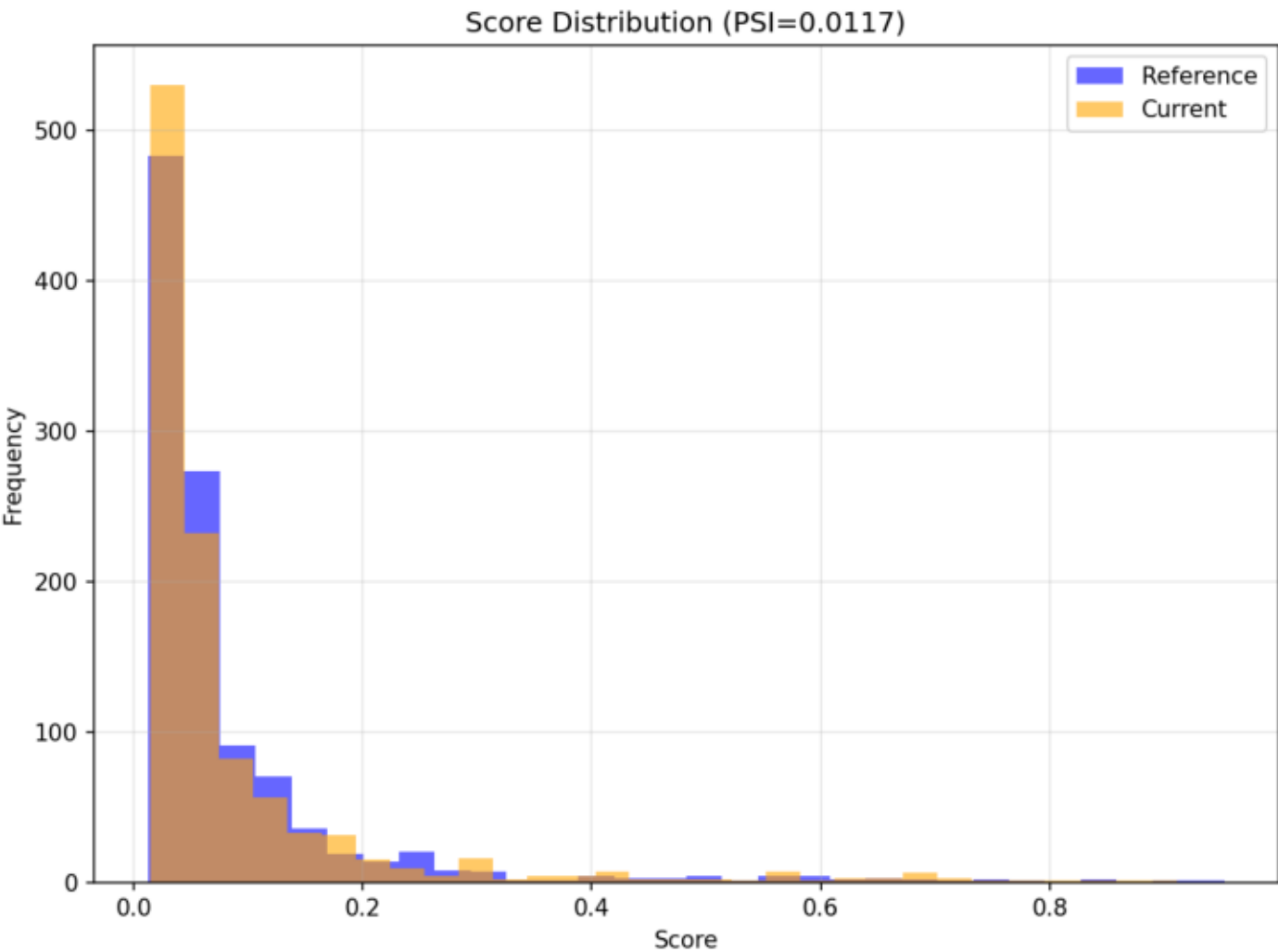
value_date_quarter=2024Q1 - 3) Model Stability (1/2)

Metric	Figure	Explanation
psi	0.0117	Measures score-distribution shift between reference and current populations; 0.0117 is negligible. Next: investigate segment mix, data drift, and recalibration needs when PSI is material.
cv_auc_roc_mean	0.8587	Average AUC-ROC across validation folds. Next: compare it with overall AUC-ROC; a large gap can indicate overfitting or unstable sampling.
cv_auc_roc_std	0.0369	Standard deviation of AUC-ROC across folds. Next: if variability is high, simplify the model or increase training stability before deployment.
cv_auc_pr_mean	0.5311	Average AUC-PR across validation folds. Next: confirm that precision-recall performance remains acceptable under resampling.
cv_auc_pr_std	0.1497	Standard deviation of AUC-PR across folds. Next: high dispersion suggests unstable alert quality and warrants deeper validation.
bootstrap_auc_roc_mean	0.8605	Bootstrap mean AUC-ROC estimates expected ranking performance under repeated sampling. Next: compare it with the point estimate to judge optimism in the current test run.
bootstrap_auc_roc_ci_lower	0.8226	Lower confidence bound for bootstrap AUC-ROC. Next: use it as a conservative performance floor in model-risk discussions.
bootstrap_auc_roc_ci_upper	0.9054	Upper confidence bound for bootstrap AUC-ROC. Next: review the interval width with the lower bound to assess uncertainty.

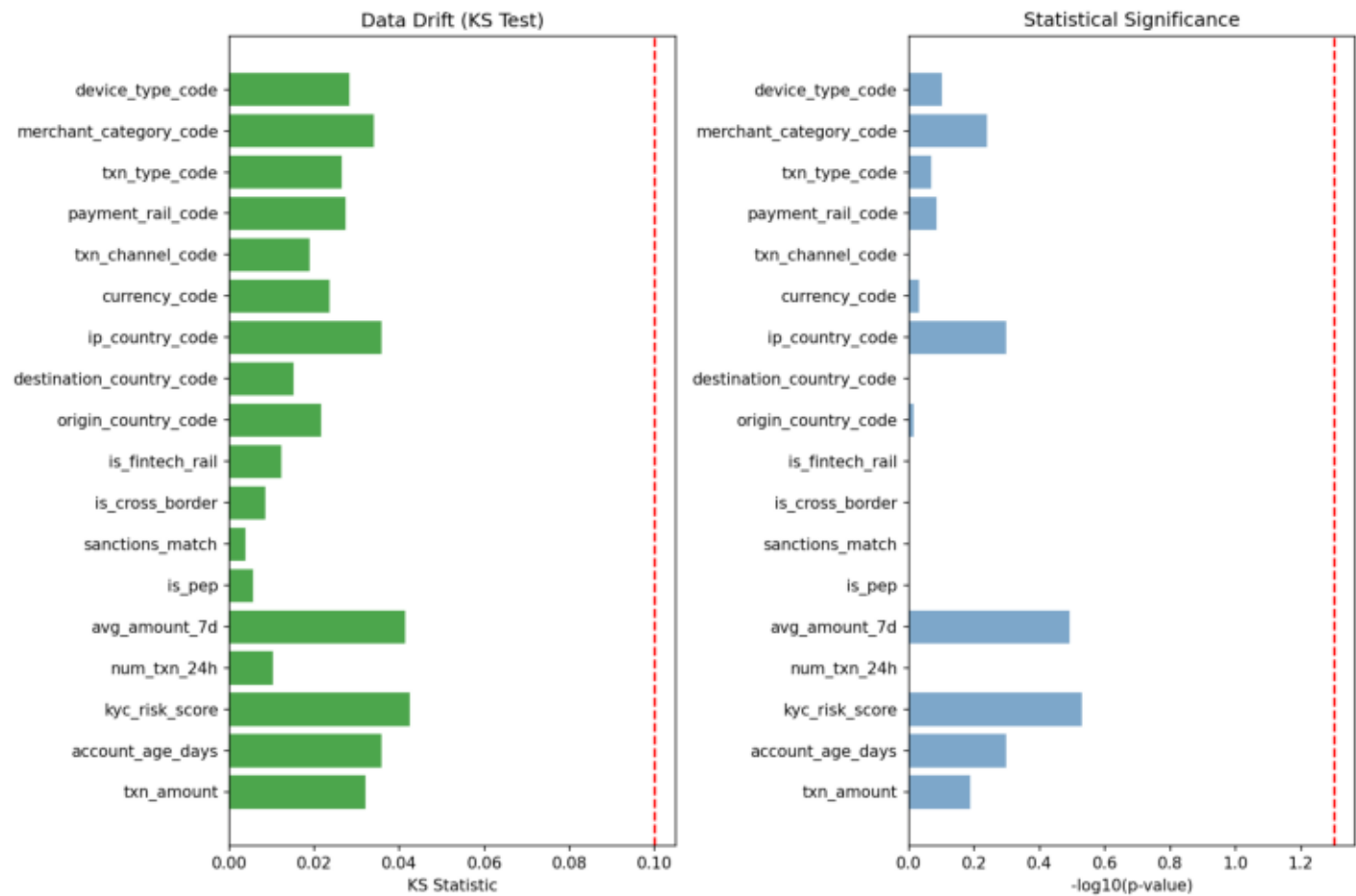
value_date_quarter=2024Q1 - 3) Model Stability (2/2)

Metric	Figure	Explanation
concept_drift_detected	False	No material concept-drift signal was detected in the evaluated slices. Next: retain ongoing monitoring because stable historical results do not guarantee future stability.
concept_drift_score	0.0441	Quantifies the magnitude of performance change across ordered slices. Next: rising scores should trigger deeper review of segment behavior and model refresh timing.

value_date_quarter=2024Q1 - 3) Model Stability

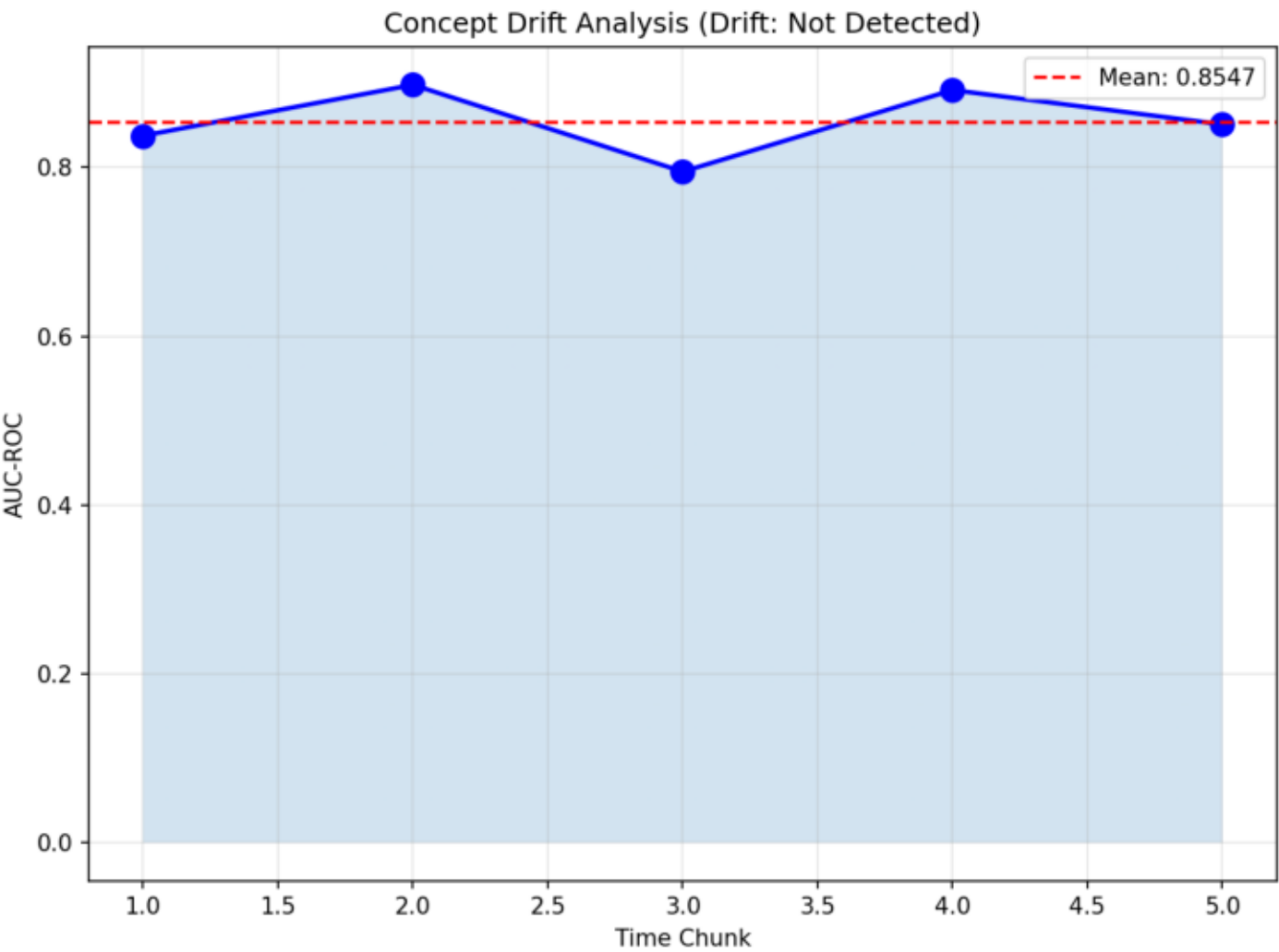


value_date_quarter=2024Q1 - 3) Model Stability



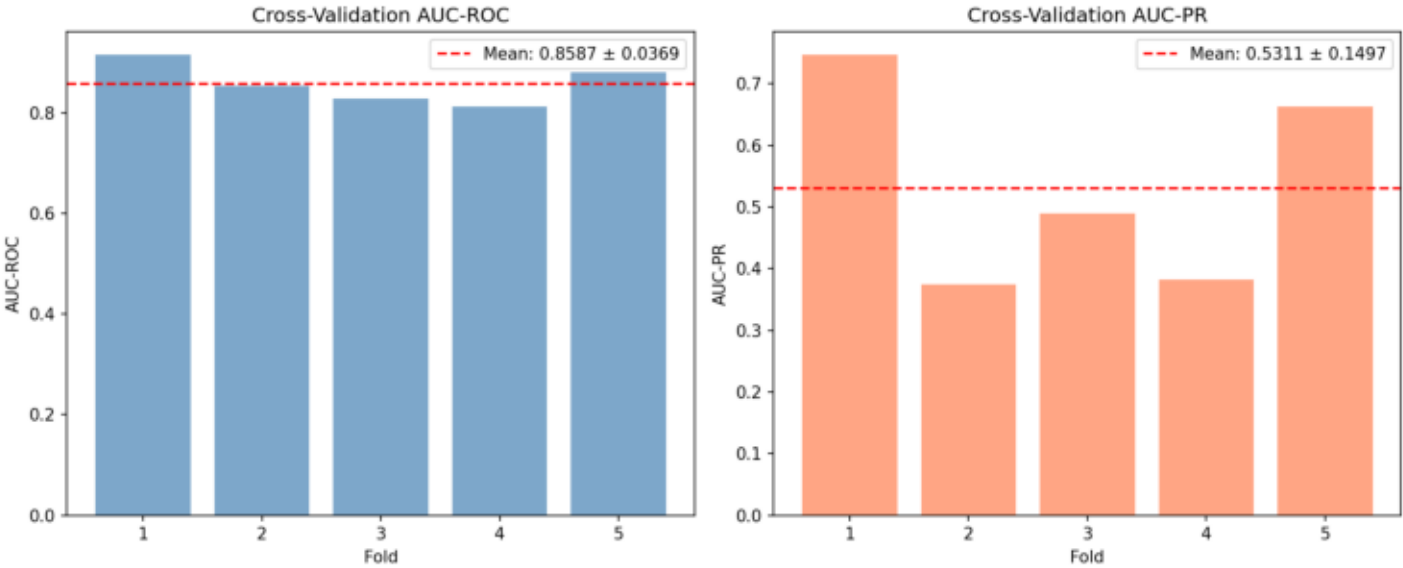
data_drift_heatmap: 0/18 features show drift; red bars highlight flagged features.

value_date_quarter=2024Q1 - 3) Model Stability

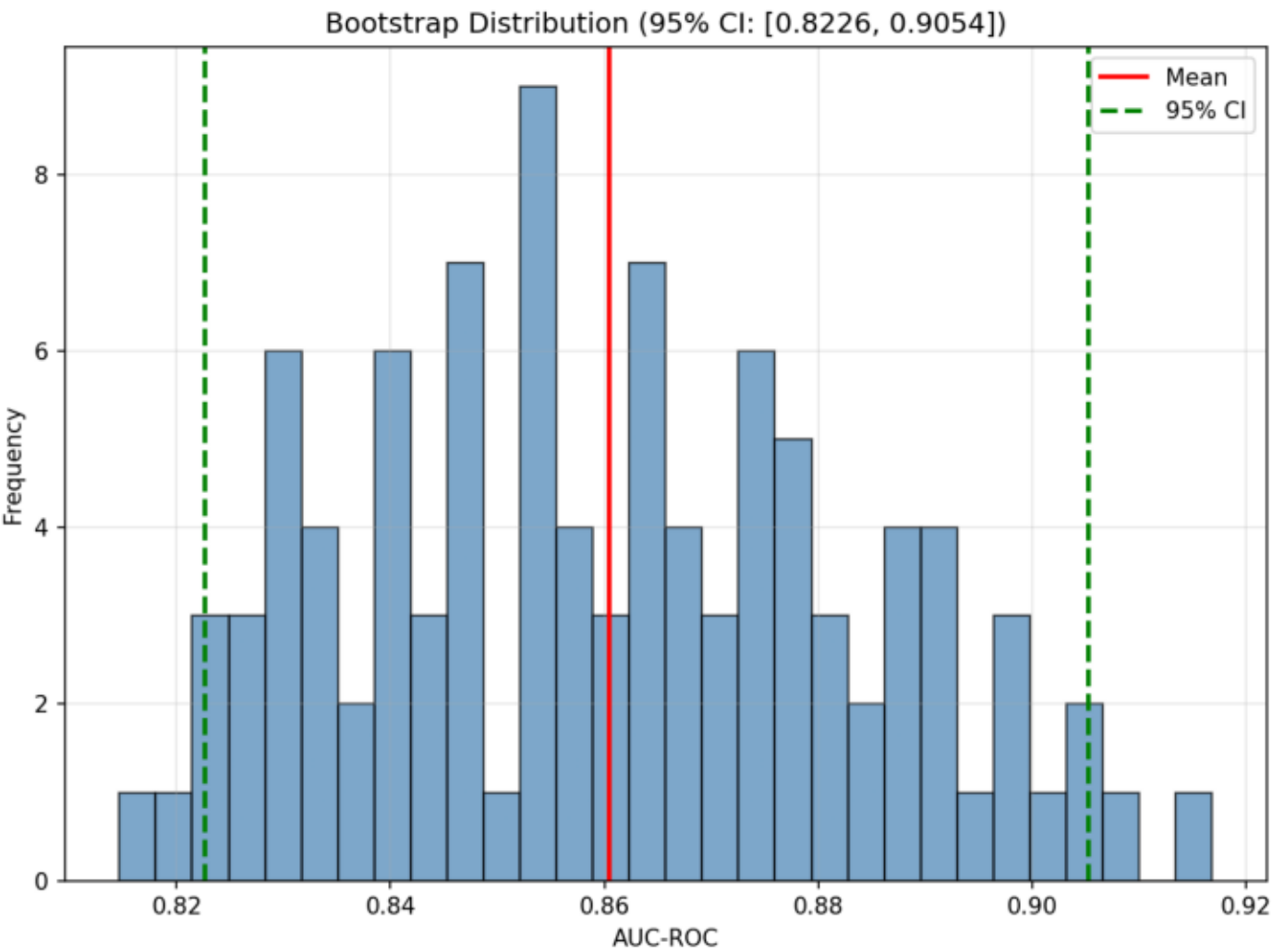


concept_drift: AUC across time chunks shows whether performance is stable over time.

value_date_quarter=2024Q1 - 3) Model Stability

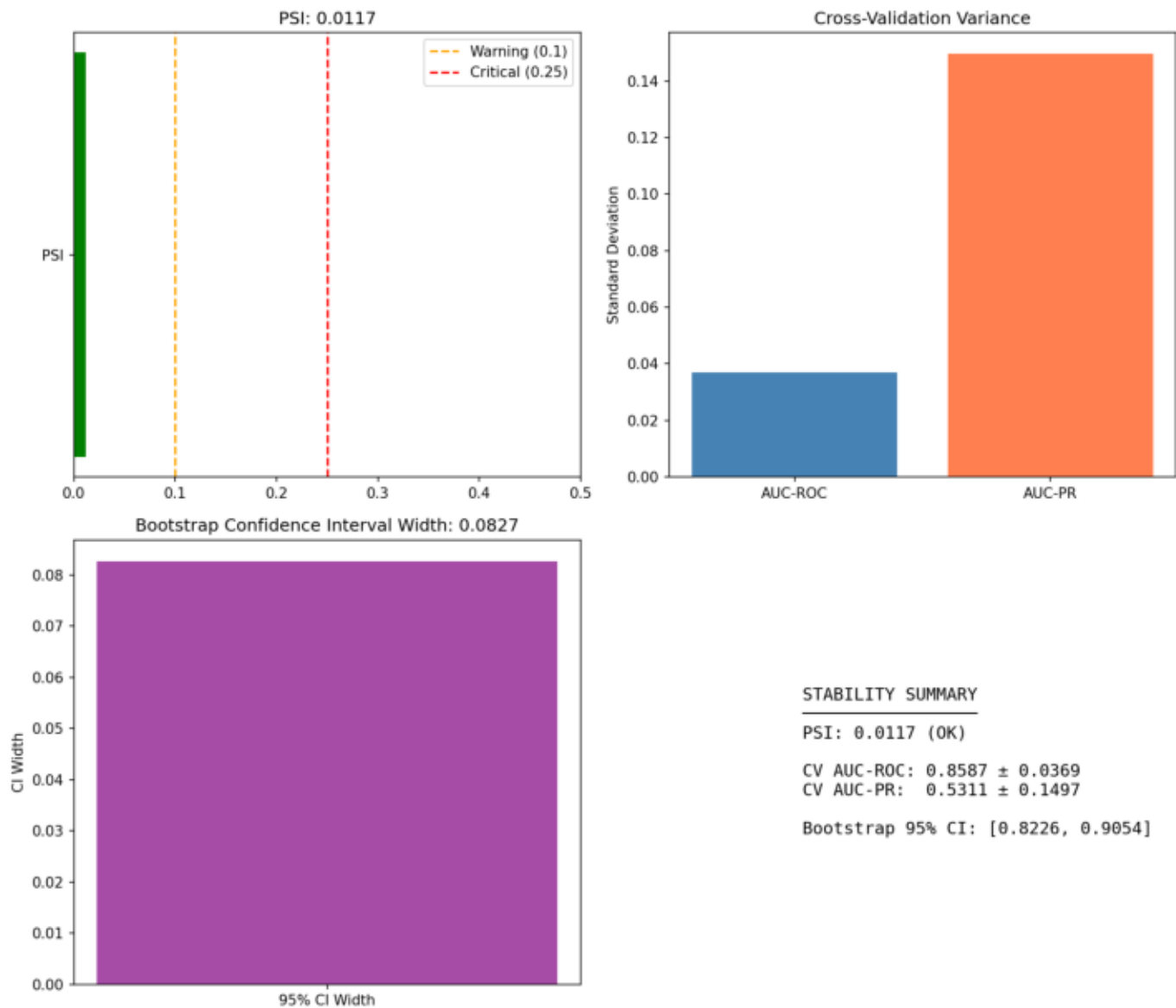


value_date_quarter=2024Q1 - 3) Model Stability



bootstrap_distribution: Bootstrap CI [0.8226, 0.9054] reflects performance stability.

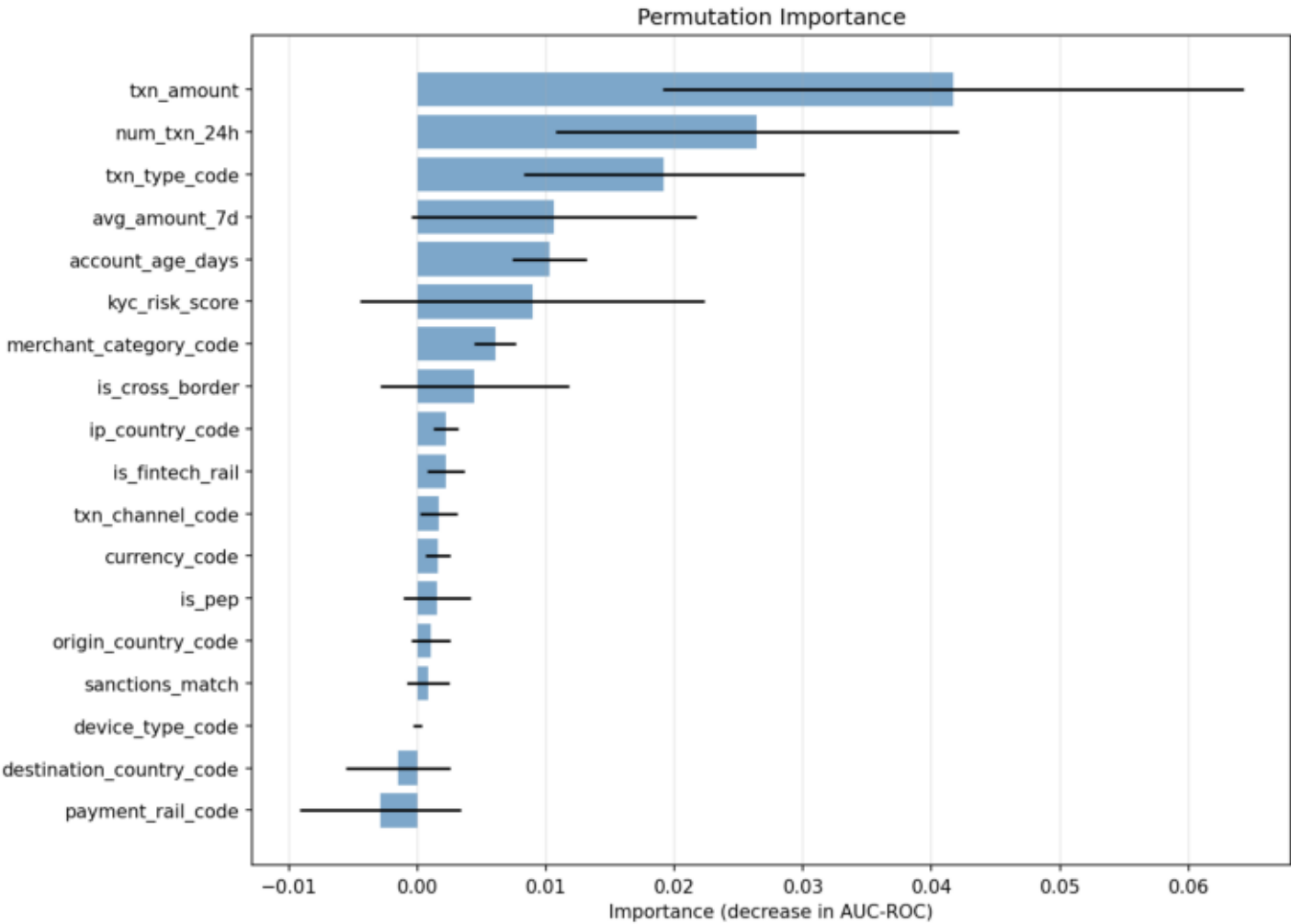
value_date_quarter=2024Q1 - 3) Model Stability



Metric	Figure	Explanation
model_type	tree	Identifies the estimator family so explanation methods can be interpreted in the right context. Next: confirm the chosen explanation methods are appropriate for this model class.
methods_used	permutation, pdp, ice, lime	Lists the interpretability methods produced in this run. Next: compare signals across methods before using them in documentation or model challenge.
shap_status	unavailable	Summarizes interpretability evidence for this run. Next: use these outputs to document model drivers and challenge unexpected dependencies.
shap_reason	No module named 'shap'	Summarizes interpretability evidence for this run. Next: use these outputs to document model drivers and challenge unexpected dependencies.
perm_top_features	txn_amount, num_txn_24h, txn_type_code, avg_amount_7d, account_age_days ...	Shows features with the largest performance drop when permuted, which highlights global importance. Next: validate that the top drivers align with business logic, feature governance, and policy expectations.
lime_instances	3	Counts the locally explained observations reviewed with LIME. Next: examine representative true-positive and false-positive cases to understand local decision logic.

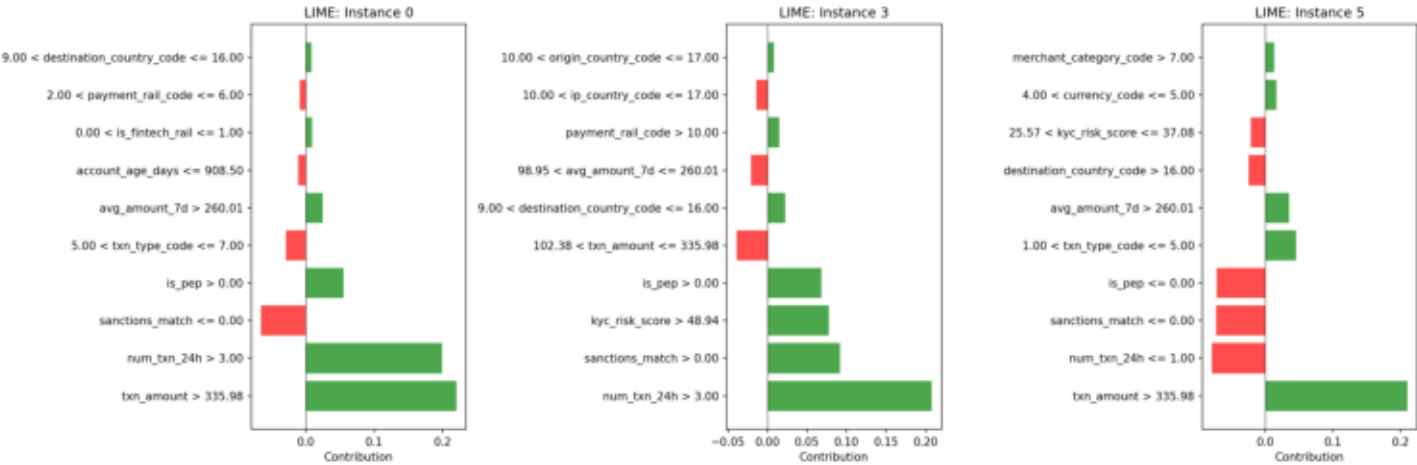
Metric	Figure	Explanation
pdp_features	avg_amount_7d, txn_amount, account_age_days, kyc_risk_score, num_txn_24h ...	Lists features with partial-dependence plots. Next: review whether average model response is monotonic, stable, and operationally sensible.
ice_features	avg_amount_7d, txn_amount, account_age_days, kyc_risk_score	Lists features with individual conditional expectation plots. Next: inspect heterogeneity and interactions that may be hidden in average-only views.

value_date_quarter=2024Q1 - 4) Model Interpretability



permutation_importance: Bars show how much AUC drops when each feature is permuted.

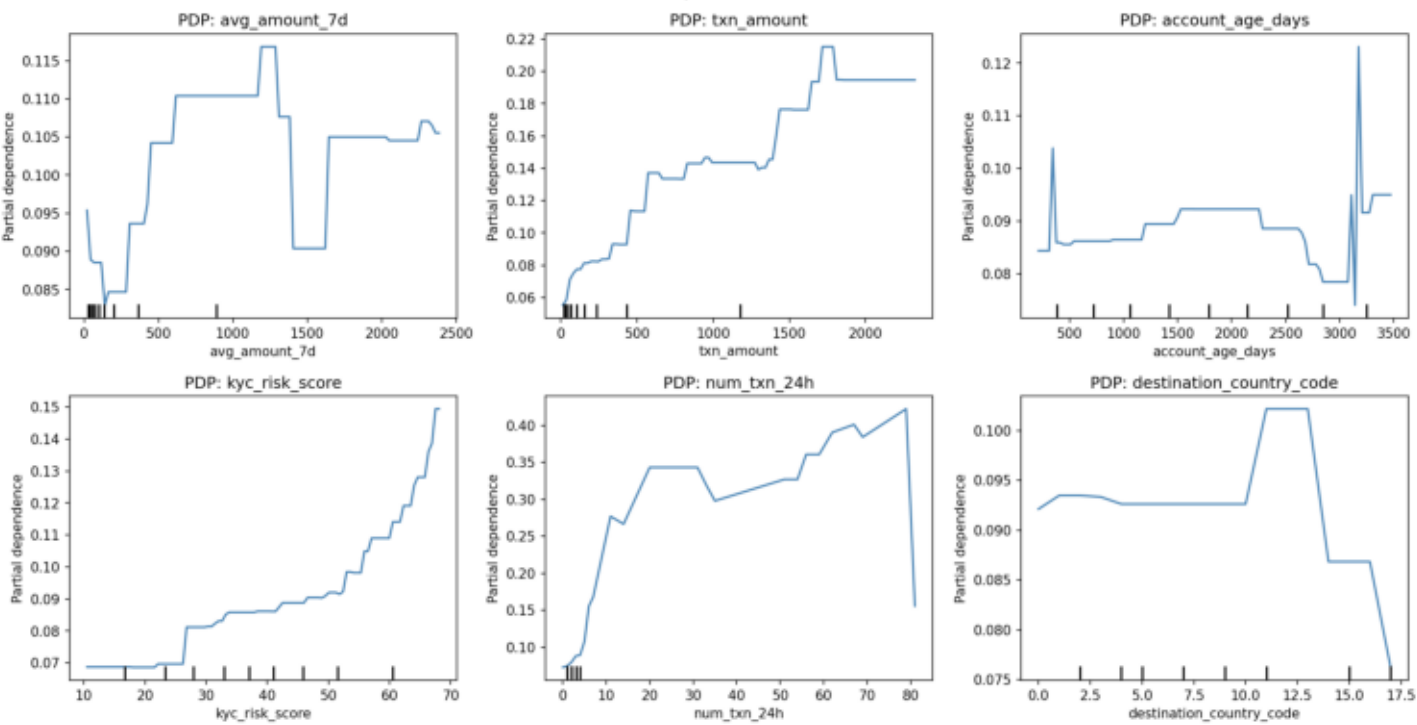
value_date_quarter=2024Q1 - 4) Model Interpretability



lime_explanation: LIME bars show local feature contributions for selected instances.

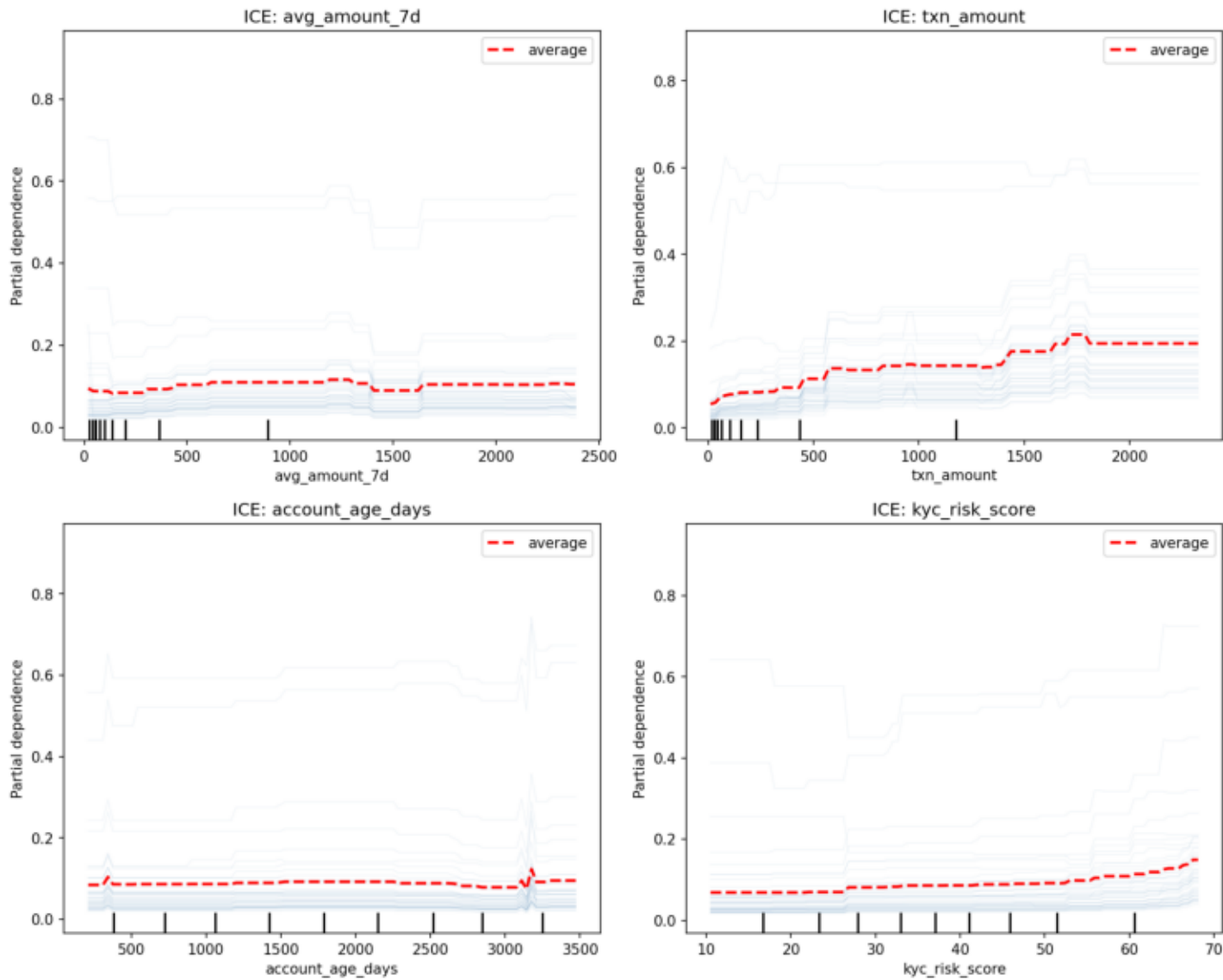
value_date_quarter=2024Q1 - 4) Model Interpretability

Partial Dependence Plots



pdp: PDP curves show average model response as a feature varies.

Individual Conditional Expectation (ICE) Plots



ice: ICE curves show per-instance responses as a feature varies.

value_date_quarter=2024Q2 - 1) Model Effectiveness (1/3)

Metric	Figure	Explanation
auc_roc	0.8420	Measures ranking power across all thresholds. 0.8420 indicates strong separation. Next: compare AUC-ROC across segments and keep threshold tuning separate from ranking validation.
auc_pr	0.4720	Measures precision-recall performance under class imbalance; the observed event rate is 7.92%. Next: use this together with alert capacity to decide whether the high-score queue is practically useful.
precision	0.7188	Shows the share of flagged cases that are true positives at the current threshold. Next: if this is below review expectations, raise the threshold or improve exclusion features.
recall	0.2371	Shows the share of true positives captured at the current threshold. Next: if missed cases are too high, lower the threshold or add stronger risk features.
f1	0.3566	Balances precision and recall in one score. Next: use it as a threshold-selection aid when both missed cases and false alerts matter.
ks_statistic	0.5052	Measures score separation between positive and negative classes. Next: low KS suggests the model needs better features or a different specification.

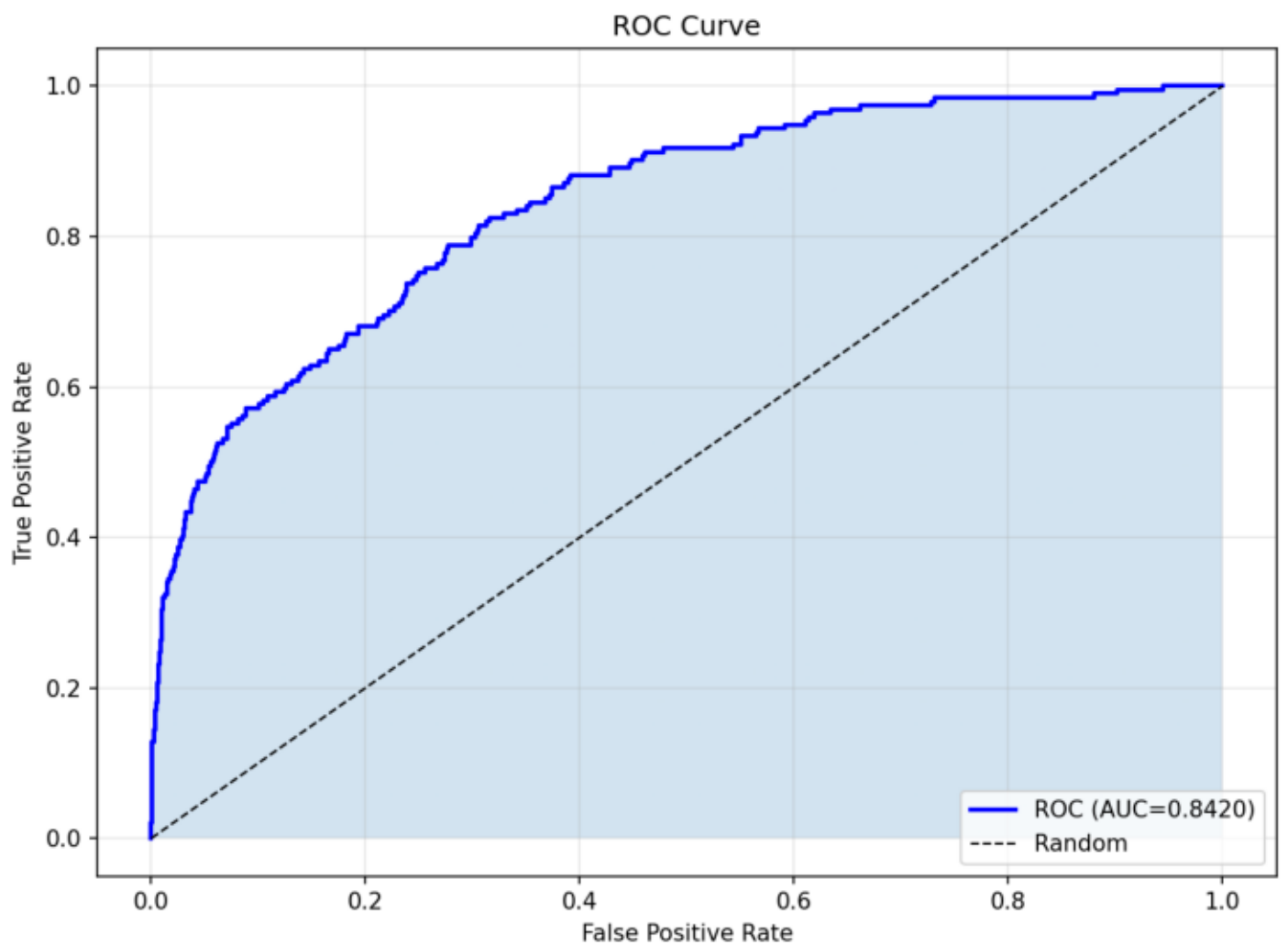
value_date_quarter=2024Q2 - 1) Model Effectiveness (2/3)

Metric	Figure	Explanation
ks_threshold	0.0700	This is the threshold that maximizes KS separation. Next: treat it as a candidate operating point, then compare it with business capacity and policy limits.
confusion_matrix[TN]	2236	True negatives correctly cleared. Next: preserve this base while tuning the alert threshold.
confusion_matrix[FP]	18	False positives create review cost without SAR conversion. Next: reduce this count with threshold tuning or sharper risk features.
confusion_matrix[FN]	148	False negatives are missed SAR cases. Next: if this count is unacceptable, lower the threshold or add features that capture suspicious behavior earlier.
confusion_matrix[TP]	46	True positives are correctly escalated cases. Next: assess whether this volume is sufficient relative to expected SAR capture goals.
precision_at_k[10]	0.9000	Shows how accurate the top-10 highest-risk queue is. Next: align K with investigator capacity and confirm that top alerts stay high quality.
precision_at_k[50]	0.7600	Shows how accurate the top-50 highest-risk queue is. Next: align K with investigator capacity and confirm that top alerts stay high quality.
precision_at_k[100]	0.6600	Shows how accurate the top-100 highest-risk queue is. Next: align K with investigator capacity and confirm that top alerts stay high quality.

value_date_quarter=2024Q2 - 1) Model Effectiveness (3/3)

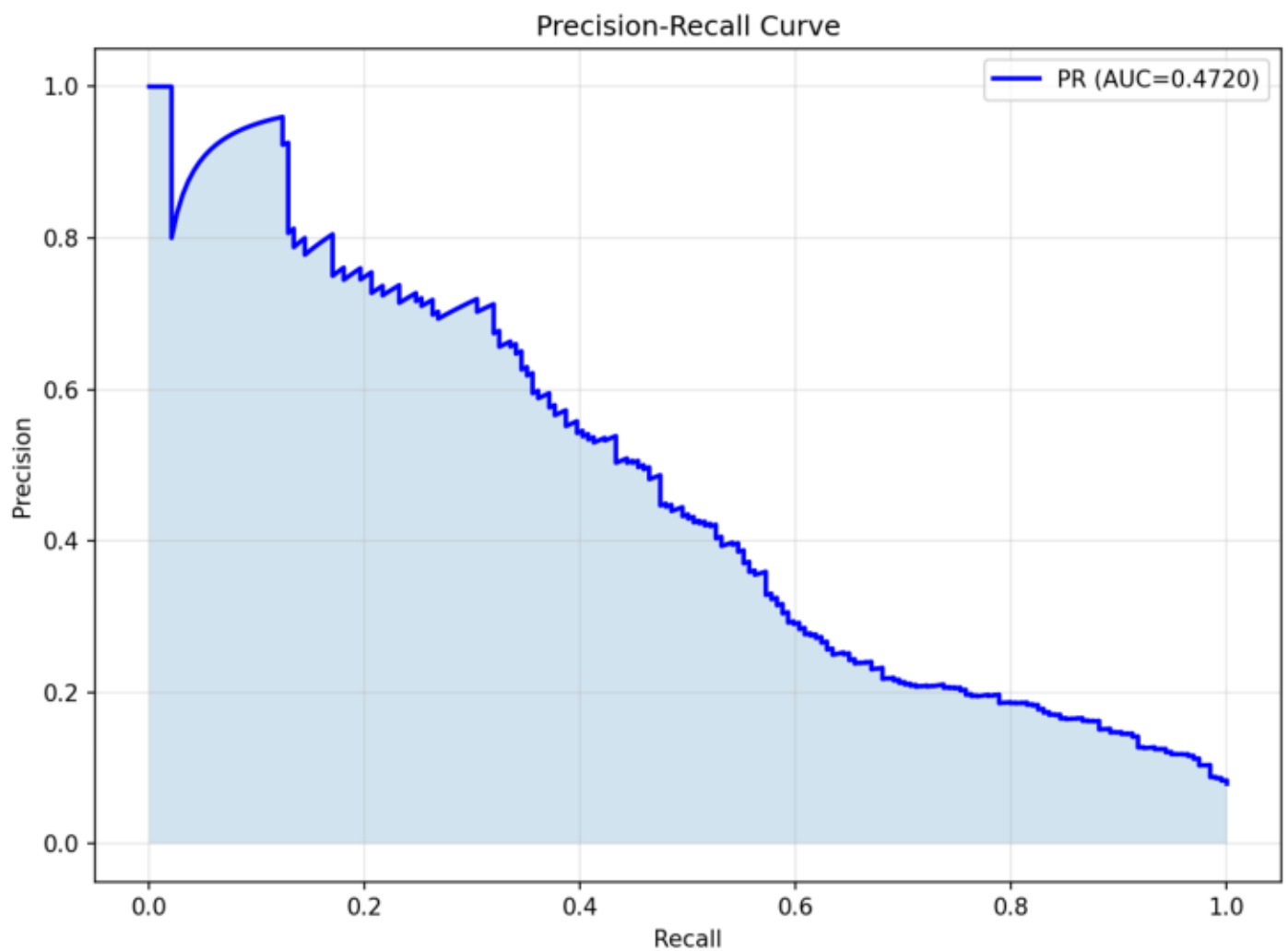
Metric	Figure	Explanation
precision_at_k[200]	0.4600	Shows how accurate the top-200 highest-risk queue is. Next: align K with investigator capacity and confirm that top alerts stay high quality.
precision_at_k[500]	0.2500	Shows how accurate the top-500 highest-risk queue is. Next: align K with investigator capacity and confirm that top alerts stay high quality.
recall_at_k[10]	0.0464	Shows how much of the positive population is captured within the top-10 queue. Next: increase K if coverage is too low, or improve ranking quality if capacity cannot expand.
recall_at_k[50]	0.1959	Shows how much of the positive population is captured within the top-50 queue. Next: increase K if coverage is too low, or improve ranking quality if capacity cannot expand.
recall_at_k[100]	0.3402	Shows how much of the positive population is captured within the top-100 queue. Next: increase K if coverage is too low, or improve ranking quality if capacity cannot expand.
recall_at_k[200]	0.4742	Shows how much of the positive population is captured within the top-200 queue. Next: increase K if coverage is too low, or improve ranking quality if capacity cannot expand.
recall_at_k[500]	0.6443	Shows how much of the positive population is captured within the top-500 queue. Next: increase K if coverage is too low, or improve ranking quality if capacity cannot expand.

value_date_quarter=2024Q2 - 1) Model Effectiveness



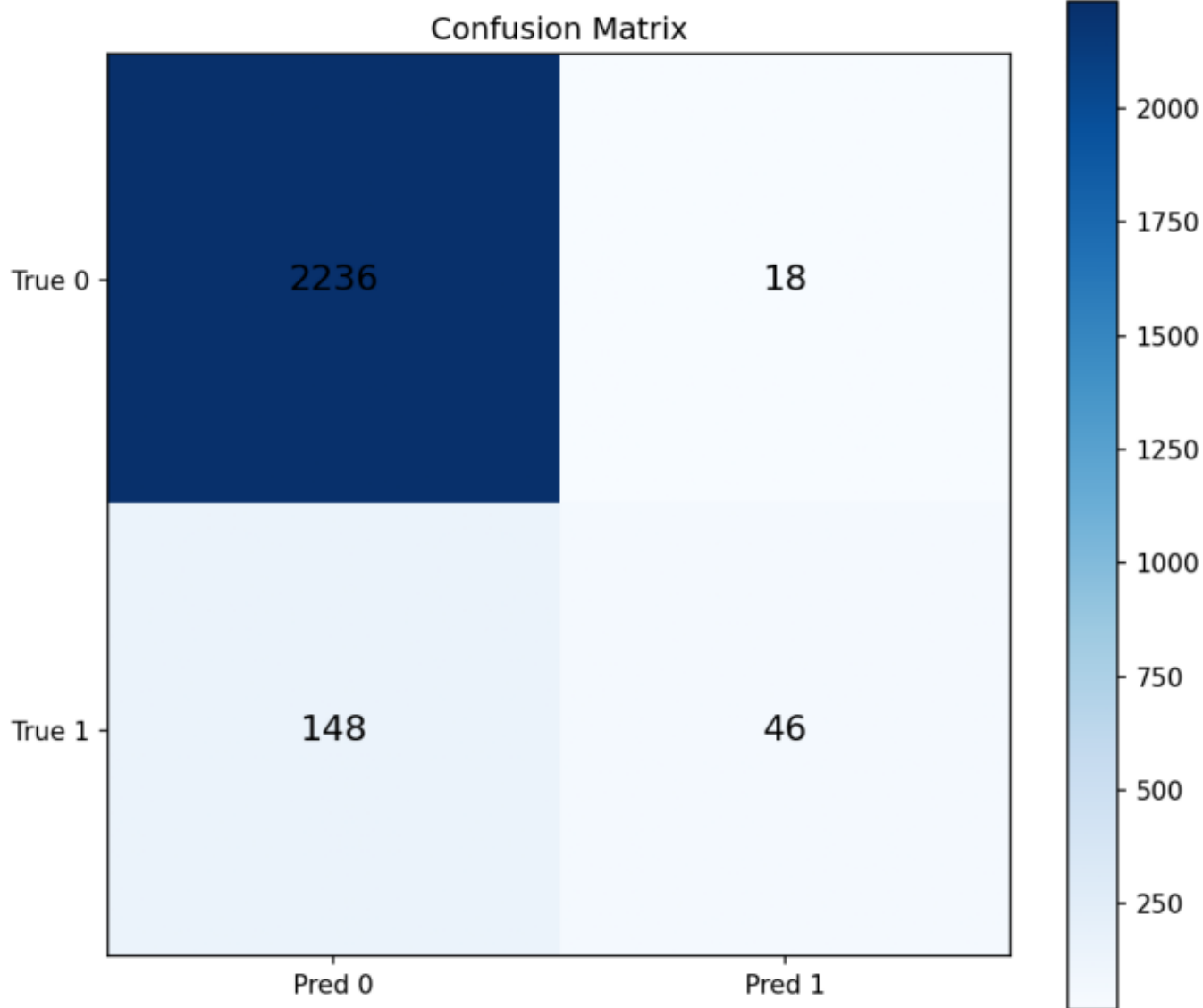
roc_curve: AUC-ROC=0.8420; curve above diagonal indicates strong separation.

value_date_quarter=2024Q2 - 1) Model Effectiveness



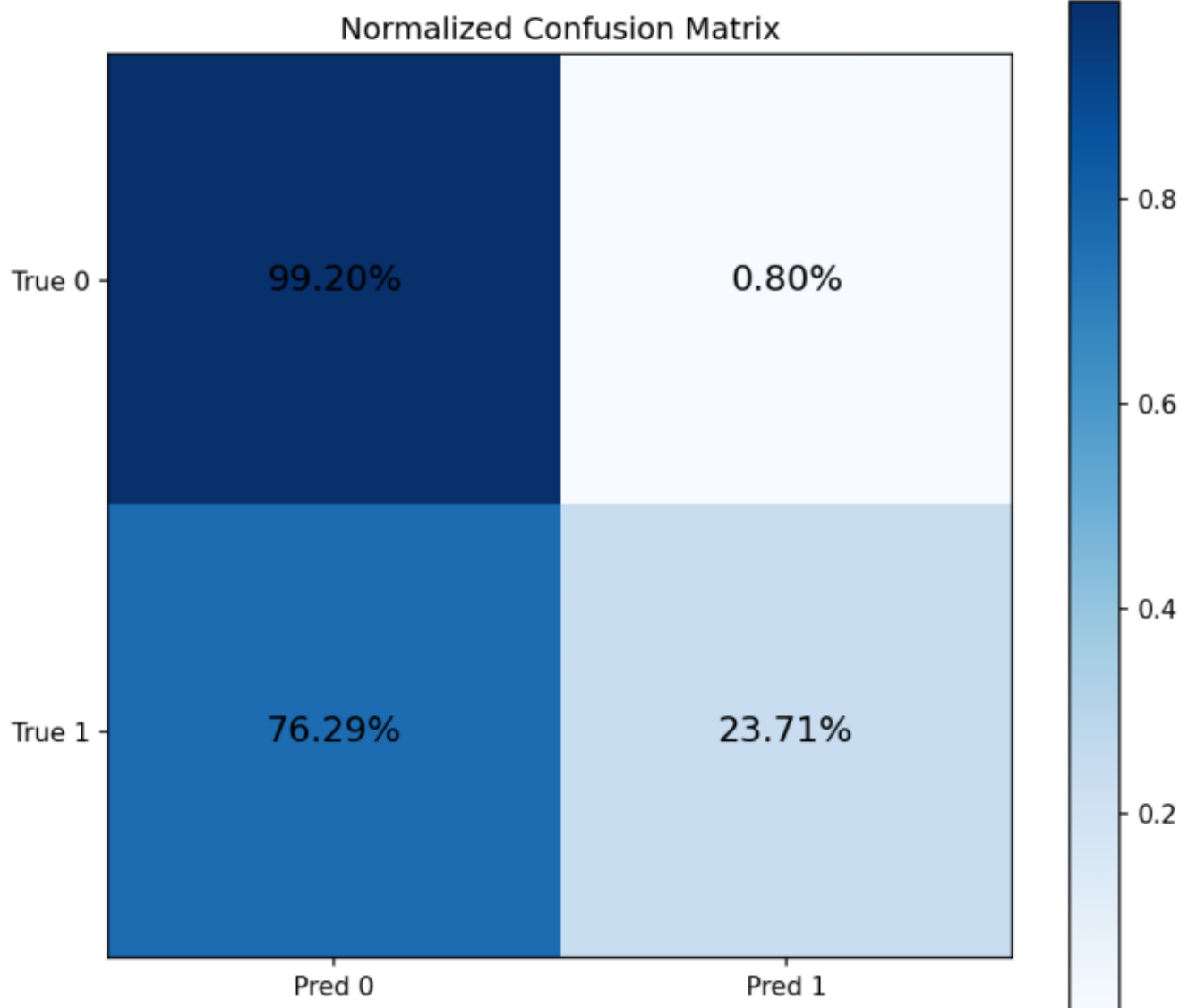
pr_curve: AUC-PR=0.4720 vs baseline 7.92%; higher curve implies better precision at recall.

value_date_quarter=2024Q2 - 1) Model Effectiveness



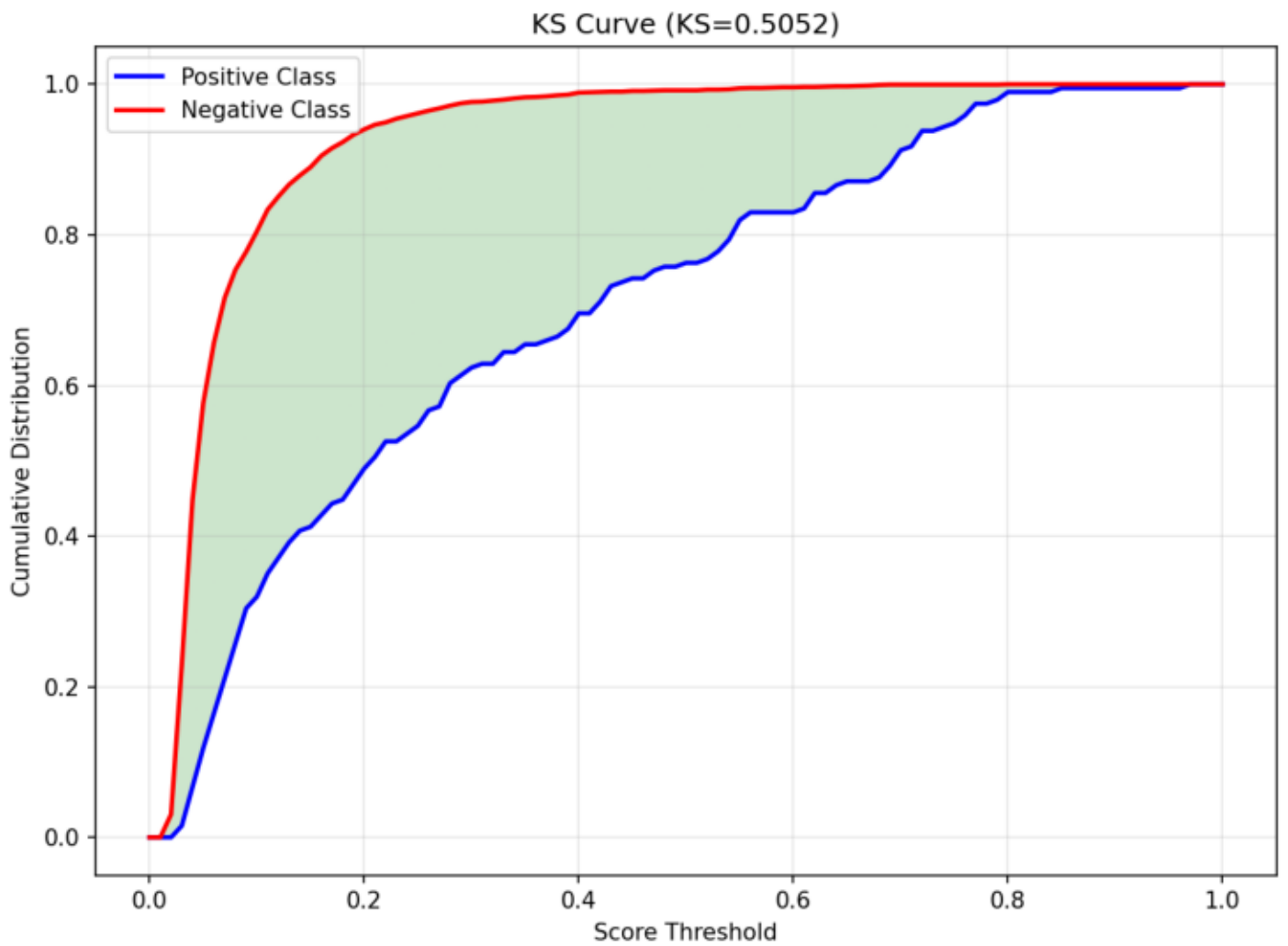
confusion_matrix: At threshold 0.50: TP=46, FP=18, FN=148, TN=2236.

value_date_quarter=2024Q2 - 1) Model Effectiveness



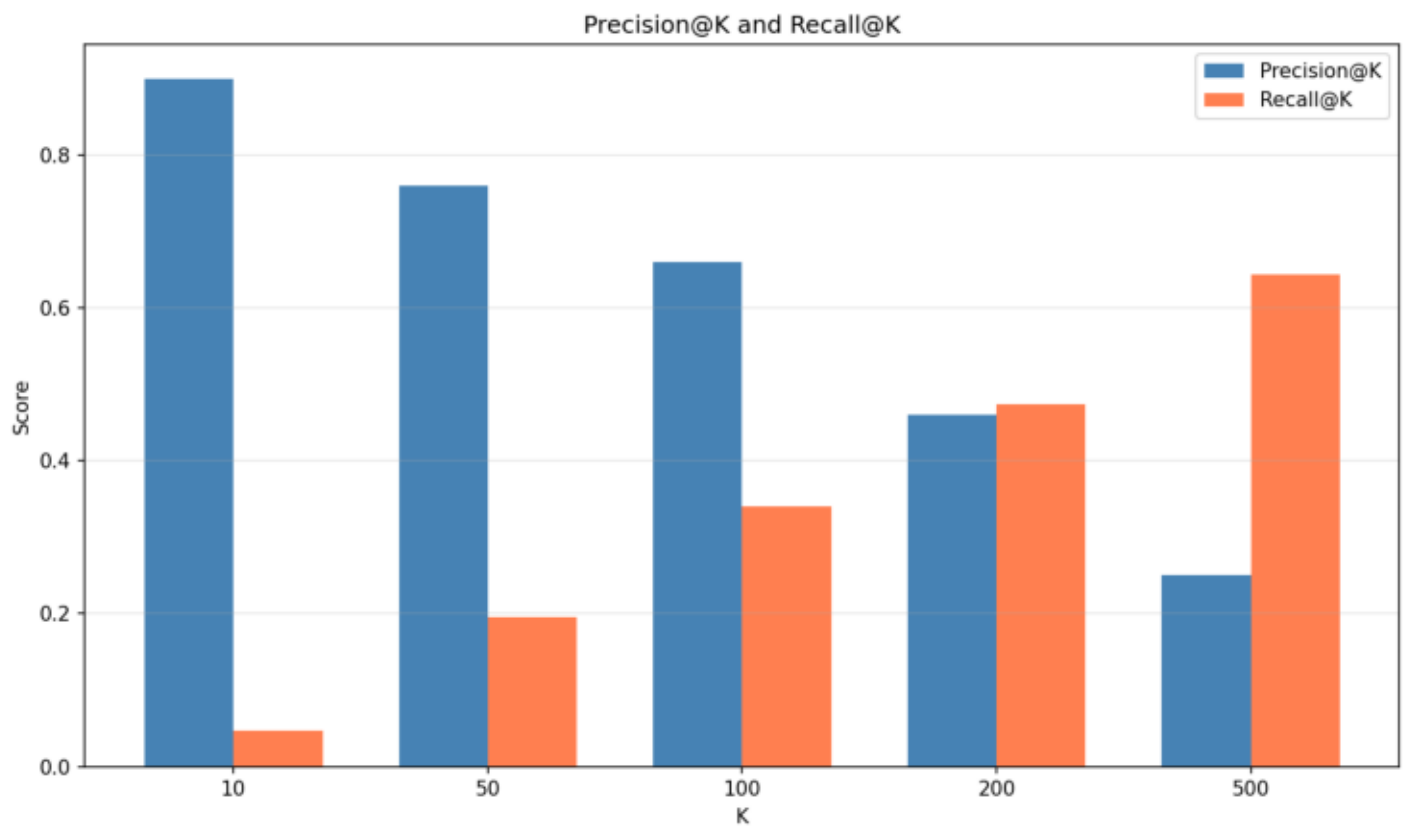
confusion_matrix_norm: Normalized rates: TPR=23.71%, TNR=99.20%.

value_date_quarter=2024Q2 - 1) Model Effectiveness



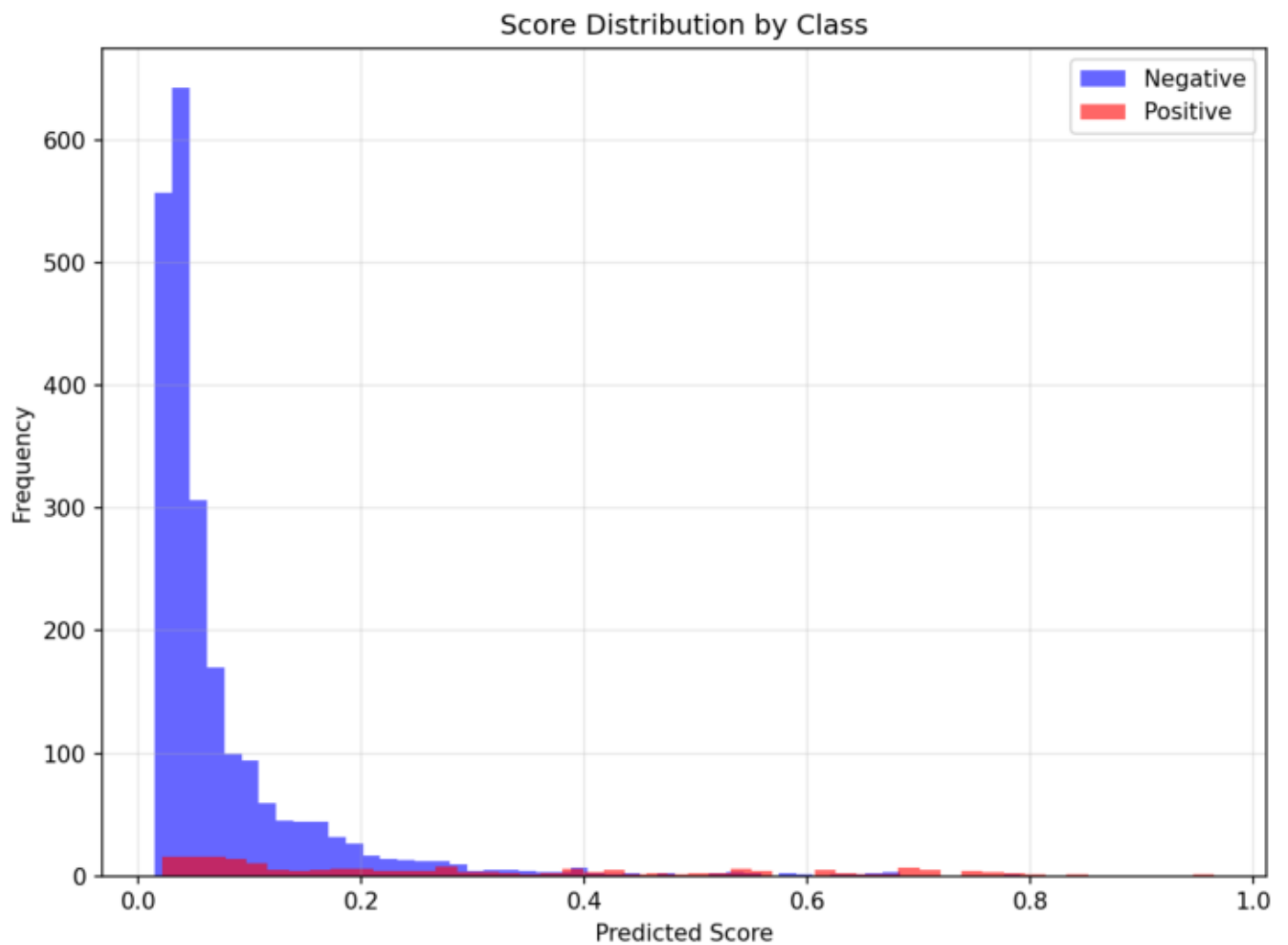
ks_curve: Maximum separation KS=0.5052 at threshold 0.07.

value_date_quarter=2024Q2 - 1) Model Effectiveness



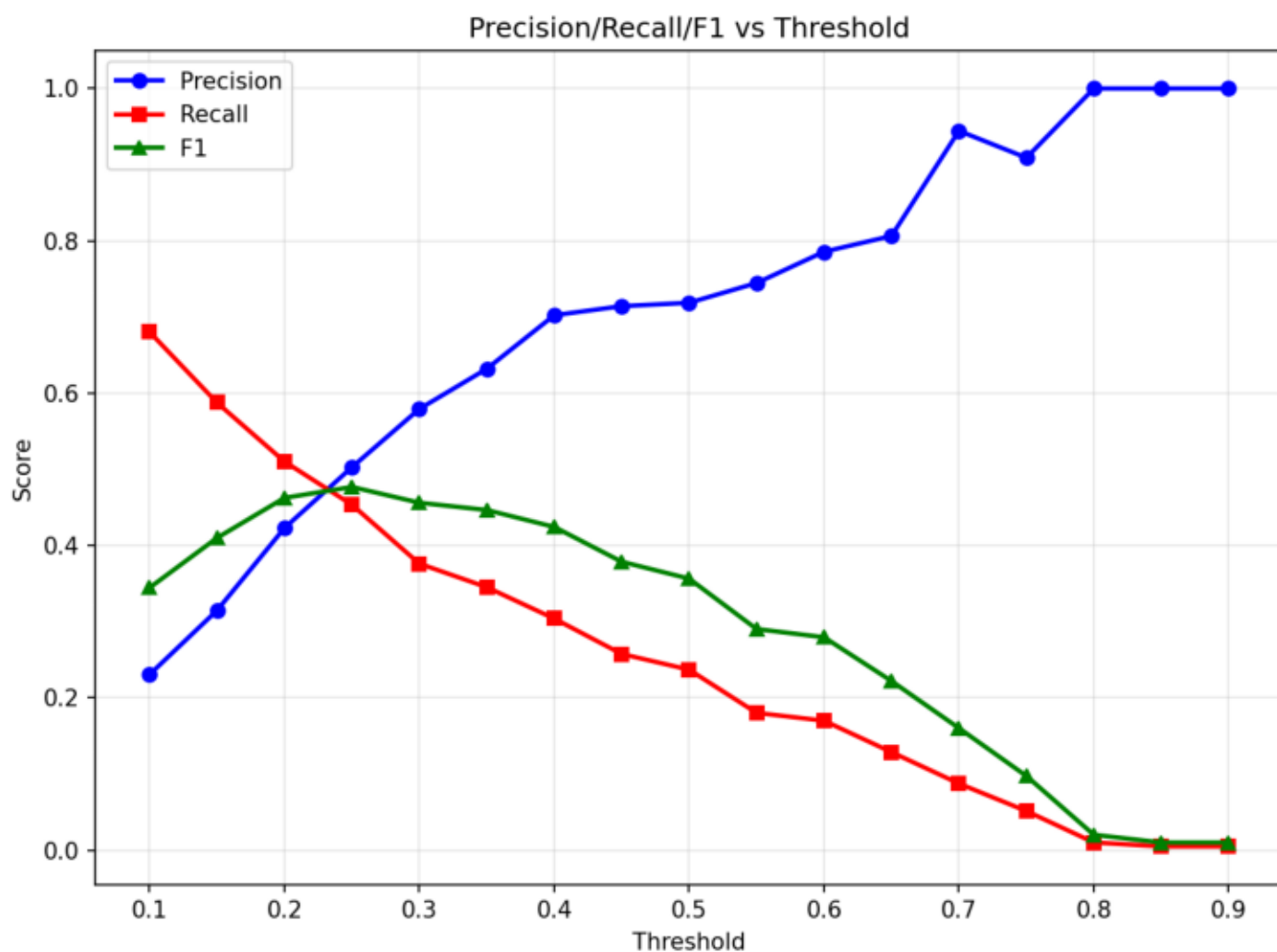
precision_recall_at_k: Bars show precision/recall as you expand the review queue.

value_date_quarter=2024Q2 - 1) Model Effectiveness



score_distribution: Mean score: positive=0.291, negative=0.073 (larger gap is better).

value_date_quarter=2024Q2 - 1) Model Effectiveness



threshold_analysis: Best F1 in tested grid is 0.4770 at threshold 0.25.

value_date_quarter=2024Q2 - 2) Model Efficiency (1/4)

Metric	Figure	Explanation
fpr	0.0080	Measures the share of negatives incorrectly flagged; 0.0080 is low. Next: if review demand is too high, raise the threshold or add features that better suppress false alerts.
tn	2236	Counts negatives correctly cleared without manual review. Next: preserve this clearance rate while checking that recall does not fall below policy expectations.
fp	18	Counts non-events that would still be sent to review. Next: use this with staffing capacity to decide whether threshold or feature refinement is required.
threshold	0.5000	This is the active operating cutoff used to flag alerts. Next: tune it jointly with recall, precision, and review capacity rather than in isolation.
fpr_at_thresholds[thresholds.t_0.05]	0.4232	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.05. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.10]	0.1952	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.10. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.

value_date_quarter=2024Q2 - 2) Model Efficiency (2/4)

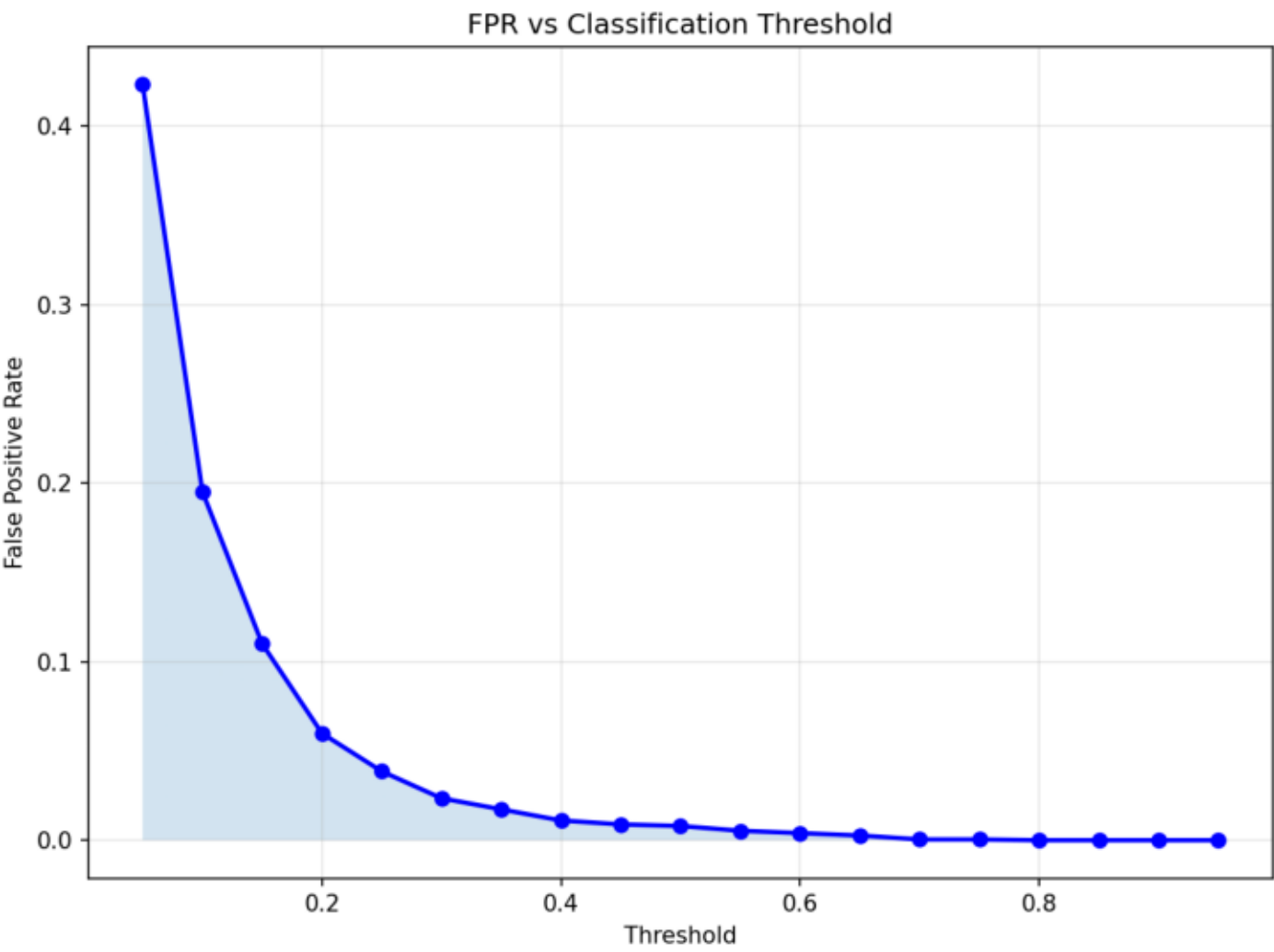
Metric	Figure	Explanation
fpr_at_thresholds[thresholds.t_0.15]	0.1100	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.15. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.20]	0.0599	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.20. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.25]	0.0386	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.25. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.30]	0.0235	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.30. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.35]	0.0173	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.35. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.40]	0.0111	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.40. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.

value_date_quarter=2024Q2 - 2) Model Efficiency (3/4)

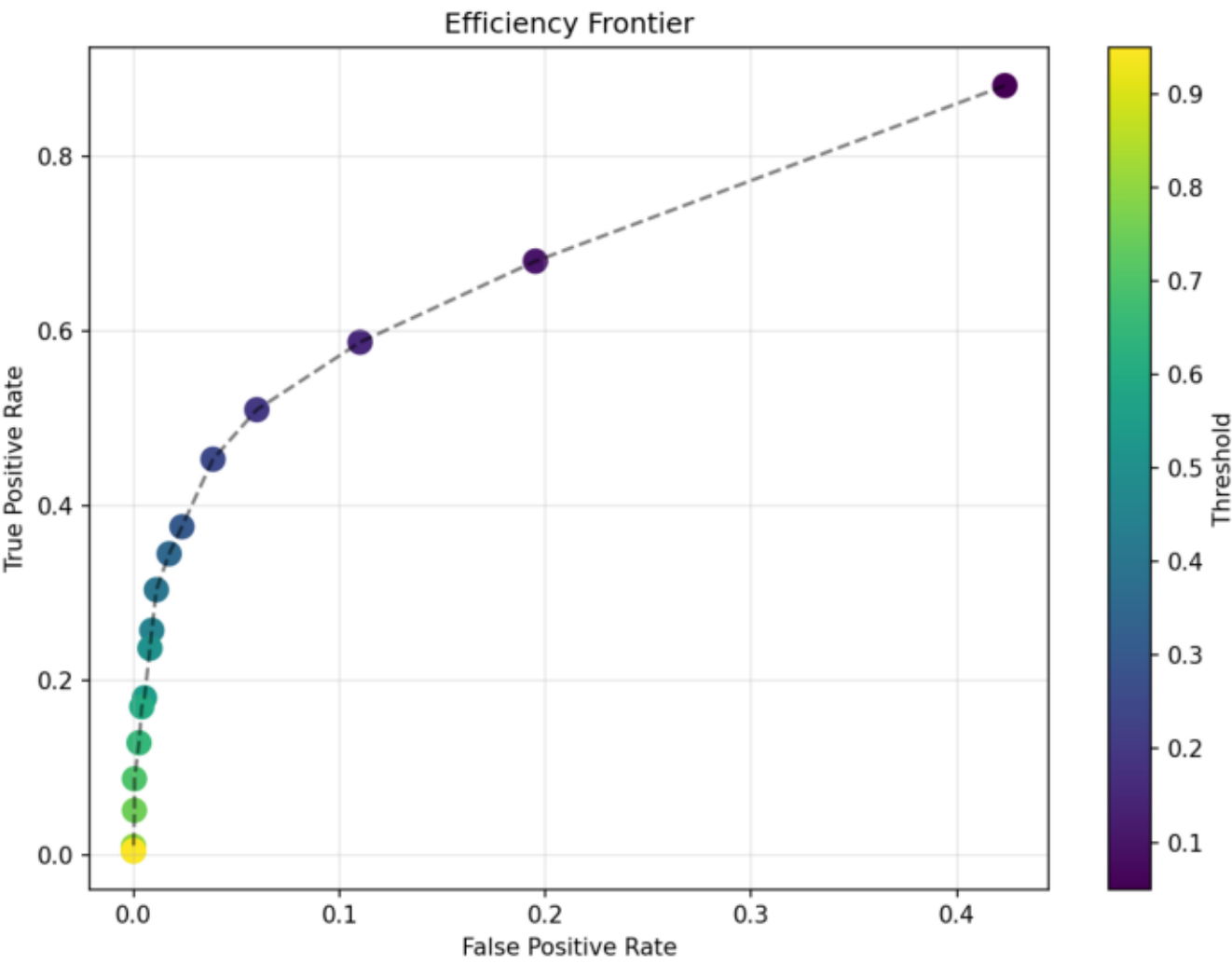
Metric	Figure	Explanation
fpr_at_thresholds[thresholds.t_0.45]	0.0089	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.45. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.50]	0.0080	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.50. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.55]	0.0053	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.55. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.60]	0.0040	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.60. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.65]	0.0027	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.65. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.70]	0.0004	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.70. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.

value_date_quarter=2024Q2 - 2) Model Efficiency (4/4)

Metric	Figure	Explanation
fpr_at_thresholds[thresholds.t_0.75]	0.0004	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.75. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.80]	0.0000	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.80. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.85]	0.0000	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.85. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.90]	0.0000	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.90. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.95]	0.0000	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.95. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.

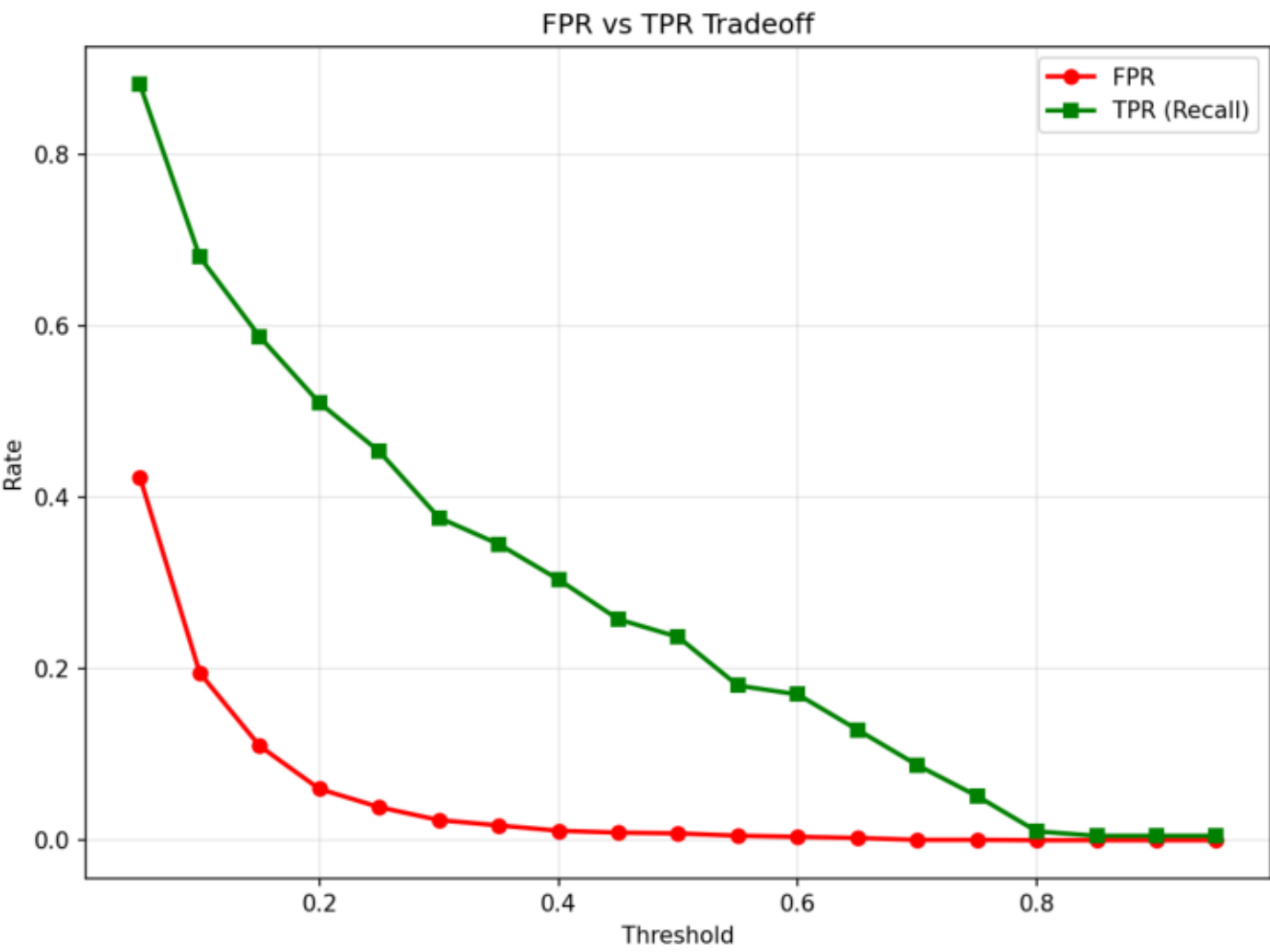


value_date_quarter=2024Q2 - 2) Model Efficiency



efficiency_frontier: Each point shows the FPR/TPR tradeoff; move toward the top-left for better efficiency.

value_date_quarter=2024Q2 - 2) Model Efficiency



fpr_tpr_tradeoff: FPR and TPR curves highlight how recall drops as you reduce false positives.

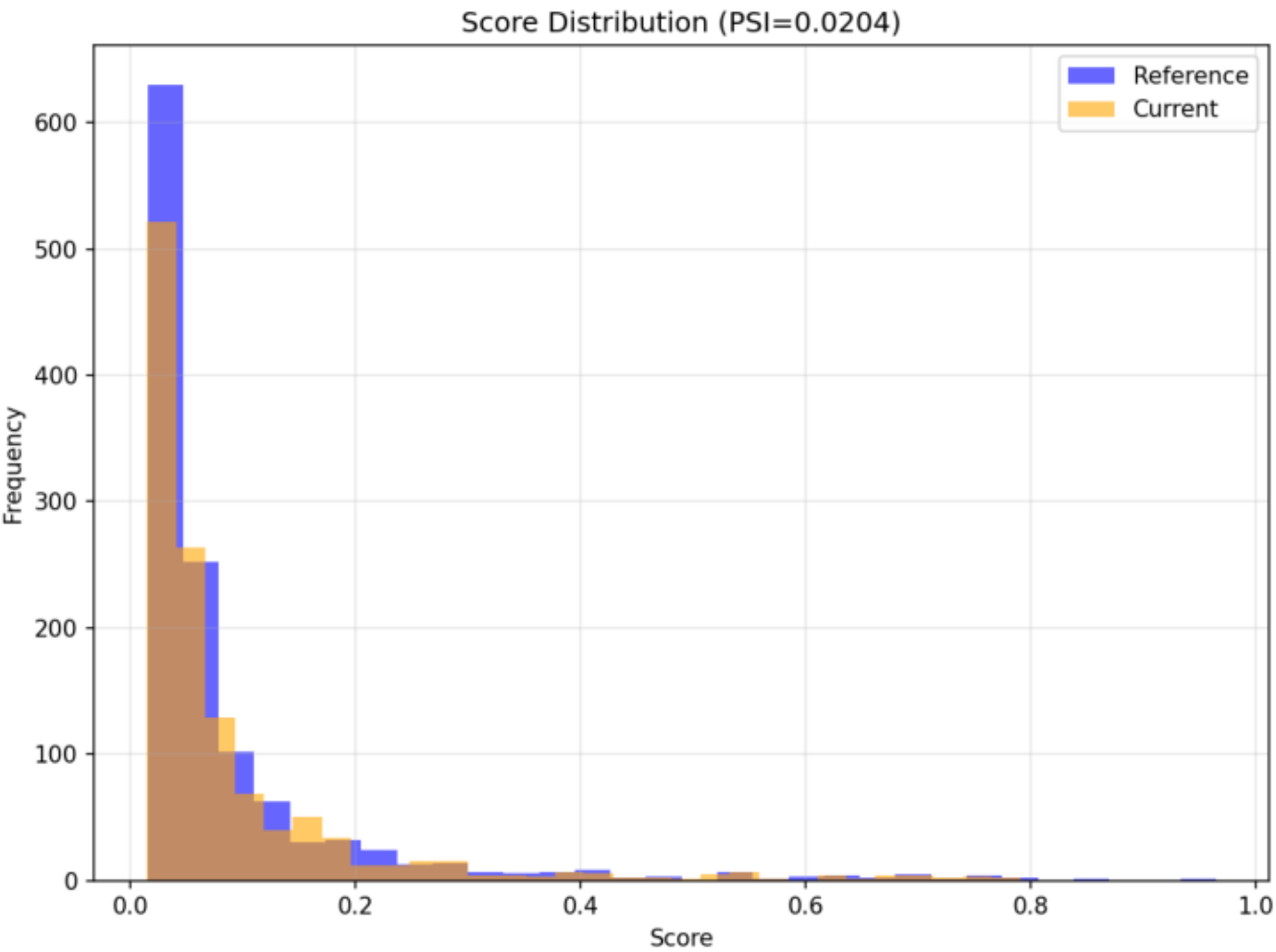
value_date_quarter=2024Q2 - 3) Model Stability (1/2)

Metric	Figure	Explanation
psi	0.0204	Measures score-distribution shift between reference and current populations; 0.0204 is negligible. Next: investigate segment mix, data drift, and recalibration needs when PSI is material.
cv_auc_roc_mean	0.8446	Average AUC-ROC across validation folds. Next: compare it with overall AUC-ROC; a large gap can indicate overfitting or unstable sampling.
cv_auc_roc_std	0.0062	Standard deviation of AUC-ROC across folds. Next: if variability is high, simplify the model or increase training stability before deployment.
cv_auc_pr_mean	0.4895	Average AUC-PR across validation folds. Next: confirm that precision-recall performance remains acceptable under resampling.
cv_auc_pr_std	0.0338	Standard deviation of AUC-PR across folds. Next: high dispersion suggests unstable alert quality and warrants deeper validation.
bootstrap_auc_roc_mean	0.8540	Bootstrap mean AUC-ROC estimates expected ranking performance under repeated sampling. Next: compare it with the point estimate to judge optimism in the current test run.
bootstrap_auc_roc_ci_lower	0.8242	Lower confidence bound for bootstrap AUC-ROC. Next: use it as a conservative performance floor in model-risk discussions.
bootstrap_auc_roc_ci_upper	0.8914	Upper confidence bound for bootstrap AUC-ROC. Next: review the interval width with the lower bound to assess uncertainty.

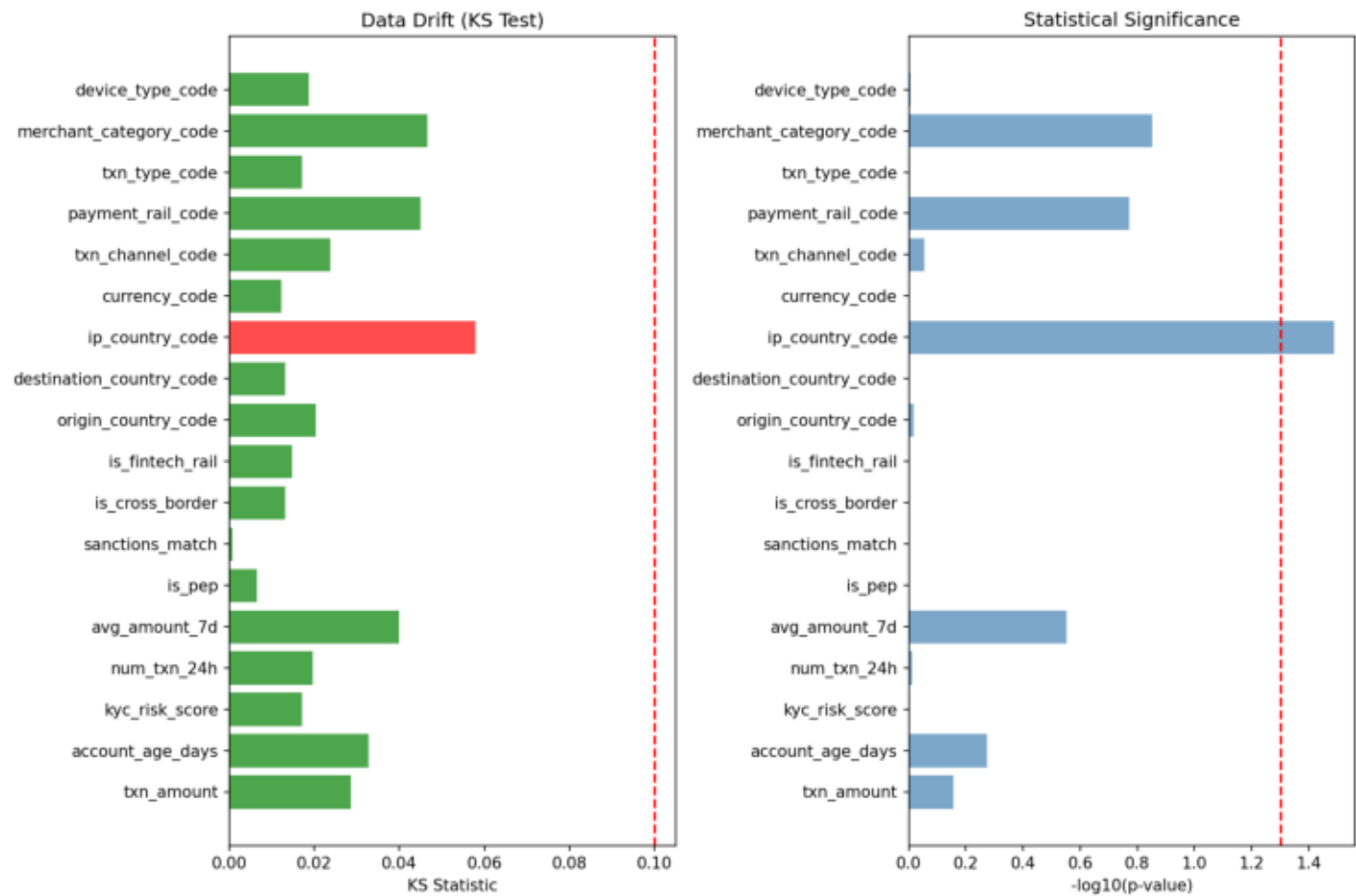
value_date_quarter=2024Q2 - 3) Model Stability (2/2)

Metric	Figure	Explanation
concept_drift_detected	False	No material concept-drift signal was detected in the evaluated slices. Next: retain ongoing monitoring because stable historical results do not guarantee future stability.
concept_drift_score	0.0206	Quantifies the magnitude of performance change across ordered slices. Next: rising scores should trigger deeper review of segment behavior and model refresh timing.

value_date_quarter=2024Q2 - 3) Model Stability

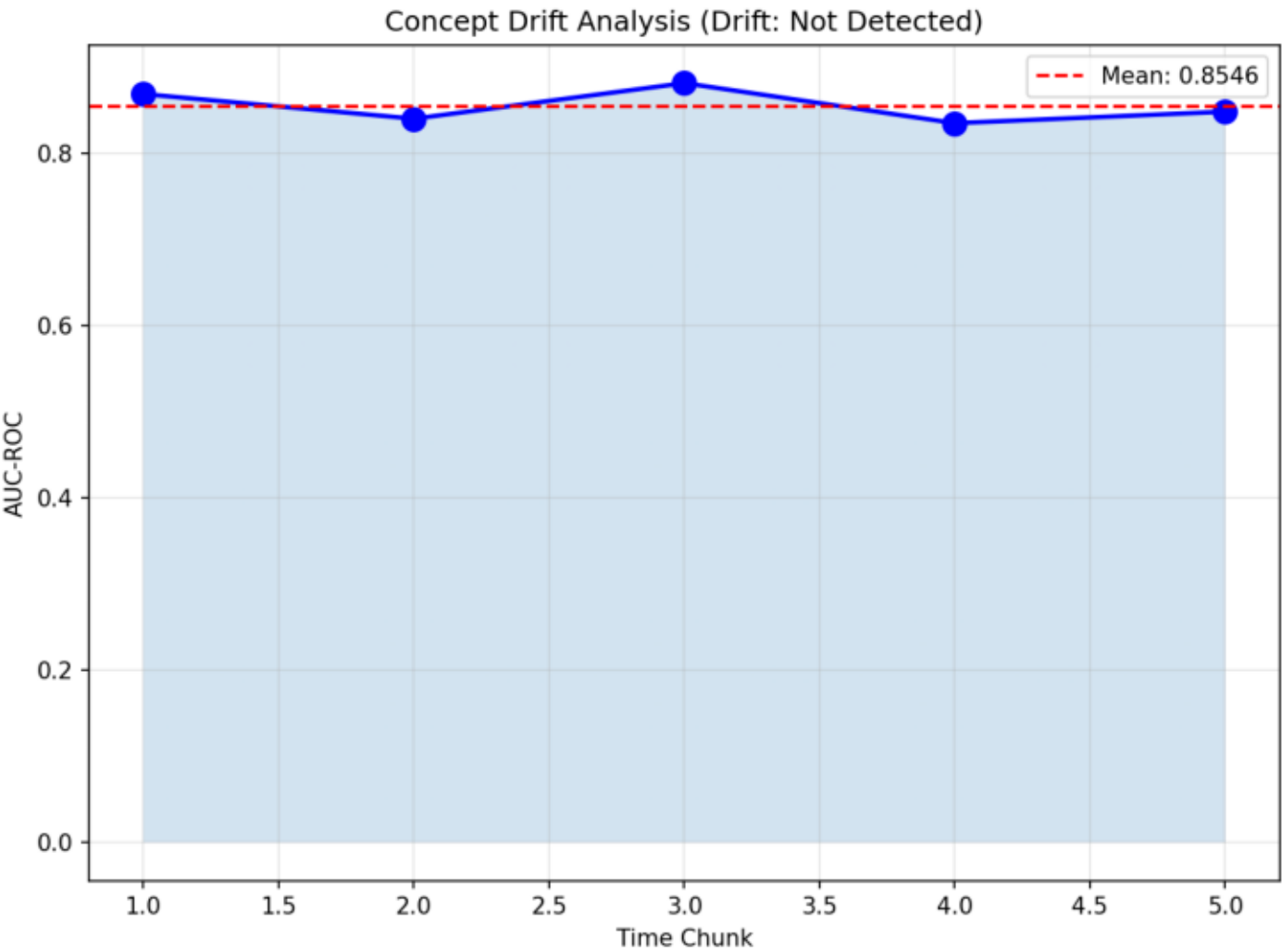


value_date_quarter=2024Q2 - 3) Model Stability



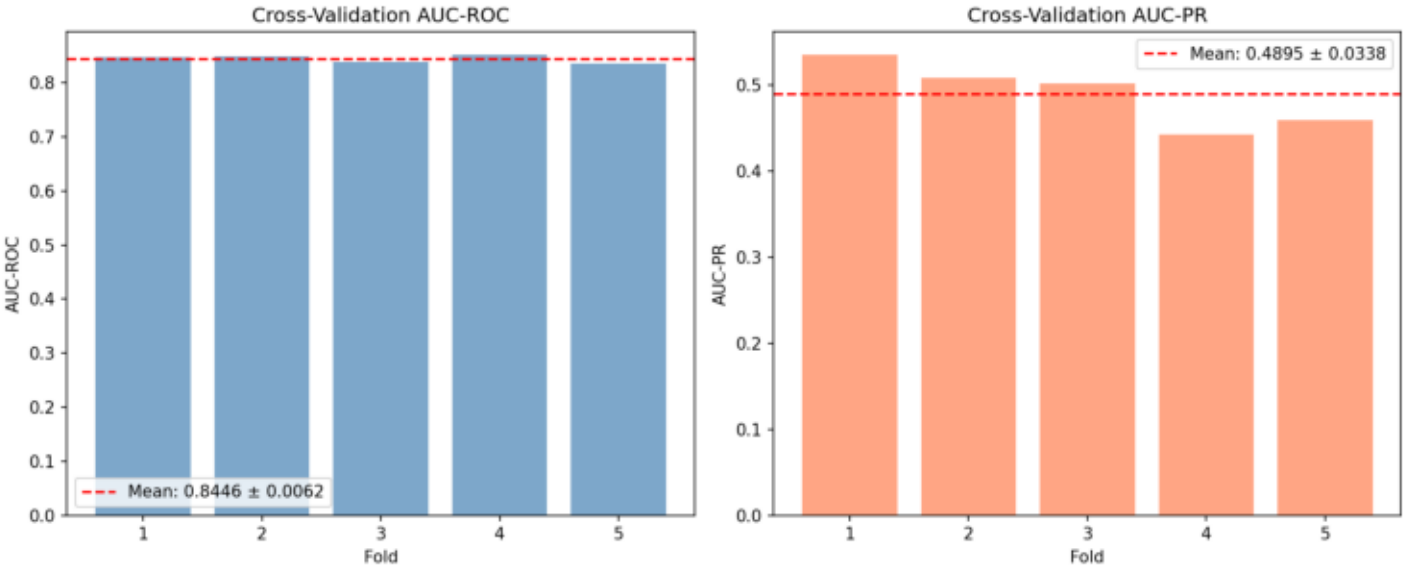
data_drift_heatmap: 1/18 features show drift; red bars highlight flagged features.

value_date_quarter=2024Q2 - 3) Model Stability



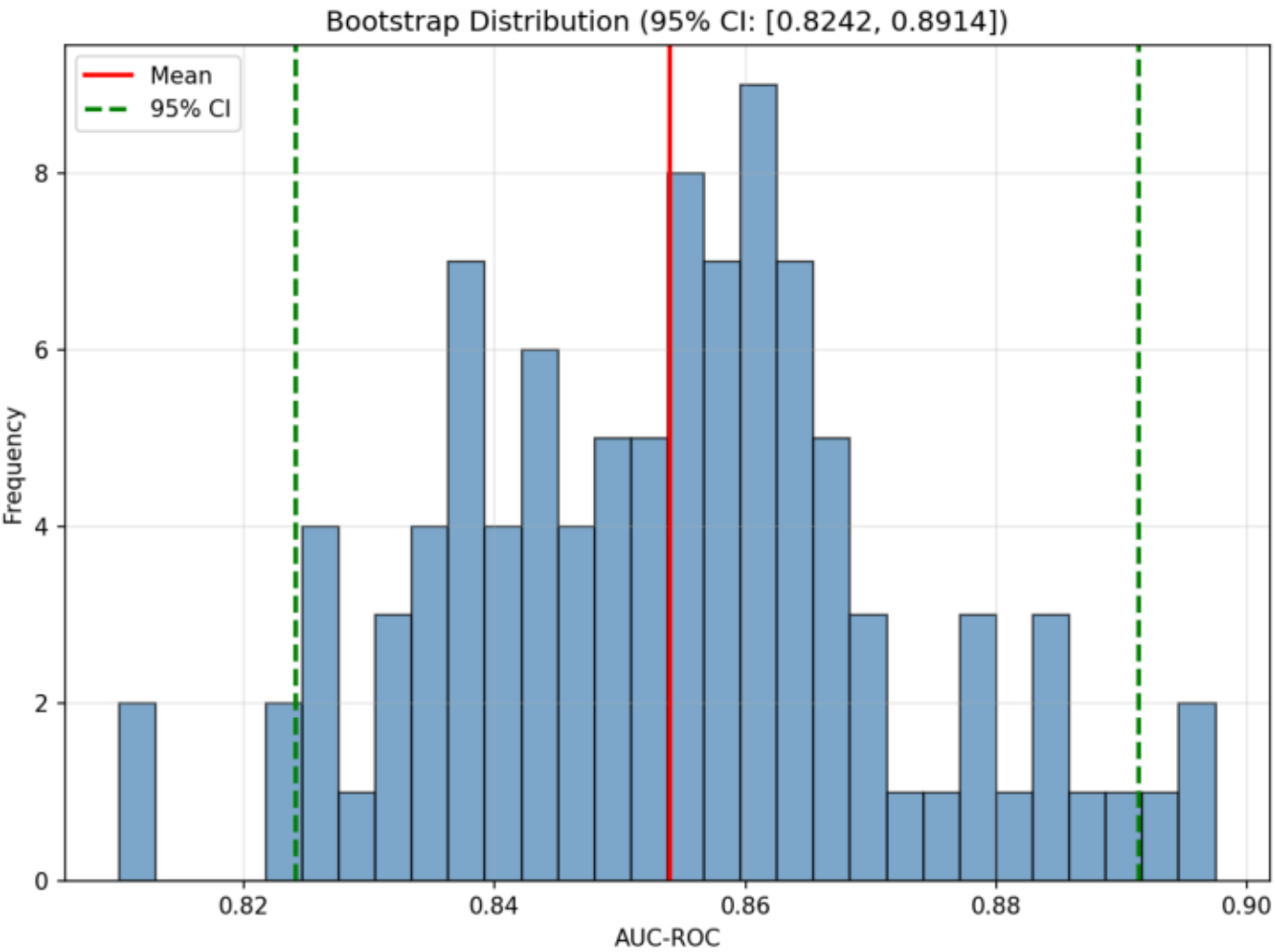
concept_drift: AUC across time chunks shows whether performance is stable over time.

value_date_quarter=2024Q2 - 3) Model Stability

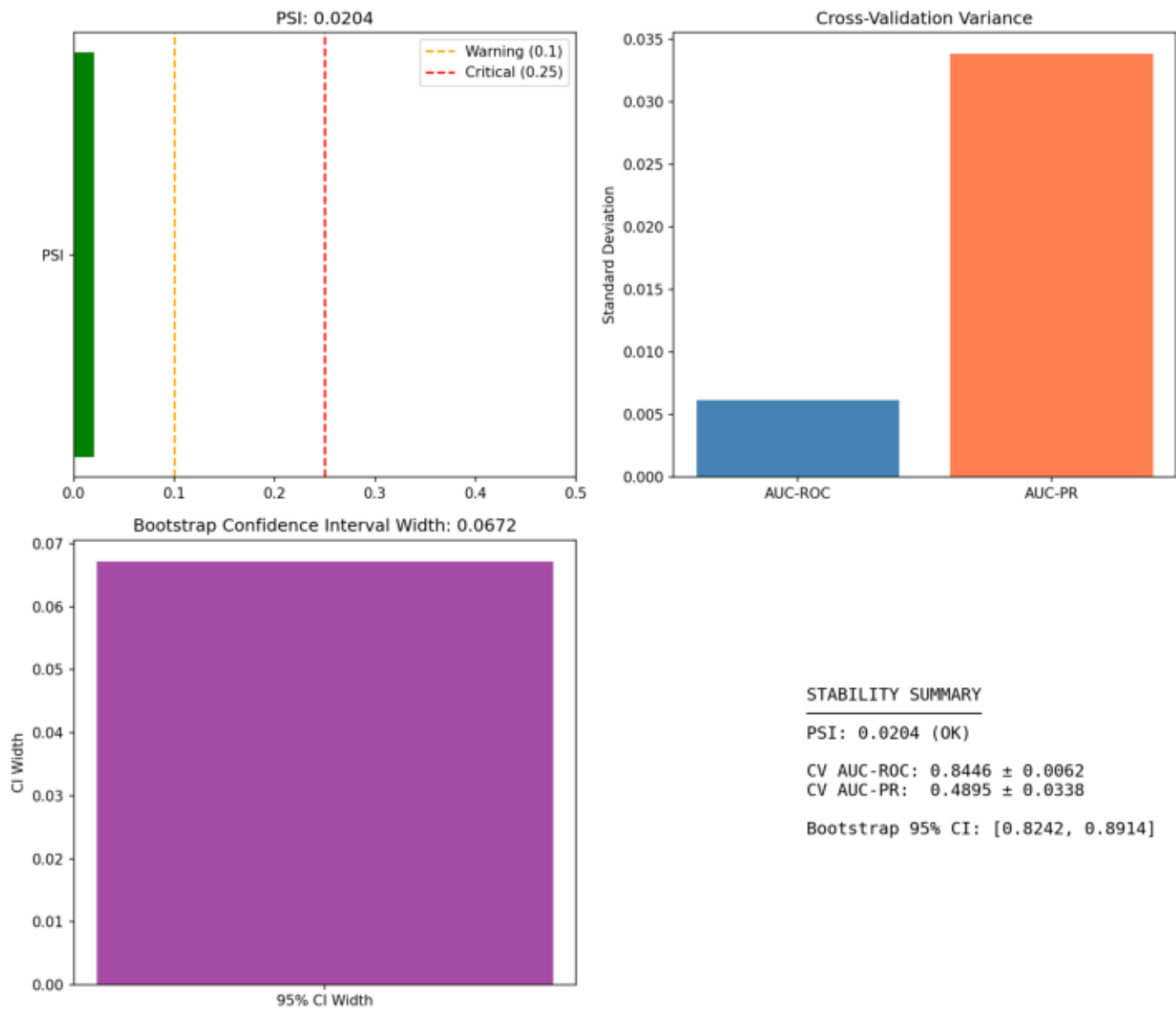


cv_results: Fold scores cluster around ROC 0.8446 and PR 0.4895.

value_date_quarter=2024Q2 - 3) Model Stability



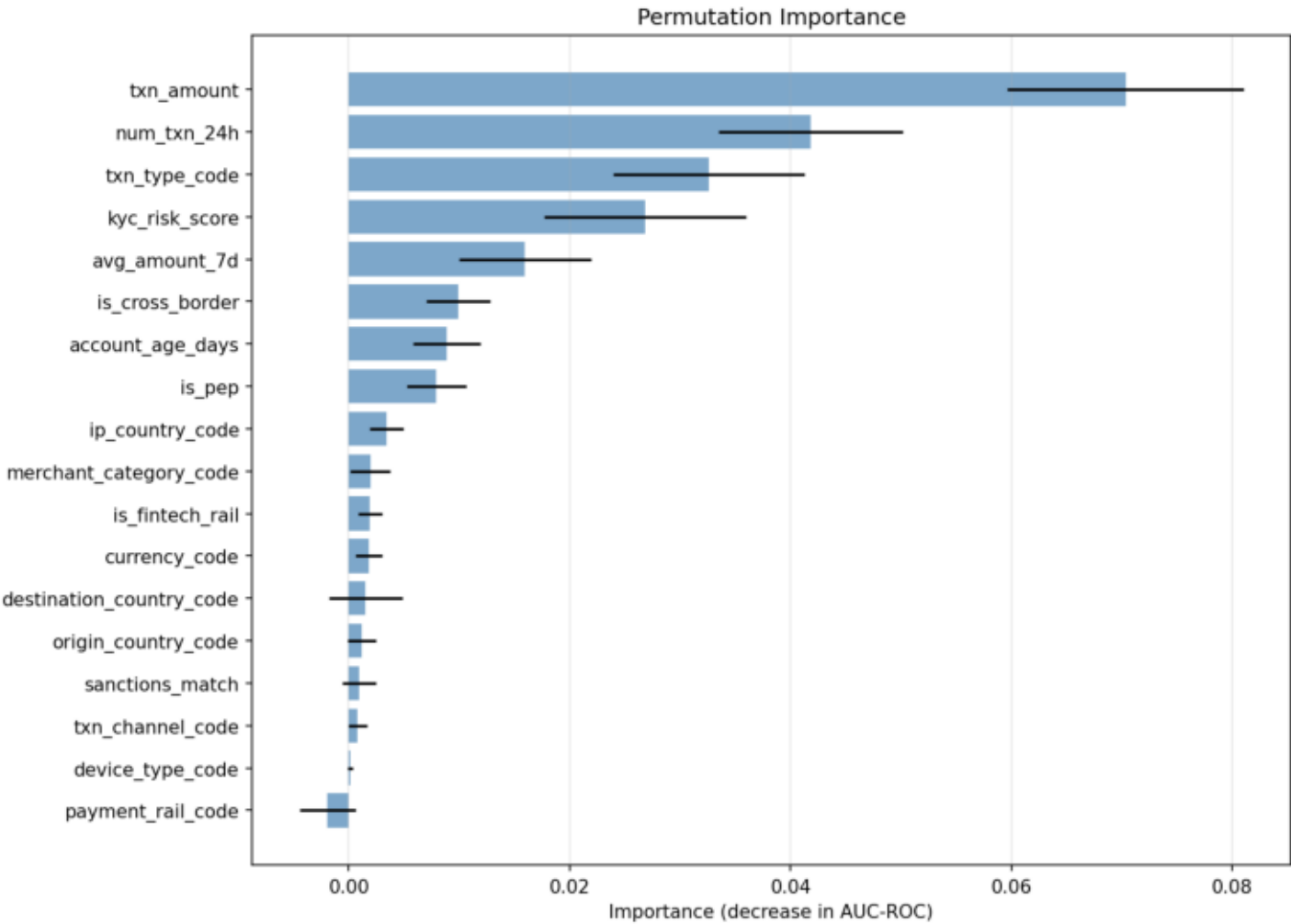
value_date_quarter=2024Q2 - 3) Model Stability



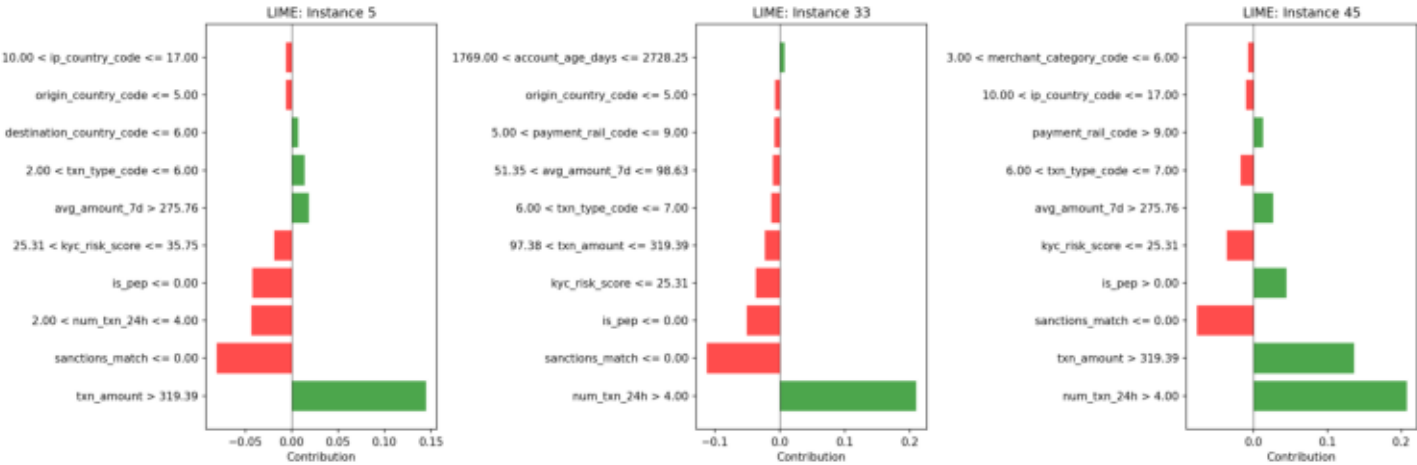
Metric	Figure	Explanation
model_type	tree	Identifies the estimator family so explanation methods can be interpreted in the right context. Next: confirm the chosen explanation methods are appropriate for this model class.
methods_used	permutation, pdp, ice, lime	Lists the interpretability methods produced in this run. Next: compare signals across methods before using them in documentation or model challenge.
shap_status	unavailable	Summarizes interpretability evidence for this run. Next: use these outputs to document model drivers and challenge unexpected dependencies.
shap_reason	No module named 'shap'	Summarizes interpretability evidence for this run. Next: use these outputs to document model drivers and challenge unexpected dependencies.
perm_top_features	txn_amount, num_txn_24h, txn_type_code, kyc_risk_score, avg_amount_7d ...	Shows features with the largest performance drop when permuted, which highlights global importance. Next: validate that the top drivers align with business logic, feature governance, and policy expectations.
lime_instances	3	Counts the locally explained observations reviewed with LIME. Next: examine representative true-positive and false-positive cases to understand local decision logic.

Metric	Figure	Explanation
pdp_features	avg_amount_7d, txn_amount, account_age_days, kyc_risk_score, num_txn_24h ...	Lists features with partial-dependence plots. Next: review whether average model response is monotonic, stable, and operationally sensible.
ice_features	avg_amount_7d, txn_amount, account_age_days, kyc_risk_score	Lists features with individual conditional expectation plots. Next: inspect heterogeneity and interactions that may be hidden in average-only views.

value_date_quarter=2024Q2 - 4) Model Interpretability

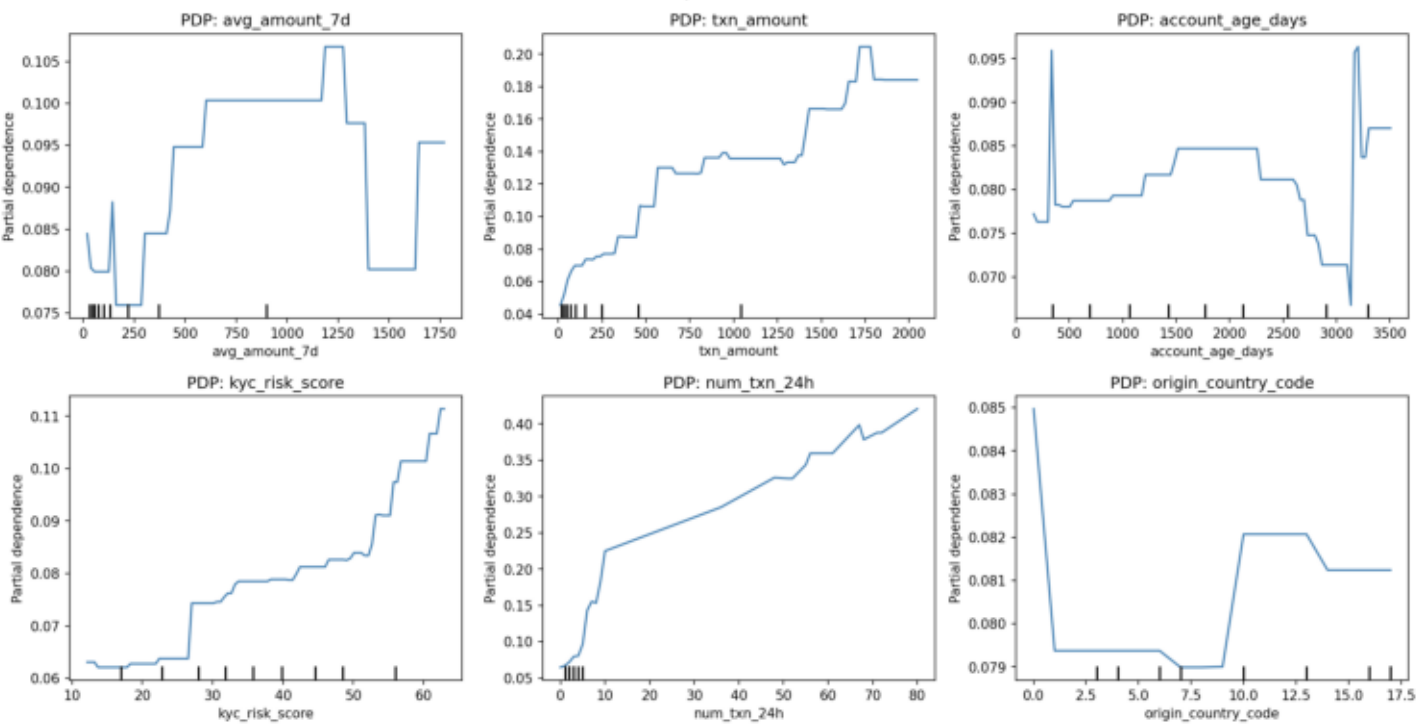


permutation_importance: Bars show how much AUC drops when each feature is permuted.



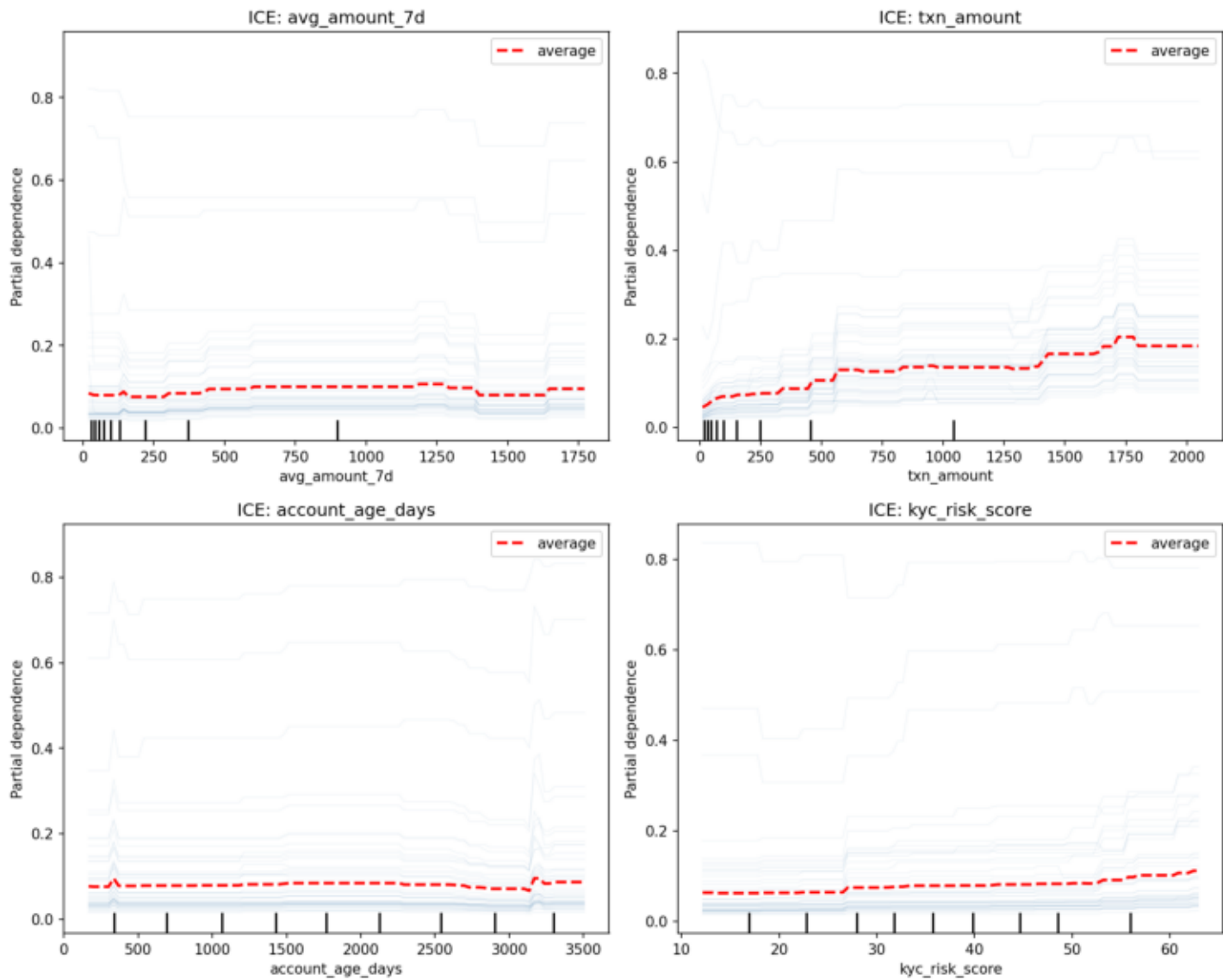
value_date_quarter=2024Q2 - 4) Model Interpretability

Partial Dependence Plots



pdp: PDP curves show average model response as a feature varies.

Individual Conditional Expectation (ICE) Plots



ice: ICE curves show per-instance responses as a feature varies.

value_date_quarter=2024Q3 - 1) Model Effectiveness (1/3)

Metric	Figure	Explanation
auc_roc	0.8432	Measures ranking power across all thresholds. 0.8432 indicates strong separation. Next: compare AUC-ROC across segments and keep threshold tuning separate from ranking validation.
auc_pr	0.5444	Measures precision-recall performance under class imbalance; the observed event rate is 16.10%. Next: use this together with alert capacity to decide whether the high-score queue is practically useful.
precision	0.6301	Shows the share of flagged cases that are true positives at the current threshold. Next: if this is below review expectations, raise the threshold or improve exclusion features.
recall	0.3232	Shows the share of true positives captured at the current threshold. Next: if missed cases are too high, lower the threshold or add stronger risk features.
f1	0.4272	Balances precision and recall in one score. Next: use it as a threshold-selection aid when both missed cases and false alerts matter.
ks_statistic	0.5416	Measures score separation between positive and negative classes. Next: low KS suggests the model needs better features or a different specification.

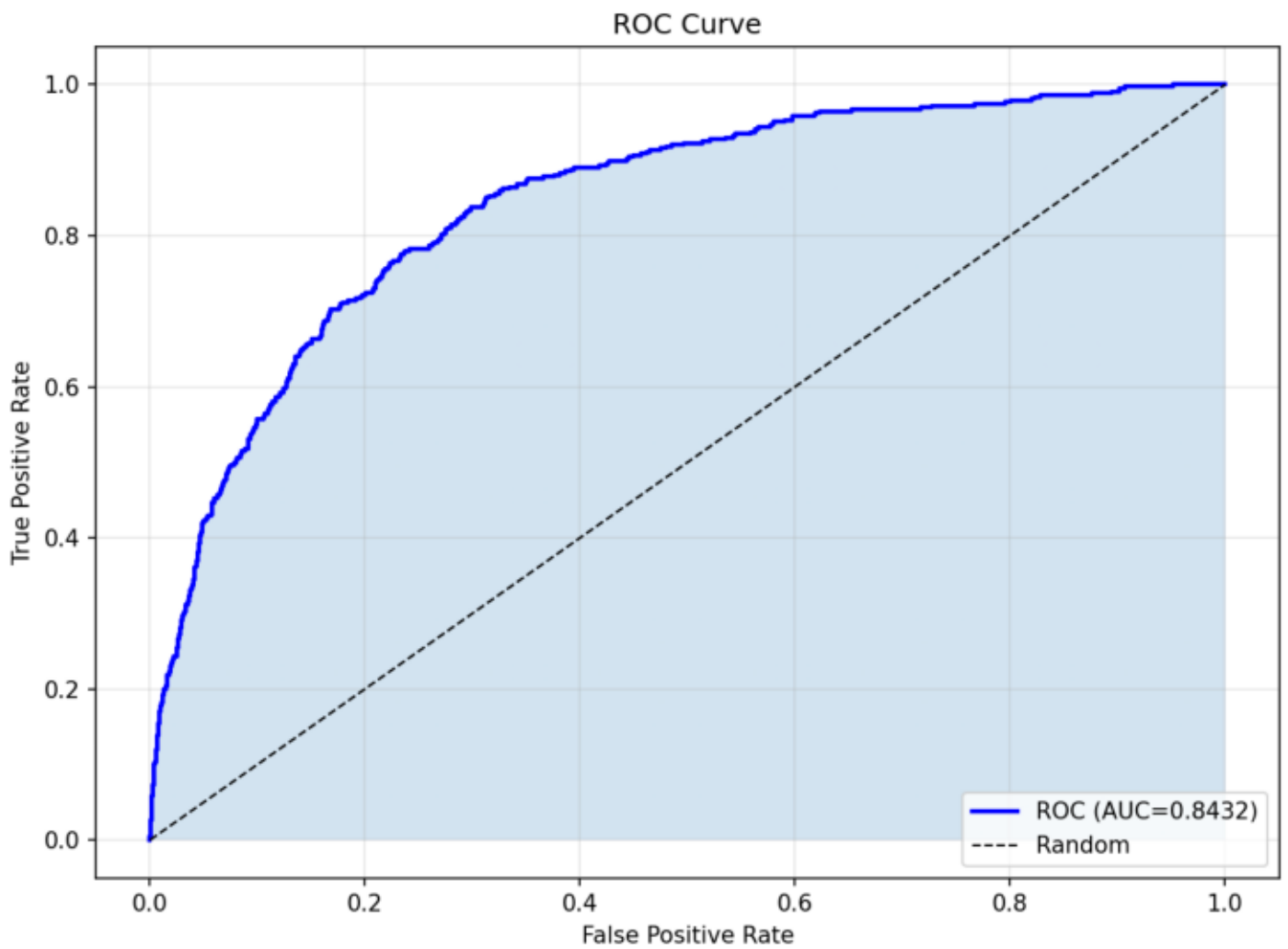
value_date_quarter=2024Q3 - 1) Model Effectiveness (2/3)

Metric	Figure	Explanation
ks_threshold	0.1600	This is the threshold that maximizes KS separation. Next: treat it as a candidate operating point, then compare it with business capacity and policy limits.
confusion_matrix[TN]	2144	True negatives correctly cleared. Next: preserve this base while tuning the alert threshold.
confusion_matrix[FP]	81	False positives create review cost without SAR conversion. Next: reduce this count with threshold tuning or sharper risk features.
confusion_matrix[FN]	289	False negatives are missed SAR cases. Next: if this count is unacceptable, lower the threshold or add features that capture suspicious behavior earlier.
confusion_matrix[TP]	138	True positives are correctly escalated cases. Next: assess whether this volume is sufficient relative to expected SAR capture goals.
precision_at_k[10]	0.8000	Shows how accurate the top-10 highest-risk queue is. Next: align K with investigator capacity and confirm that top alerts stay high quality.
precision_at_k[50]	0.8400	Shows how accurate the top-50 highest-risk queue is. Next: align K with investigator capacity and confirm that top alerts stay high quality.
precision_at_k[100]	0.7600	Shows how accurate the top-100 highest-risk queue is. Next: align K with investigator capacity and confirm that top alerts stay high quality.

value_date_quarter=2024Q3 - 1) Model Effectiveness (3/3)

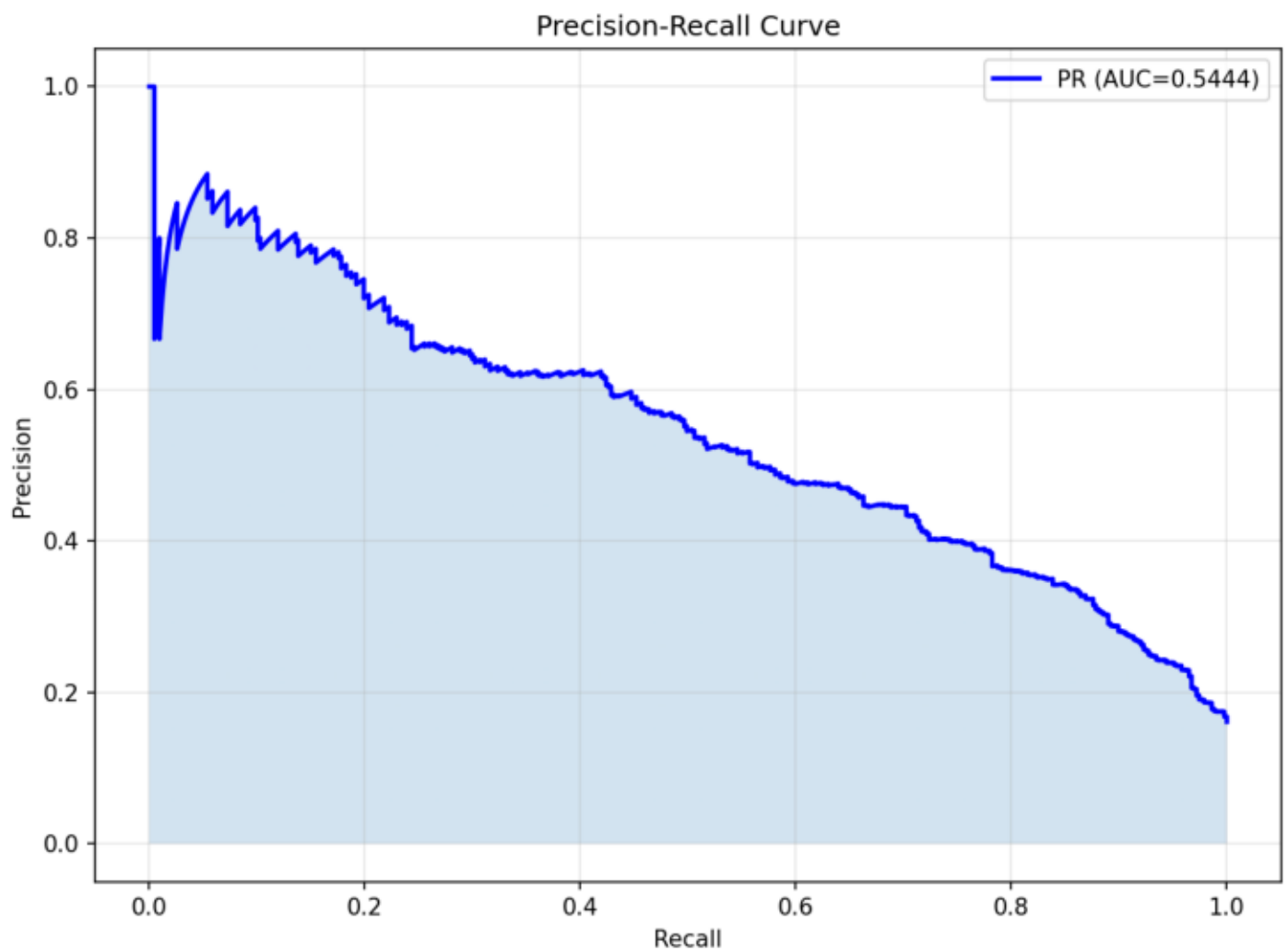
Metric	Figure	Explanation
precision_at_k[200]	0.6400	Shows how accurate the top-200 highest-risk queue is. Next: align K with investigator capacity and confirm that top alerts stay high quality.
precision_at_k[500]	0.4940	Shows how accurate the top-500 highest-risk queue is. Next: align K with investigator capacity and confirm that top alerts stay high quality.
recall_at_k[10]	0.0187	Shows how much of the positive population is captured within the top-10 queue. Next: increase K if coverage is too low, or improve ranking quality if capacity cannot expand.
recall_at_k[50]	0.0984	Shows how much of the positive population is captured within the top-50 queue. Next: increase K if coverage is too low, or improve ranking quality if capacity cannot expand.
recall_at_k[100]	0.1780	Shows how much of the positive population is captured within the top-100 queue. Next: increase K if coverage is too low, or improve ranking quality if capacity cannot expand.
recall_at_k[200]	0.2998	Shows how much of the positive population is captured within the top-200 queue. Next: increase K if coverage is too low, or improve ranking quality if capacity cannot expand.
recall_at_k[500]	0.5785	Shows how much of the positive population is captured within the top-500 queue. Next: increase K if coverage is too low, or improve ranking quality if capacity cannot expand.

value_date_quarter=2024Q3 - 1) Model Effectiveness



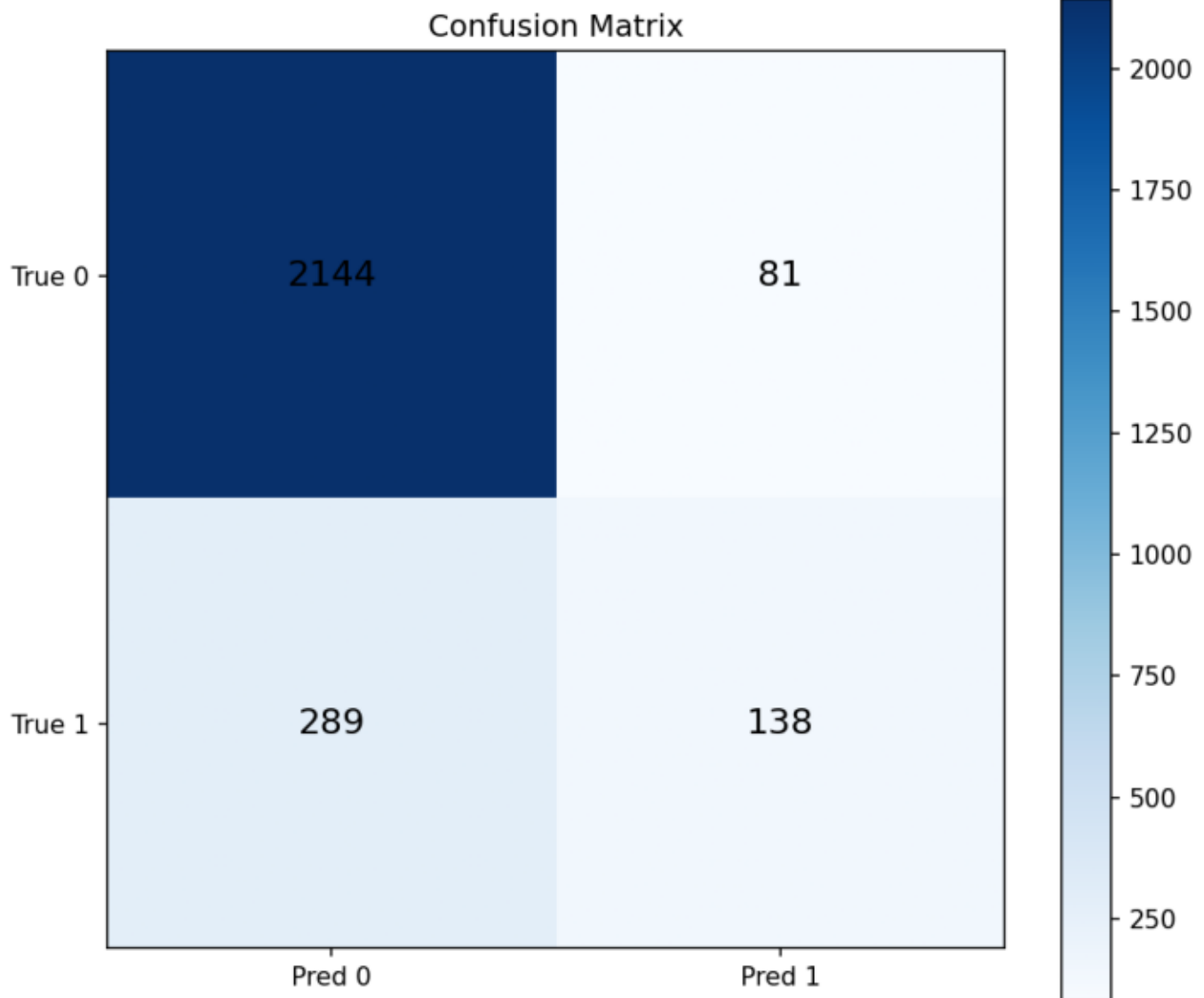
roc_curve: AUC-ROC=0.8432; curve above diagonal indicates strong separation.

value_date_quarter=2024Q3 - 1) Model Effectiveness



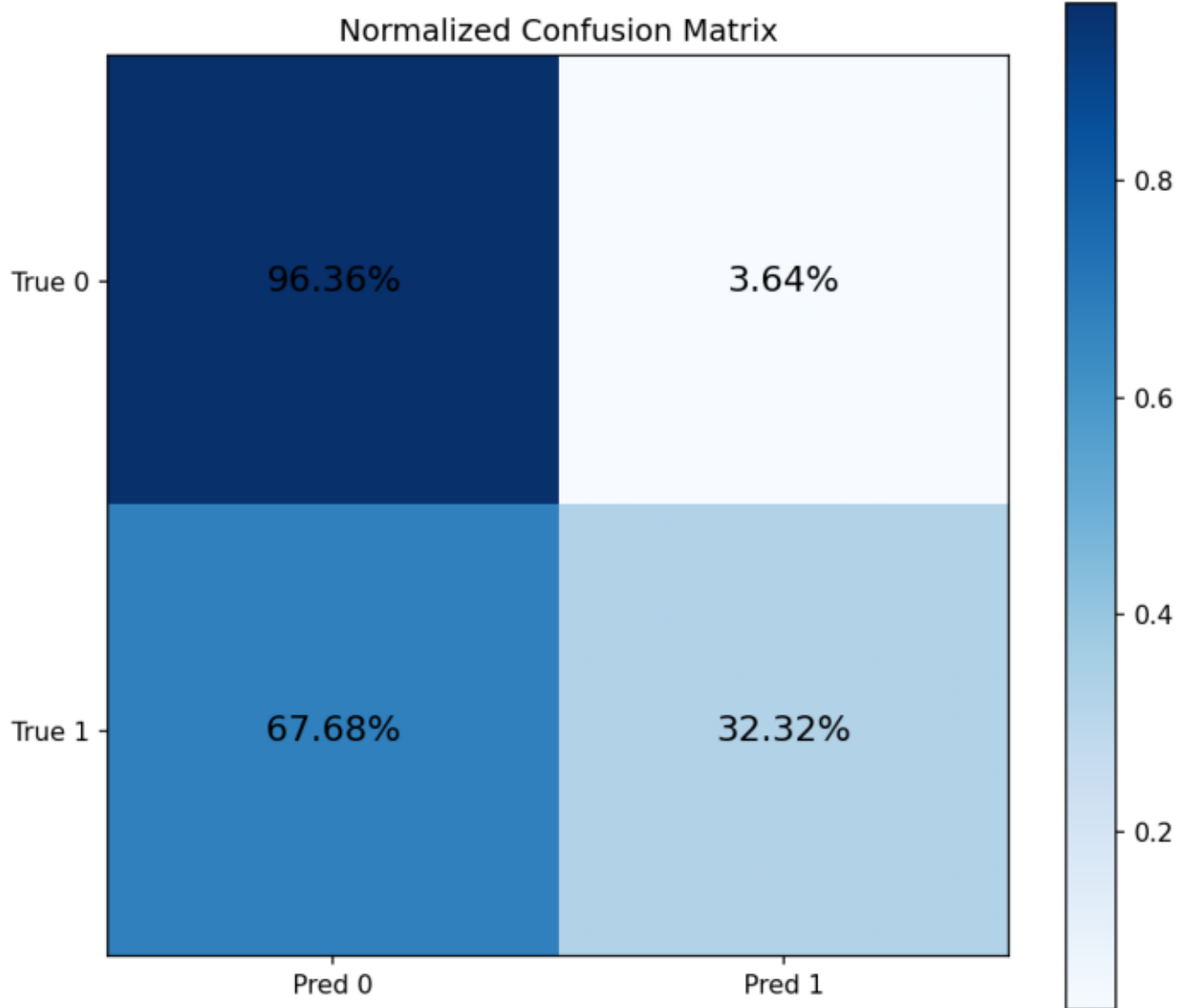
pr_curve: AUC-PR=0.5444 vs baseline 16.10%; higher curve implies better precision at recall.

value_date_quarter=2024Q3 - 1) Model Effectiveness



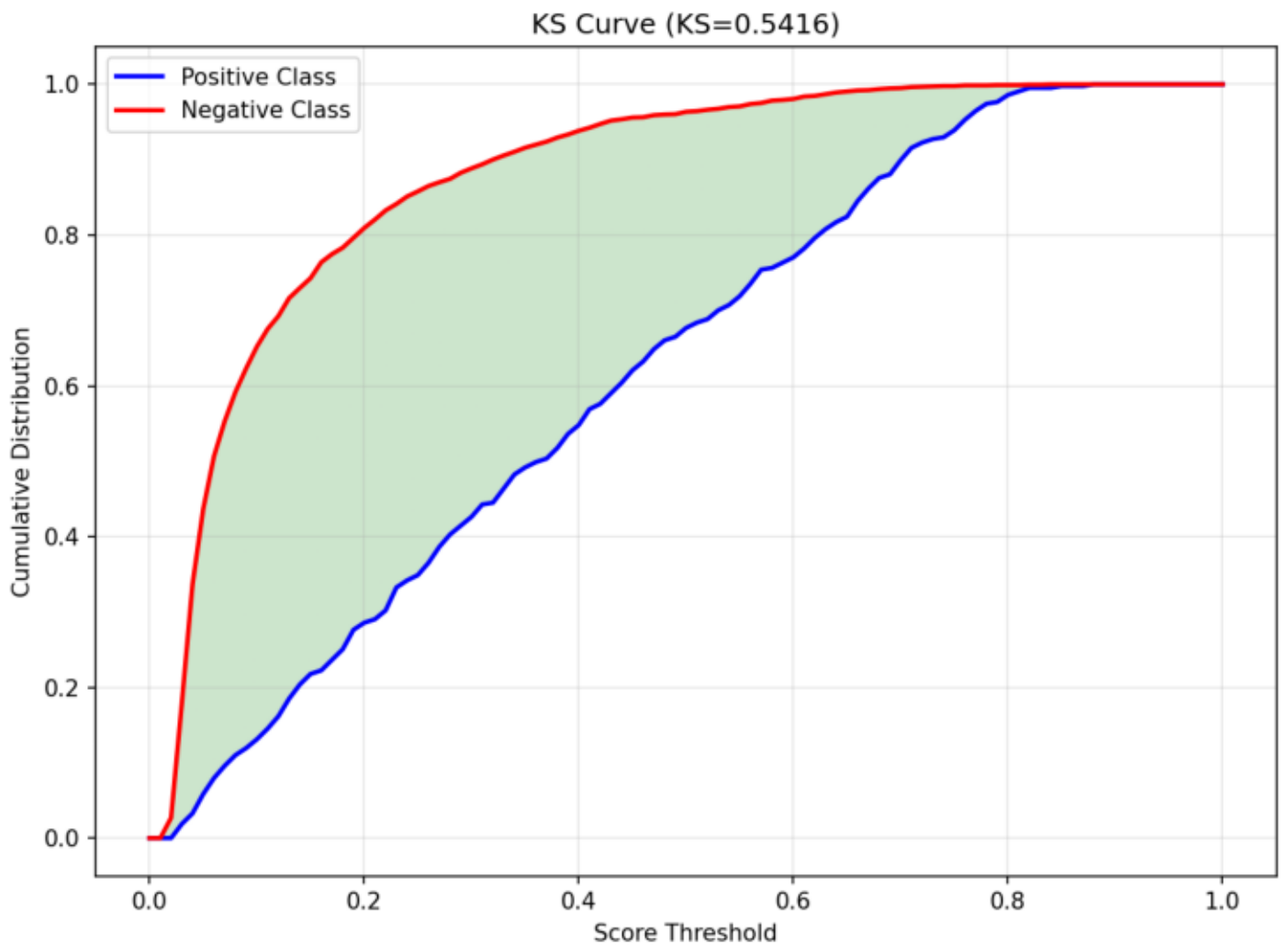
confusion_matrix: At threshold 0.50: TP=138, FP=81, FN=289, TN=2144.

value_date_quarter=2024Q3 - 1) Model Effectiveness



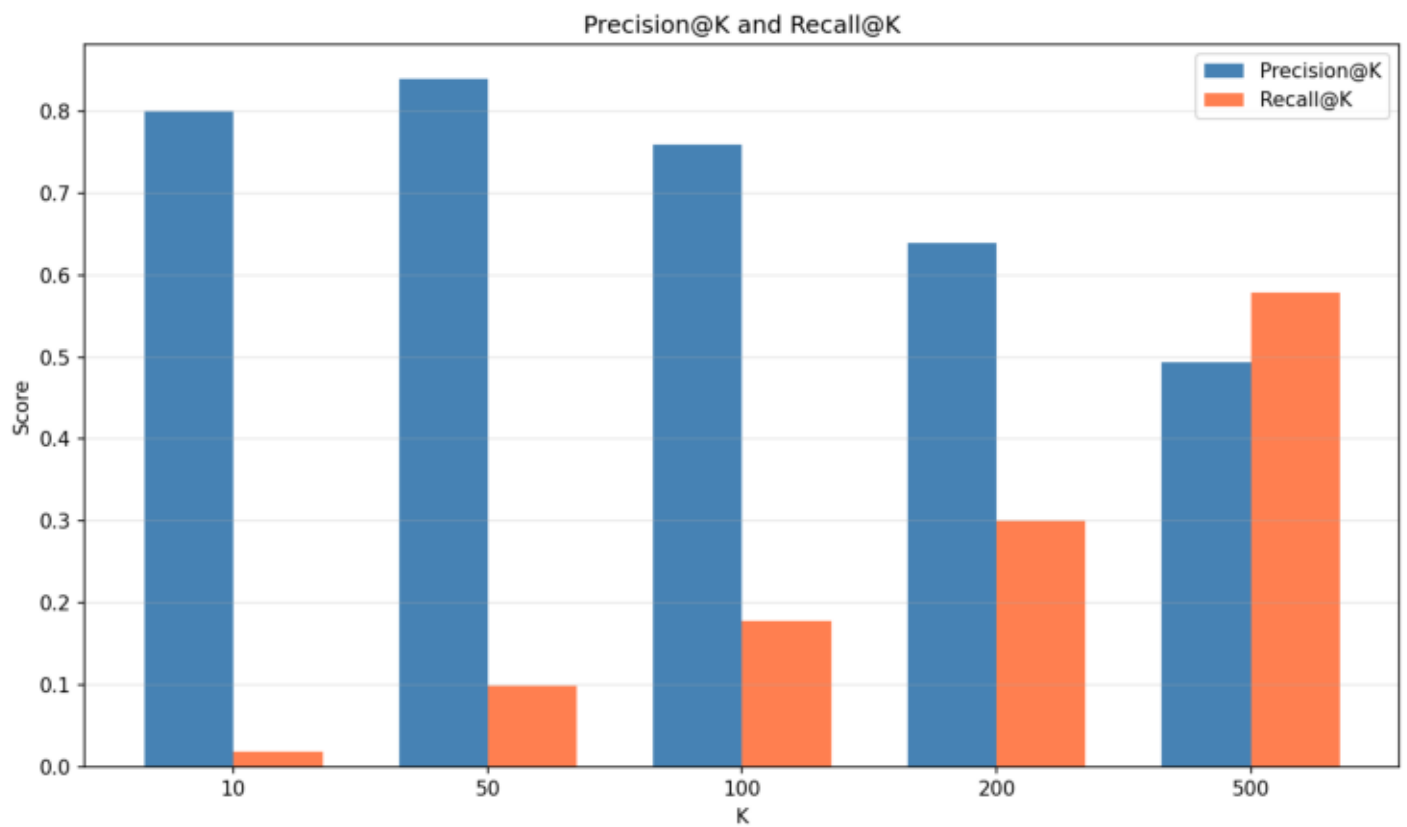
confusion_matrix_norm: Normalized rates: TPR=32.32%, TNR=96.36%.

value_date_quarter=2024Q3 - 1) Model Effectiveness



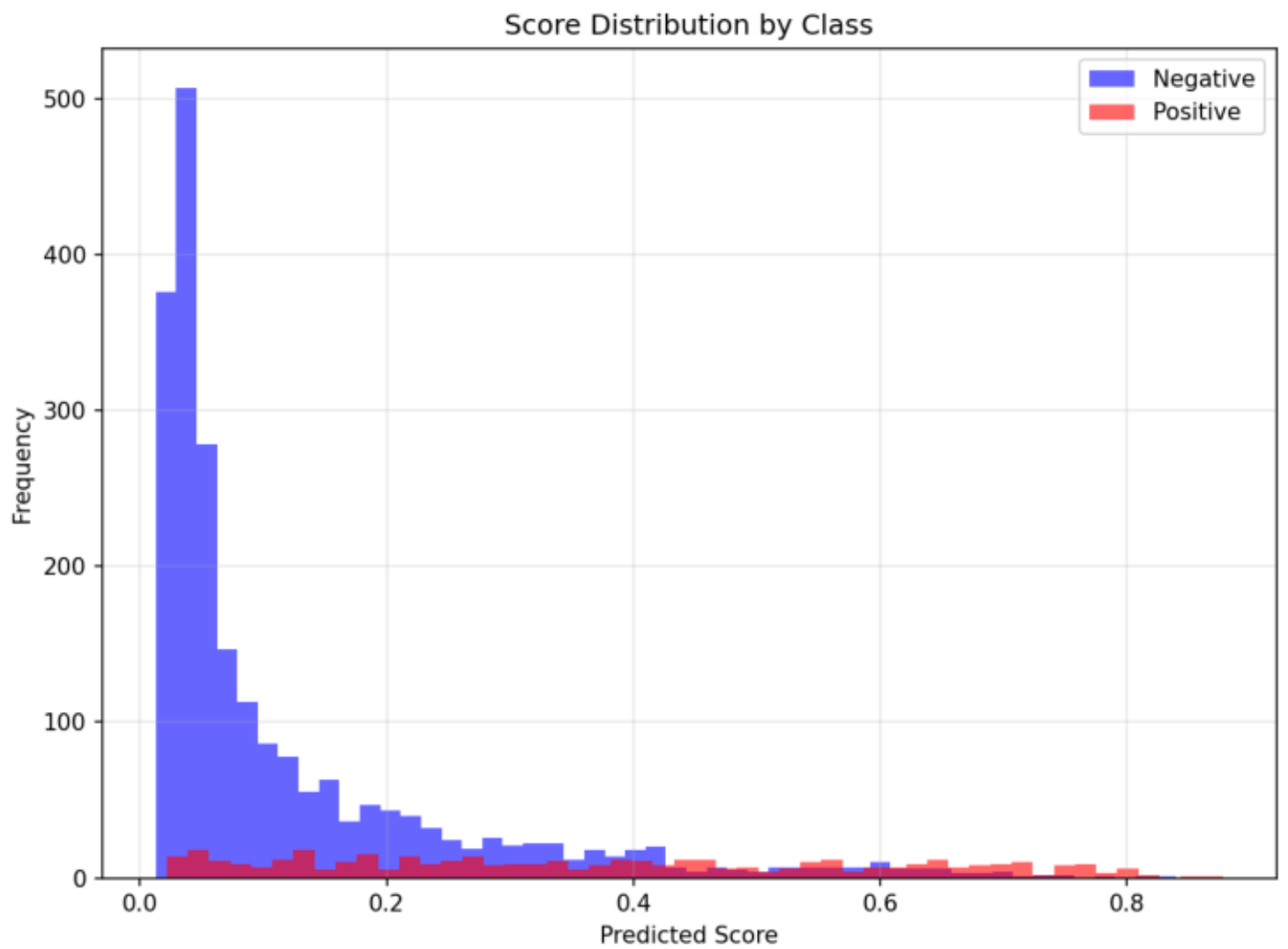
ks_curve: Maximum separation KS=0.5416 at threshold 0.16.

value_date_quarter=2024Q3 - 1) Model Effectiveness



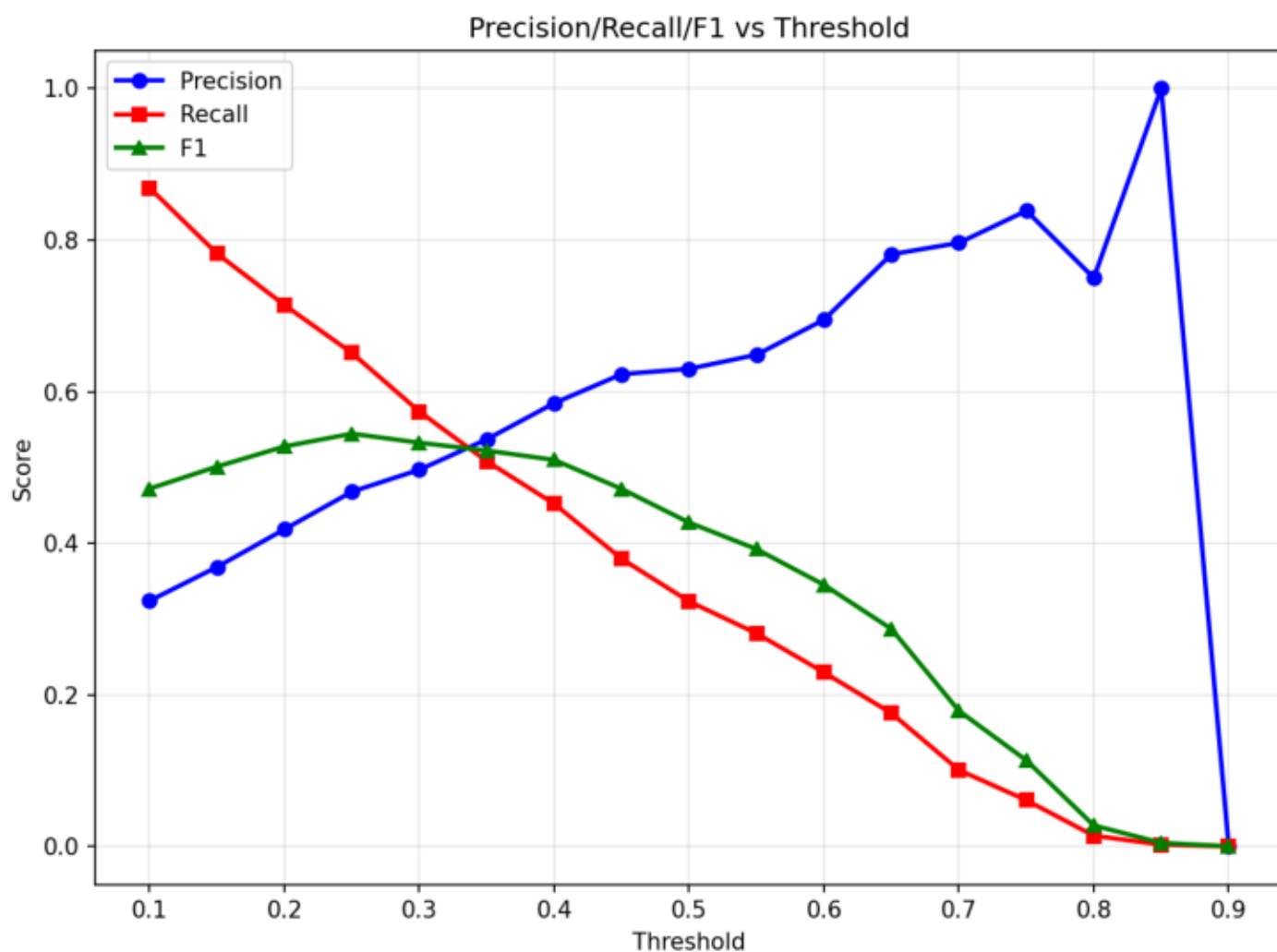
precision_recall_at_k: Bars show precision/recall as you expand the review queue.

value_date_quarter=2024Q3 - 1) Model Effectiveness



score_distribution: Mean score: positive=0.378, negative=0.122 (larger gap is better).

value_date_quarter=2024Q3 - 1) Model Effectiveness



threshold_analysis: Best F1 in tested grid is 0.5446 at threshold 0.25.

value_date_quarter=2024Q3 - 2) Model Efficiency (1/4)

Metric	Figure	Explanation
fpr	0.0364	Measures the share of negatives incorrectly flagged; 0.0364 is low. Next: if review demand is too high, raise the threshold or add features that better suppress false alerts.
tn	2144	Counts negatives correctly cleared without manual review. Next: preserve this clearance rate while checking that recall does not fall below policy expectations.
fp	81	Counts non-events that would still be sent to review. Next: use this with staffing capacity to decide whether threshold or feature refinement is required.
threshold	0.5000	This is the active operating cutoff used to flag alerts. Next: tune it jointly with recall, precision, and review capacity rather than in isolation.
fpr_at_thresholds[thresholds.t_0.05]	0.5631	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.05. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.10]	0.3479	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.10. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.

value_date_quarter=2024Q3 - 2) Model Efficiency (2/4)

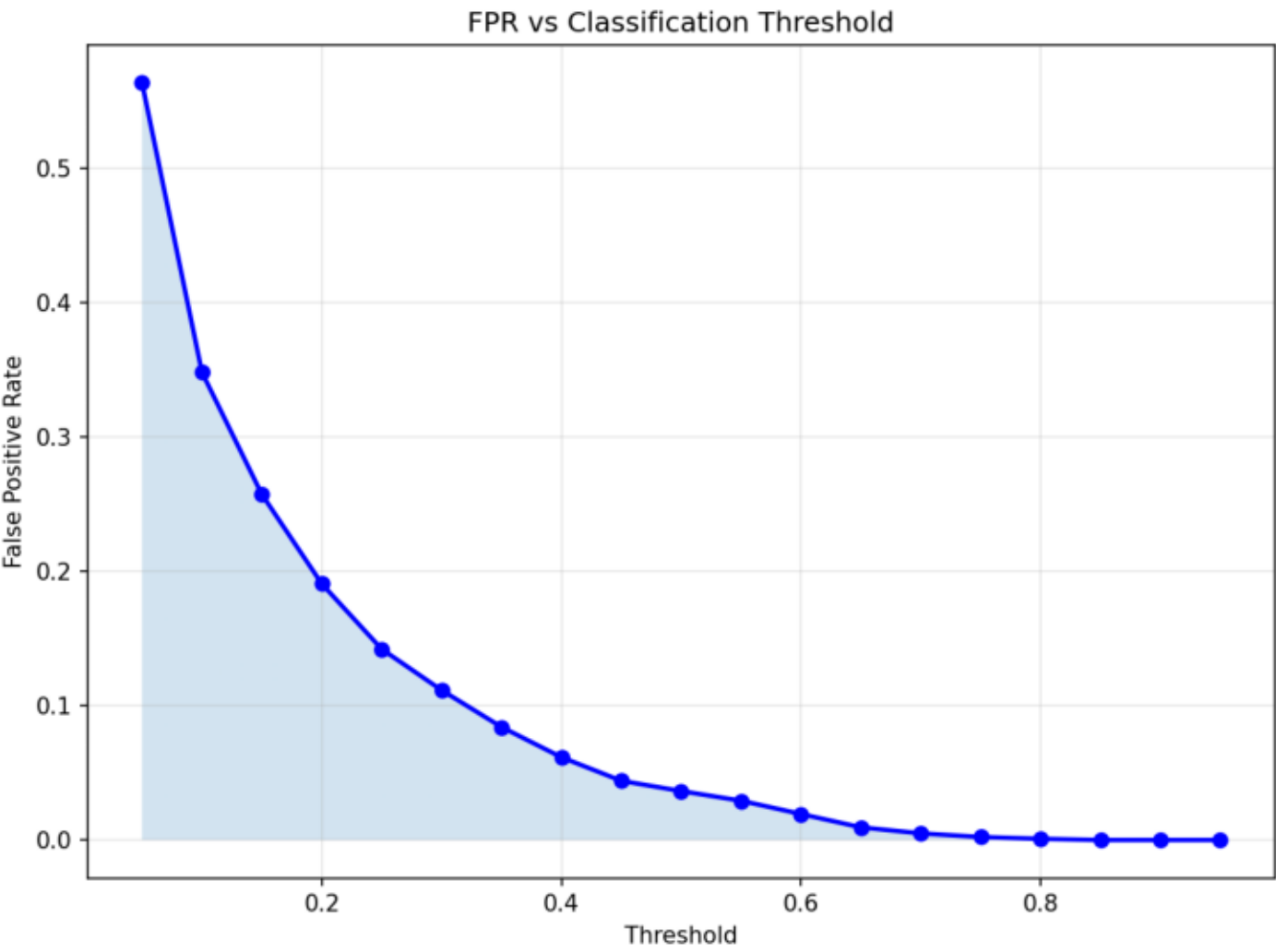
Metric	Figure	Explanation
fpr_at_thresholds[thresholds.t_0.15]	0.2571	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.15. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.20]	0.1906	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.20. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.25]	0.1420	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.25. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.30]	0.1115	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.30. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.35]	0.0840	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.35. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.40]	0.0616	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.40. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.

value_date_quarter=2024Q3 - 2) Model Efficiency (3/4)

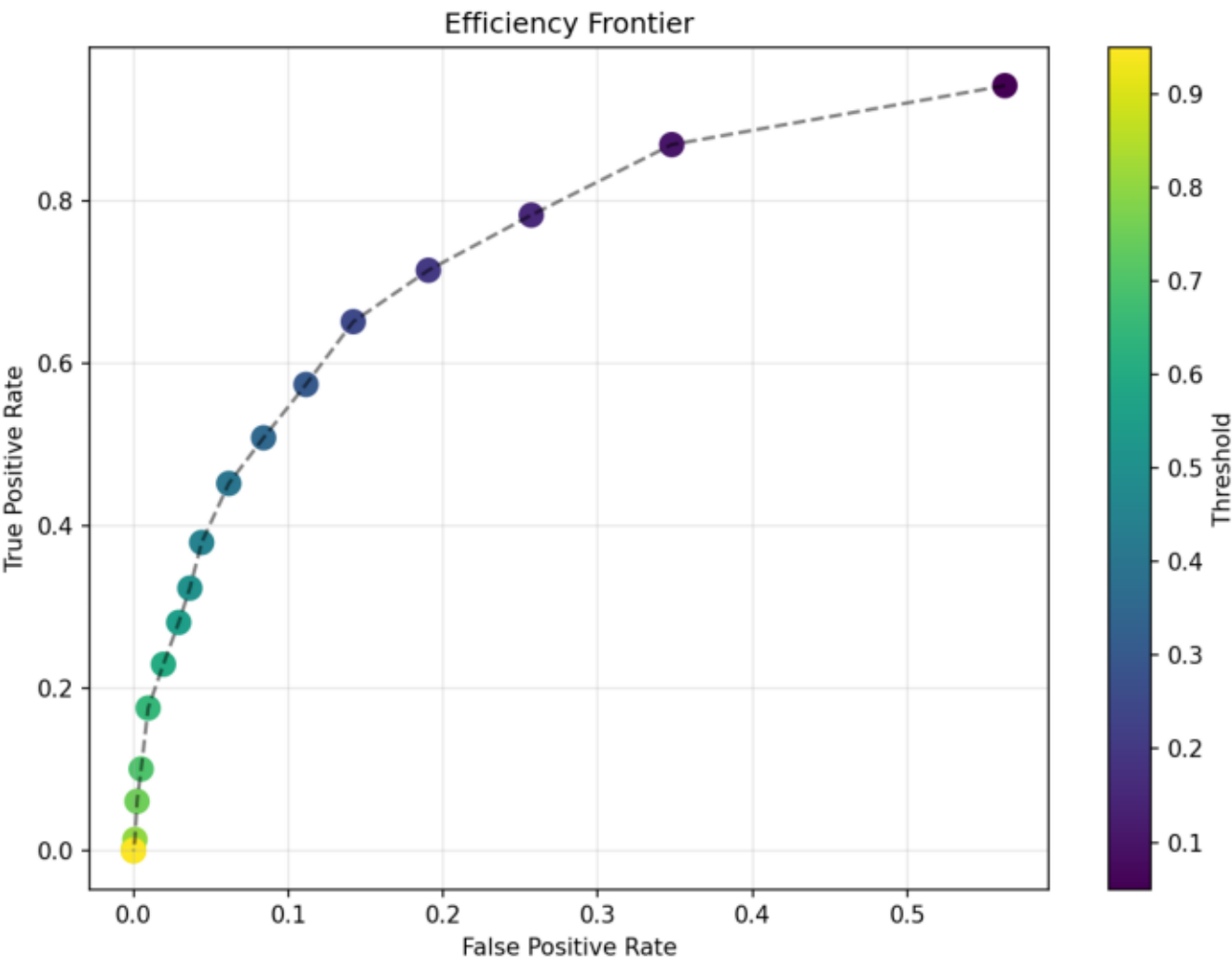
Metric	Figure	Explanation
fpr_at_thresholds[thresholds.t_0.45]	0.0440	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.45. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.50]	0.0364	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.50. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.55]	0.0292	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.55. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.60]	0.0193	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.60. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.65]	0.0094	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.65. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.70]	0.0049	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.70. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.

value_date_quarter=2024Q3 - 2) Model Efficiency (4/4)

Metric	Figure	Explanation
fpr_at_thresholds[thresholds.t_0.75]	0.0022	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.75. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.80]	0.0009	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.80. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.85]	0.0000	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.85. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.90]	0.0000	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.90. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.95]	0.0000	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.95. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.

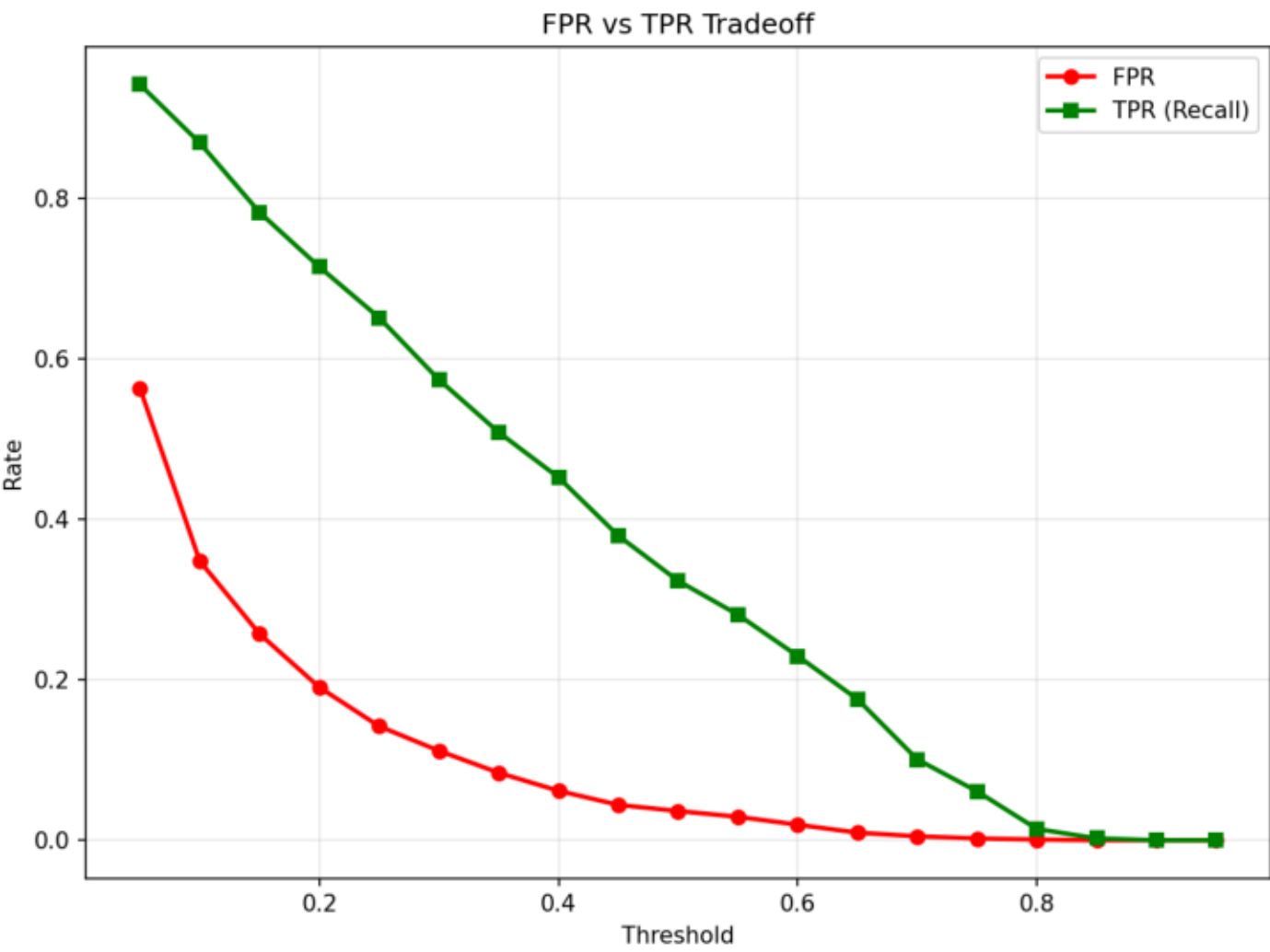


value_date_quarter=2024Q3 - 2) Model Efficiency



efficiency_frontier: Each point shows the FPR/TPR tradeoff; move toward the top-left for better efficiency.

value_date_quarter=2024Q3 - 2) Model Efficiency

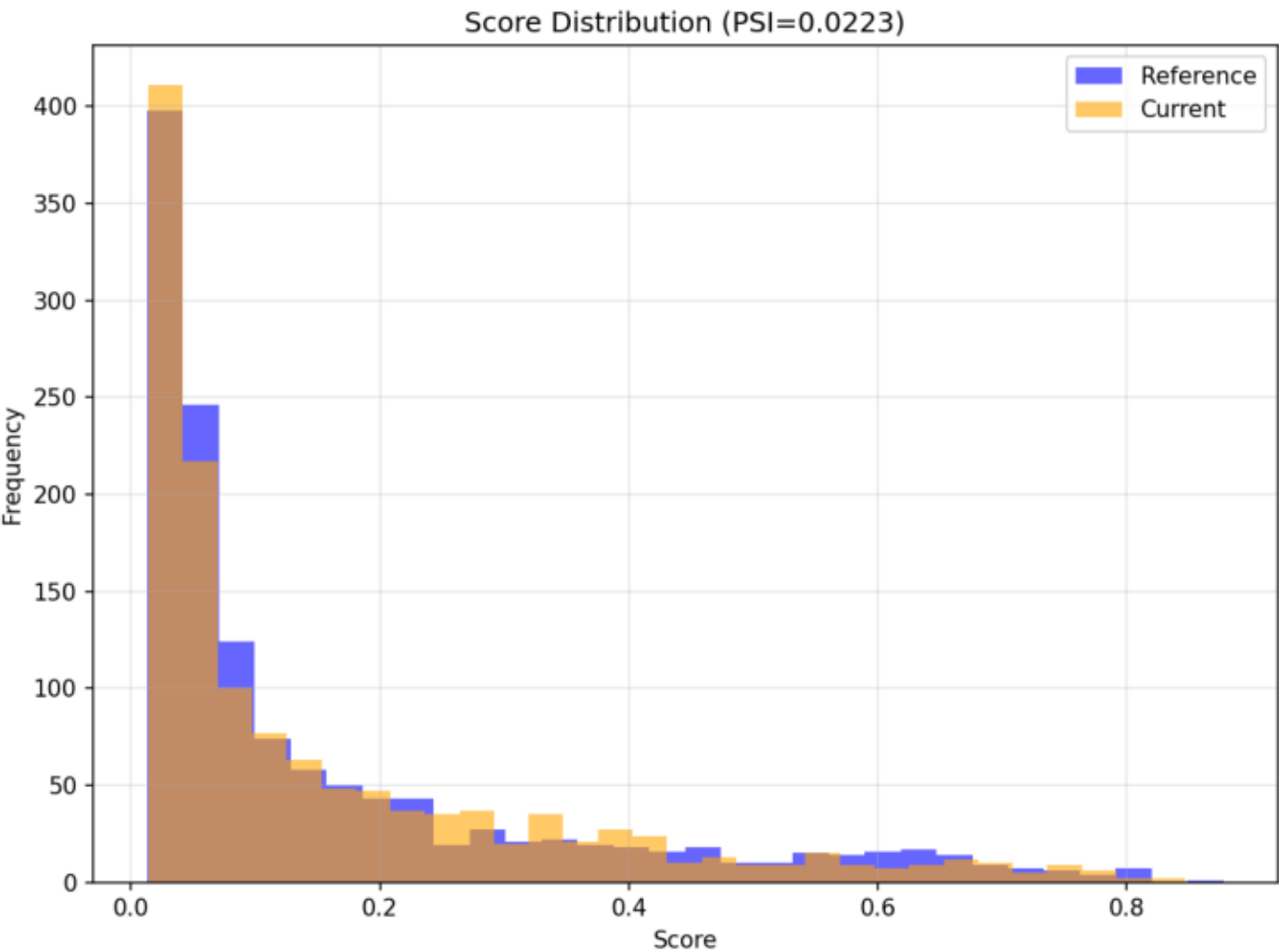


value_date_quarter=2024Q3 - 3) Model Stability (1/2)

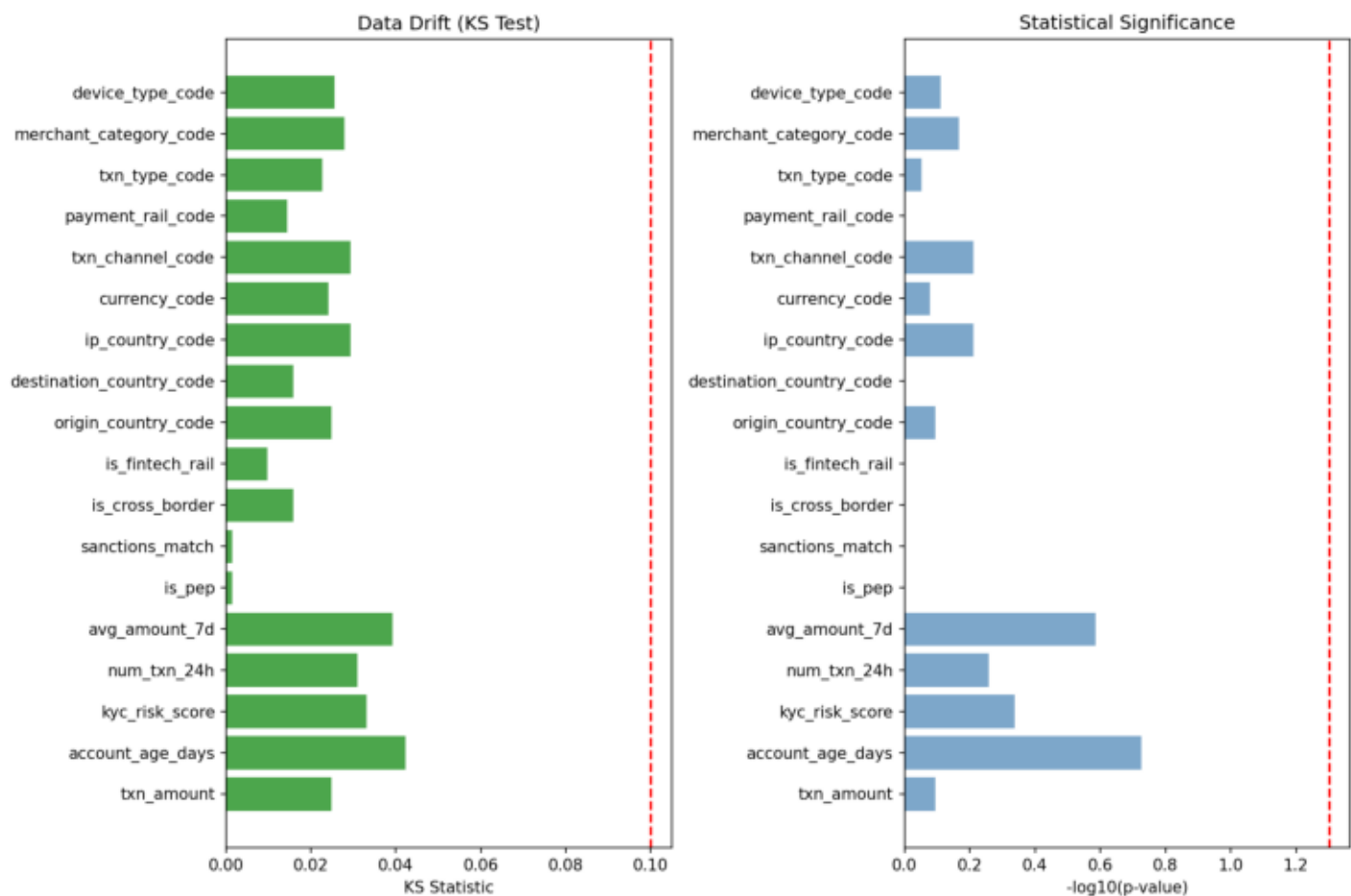
Metric	Figure	Explanation
psi	0.0223	Measures score-distribution shift between reference and current populations; 0.0223 is negligible. Next: investigate segment mix, data drift, and recalibration needs when PSI is material.
cv_auc_roc_mean	0.8434	Average AUC-ROC across validation folds. Next: compare it with overall AUC-ROC; a large gap can indicate overfitting or unstable sampling.
cv_auc_roc_std	0.0139	Standard deviation of AUC-ROC across folds. Next: if variability is high, simplify the model or increase training stability before deployment.
cv_auc_pr_mean	0.5532	Average AUC-PR across validation folds. Next: confirm that precision-recall performance remains acceptable under resampling.
cv_auc_pr_std	0.0536	Standard deviation of AUC-PR across folds. Next: high dispersion suggests unstable alert quality and warrants deeper validation.
bootstrap_auc_roc_mean	0.8466	Bootstrap mean AUC-ROC estimates expected ranking performance under repeated sampling. Next: compare it with the point estimate to judge optimism in the current test run.
bootstrap_auc_roc_ci_lower	0.8205	Lower confidence bound for bootstrap AUC-ROC. Next: use it as a conservative performance floor in model-risk discussions.
bootstrap_auc_roc_ci_upper	0.8711	Upper confidence bound for bootstrap AUC-ROC. Next: review the interval width with the lower bound to assess uncertainty.

value_date_quarter=2024Q3 - 3) Model Stability (2/2)

Metric	Figure	Explanation
concept_drift_detected	False	No material concept-drift signal was detected in the evaluated slices. Next: retain ongoing monitoring because stable historical results do not guarantee future stability.
concept_drift_score	0.0112	Quantifies the magnitude of performance change across ordered slices. Next: rising scores should trigger deeper review of segment behavior and model refresh timing.

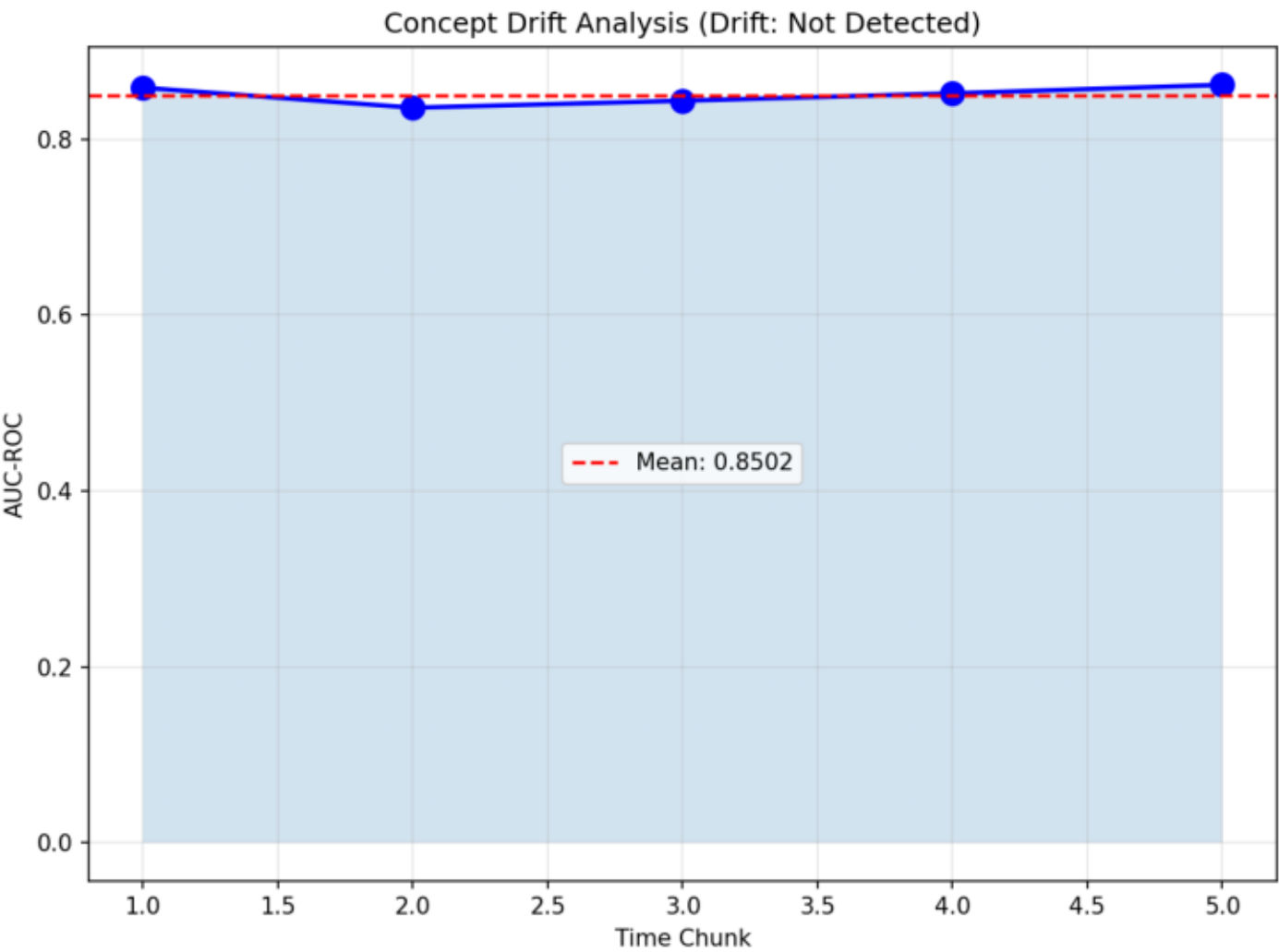


value_date_quarter=2024Q3 - 3) Model Stability



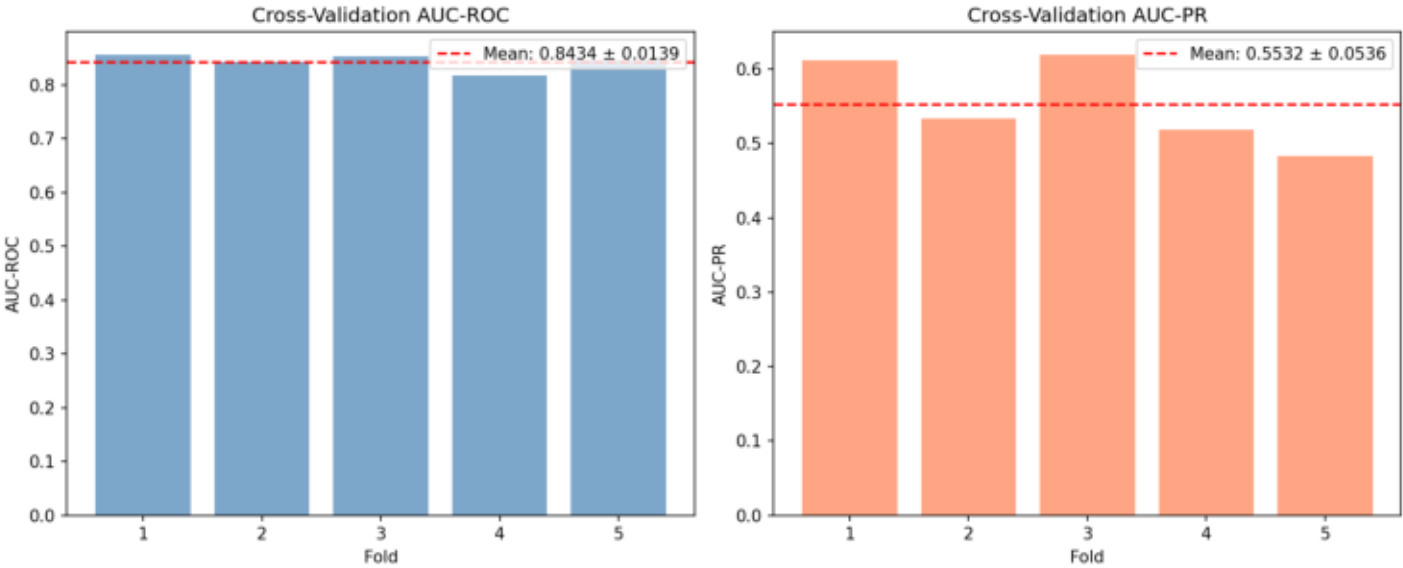
data_drift_heatmap: 0/18 features show drift; red bars highlight flagged features.

value_date_quarter=2024Q3 - 3) Model Stability



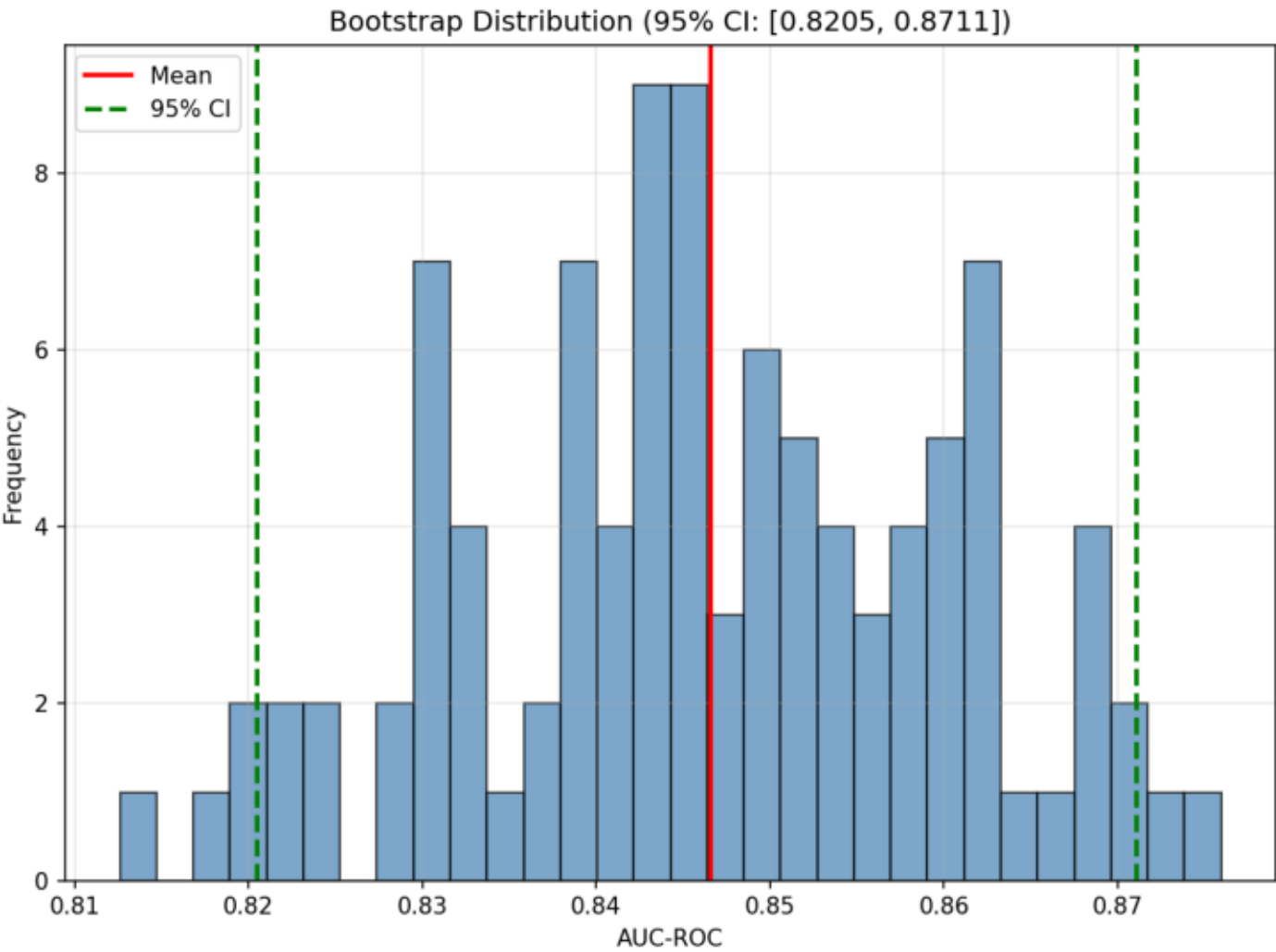
concept_drift: AUC across time chunks shows whether performance is stable over time.

value_date_quarter=2024Q3 - 3) Model Stability

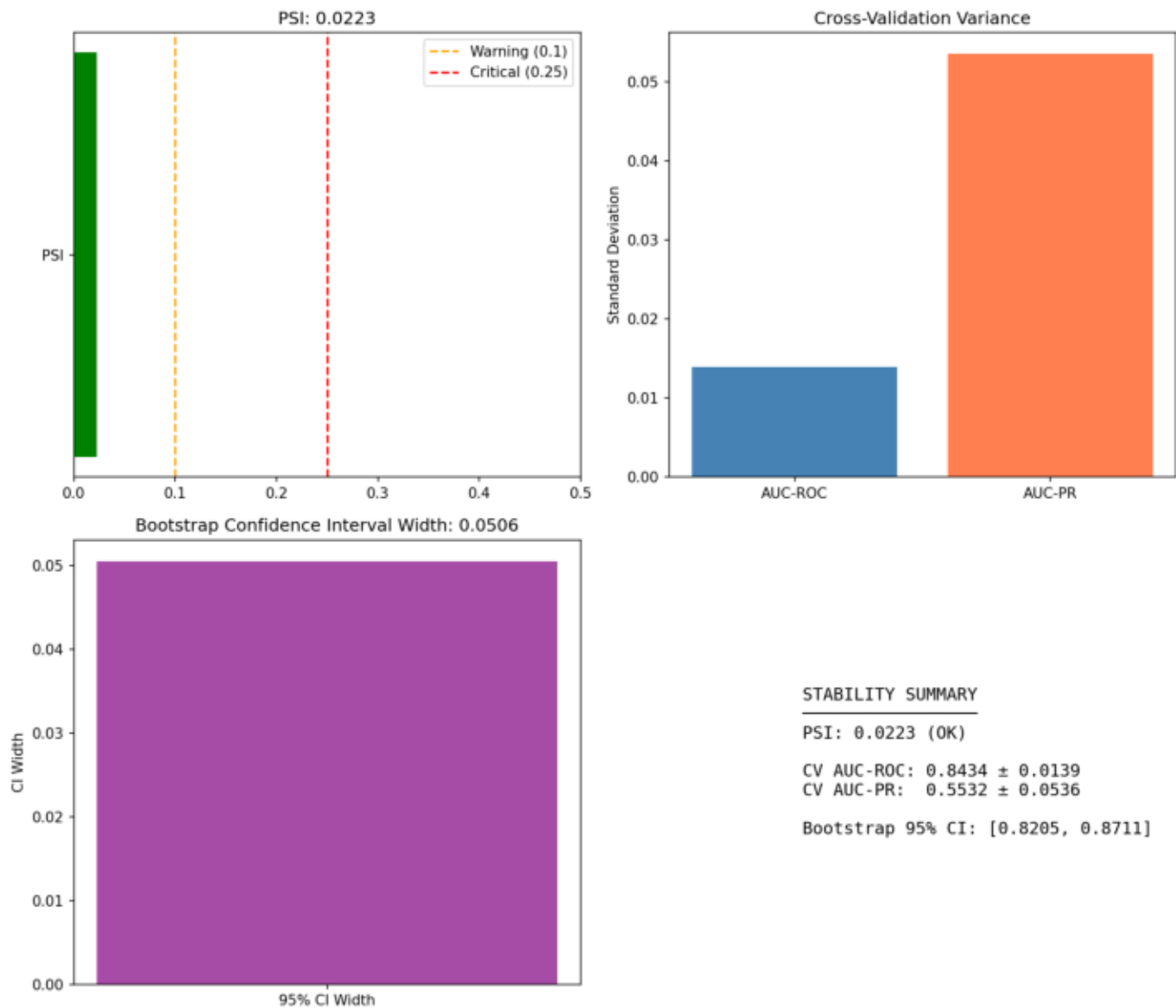


cv_results: Fold scores cluster around ROC 0.8434 and PR 0.5532.

value_date_quarter=2024Q3 - 3) Model Stability

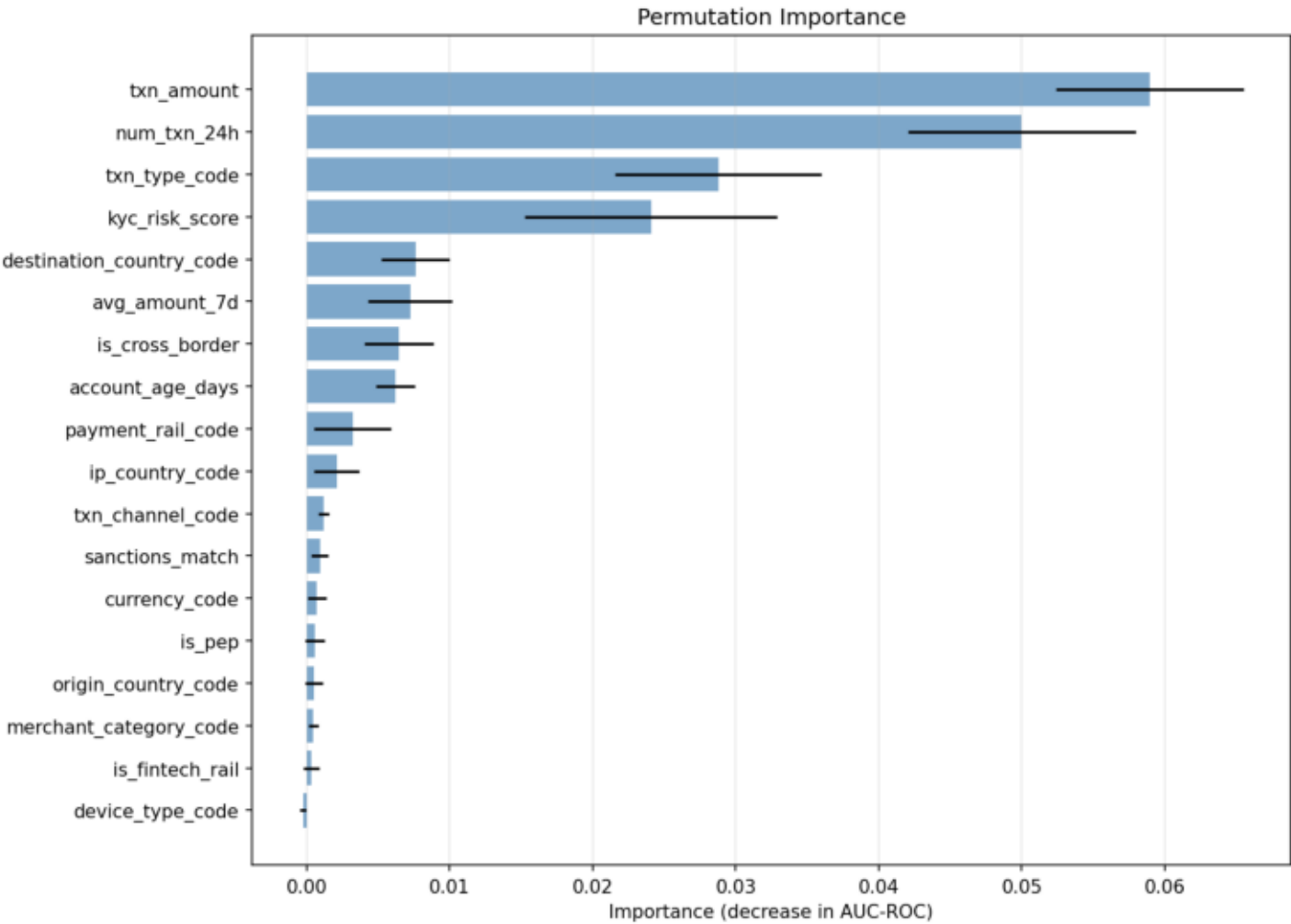


value_date_quarter=2024Q3 - 3) Model Stability



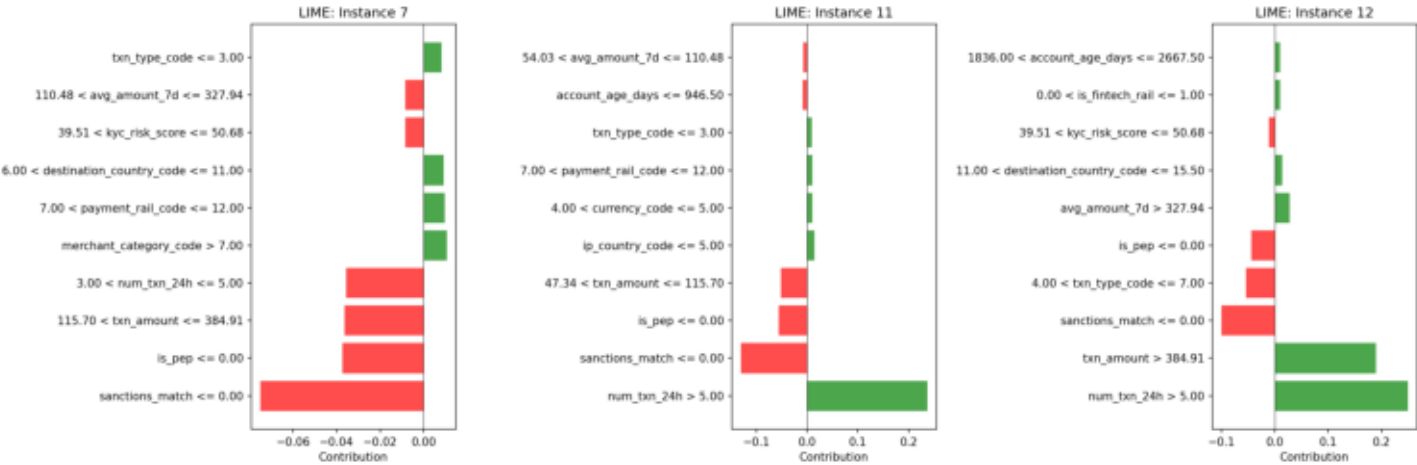
Metric	Figure	Explanation
model_type	tree	Identifies the estimator family so explanation methods can be interpreted in the right context. Next: confirm the chosen explanation methods are appropriate for this model class.
methods_used	permutation, pdp, ice, lime	Lists the interpretability methods produced in this run. Next: compare signals across methods before using them in documentation or model challenge.
shap_status	unavailable	Summarizes interpretability evidence for this run. Next: use these outputs to document model drivers and challenge unexpected dependencies.
shap_reason	No module named 'shap'	Summarizes interpretability evidence for this run. Next: use these outputs to document model drivers and challenge unexpected dependencies.
perm_top_features	txn_amount, num_txn_24h, txn_type_code, kyc_risk_score, destination_country_code ...	Shows features with the largest performance drop when permuted, which highlights global importance. Next: validate that the top drivers align with business logic, feature governance, and policy expectations.
lime_instances	3	Counts the locally explained observations reviewed with LIME. Next: examine representative true-positive and false-positive cases to understand local decision logic.

Metric	Figure	Explanation
pdp_features	avg_amount_7d, txn_amount, account_age_days, kyc_risk_score, num_txn_24h ...	Lists features with partial-dependence plots. Next: review whether average model response is monotonic, stable, and operationally sensible.
ice_features	avg_amount_7d, txn_amount, account_age_days, kyc_risk_score	Lists features with individual conditional expectation plots. Next: inspect heterogeneity and interactions that may be hidden in average-only views.



permutation_importance: Bars show how much AUC drops when each feature is permuted.

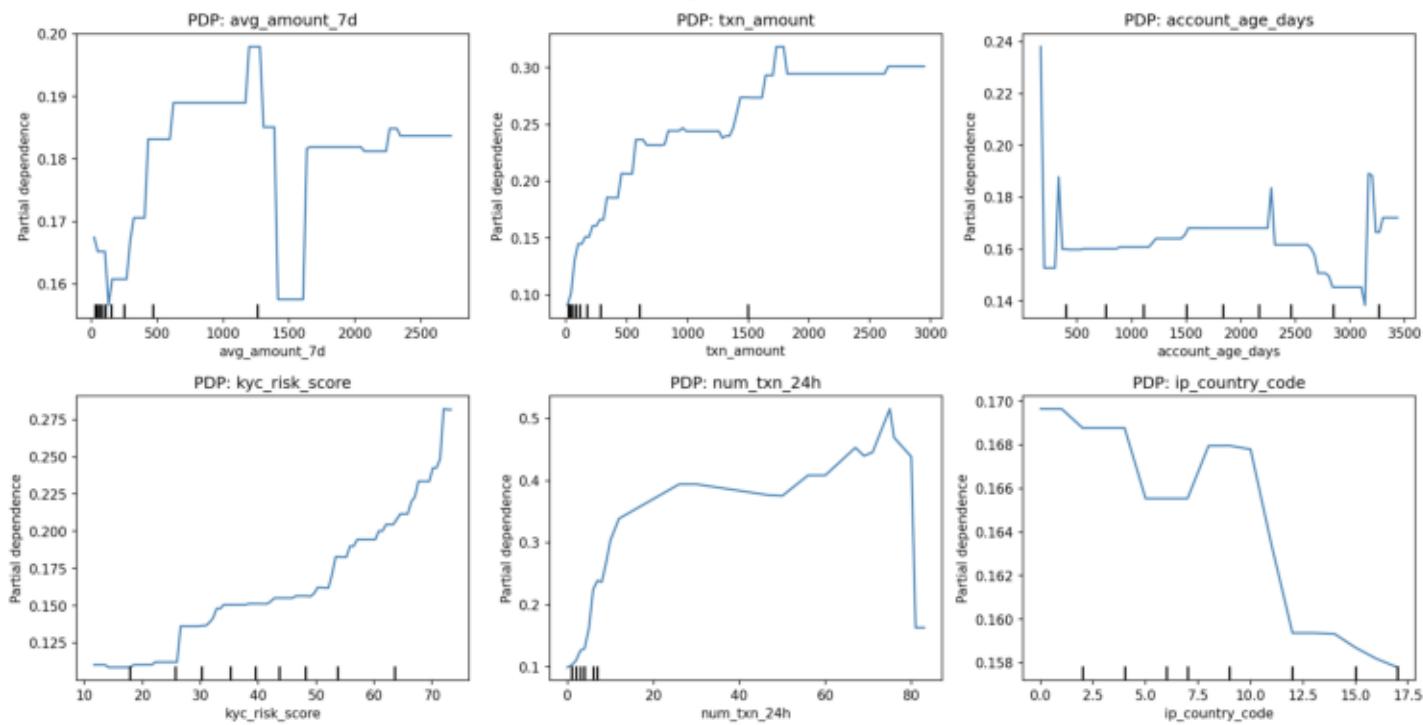
value_date_quarter=2024Q3 - 4) Model Interpretability



lime_explanation: LIME bars show local feature contributions for selected instances.

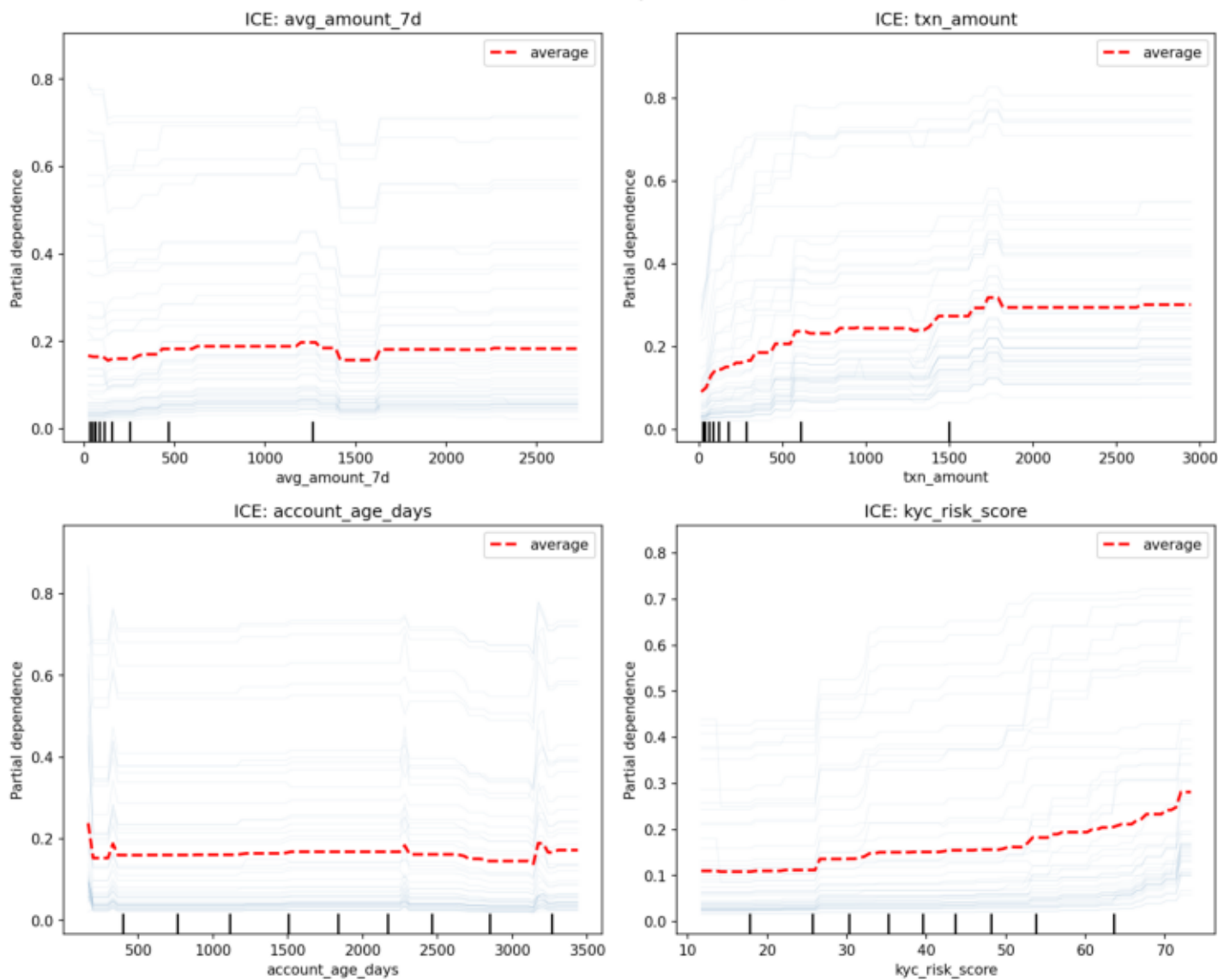
value_date_quarter=2024Q3 - 4) Model Interpretability

Partial Dependence Plots



pdp: PDP curves show average model response as a feature varies.

Individual Conditional Expectation (ICE) Plots



ice: ICE curves show per-instance responses as a feature varies.

value_date_quarter=2024Q4 - 1) Model Effectiveness (1/3)

Metric	Figure	Explanation
auc_roc	0.8401	Measures ranking power across all thresholds. 0.8401 indicates strong separation. Next: compare AUC-ROC across segments and keep threshold tuning separate from ranking validation.
auc_pr	0.6554	Measures precision-recall performance under class imbalance; the observed event rate is 28.98%. Next: use this together with alert capacity to decide whether the high-score queue is practically useful.
precision	0.6536	Shows the share of flagged cases that are true positives at the current threshold. Next: if this is below review expectations, raise the threshold or improve exclusion features.
recall	0.5366	Shows the share of true positives captured at the current threshold. Next: if missed cases are too high, lower the threshold or add stronger risk features.
f1	0.5894	Balances precision and recall in one score. Next: use it as a threshold-selection aid when both missed cases and false alerts matter.
ks_statistic	0.5595	Measures score separation between positive and negative classes. Next: low KS suggests the model needs better features or a different specification.

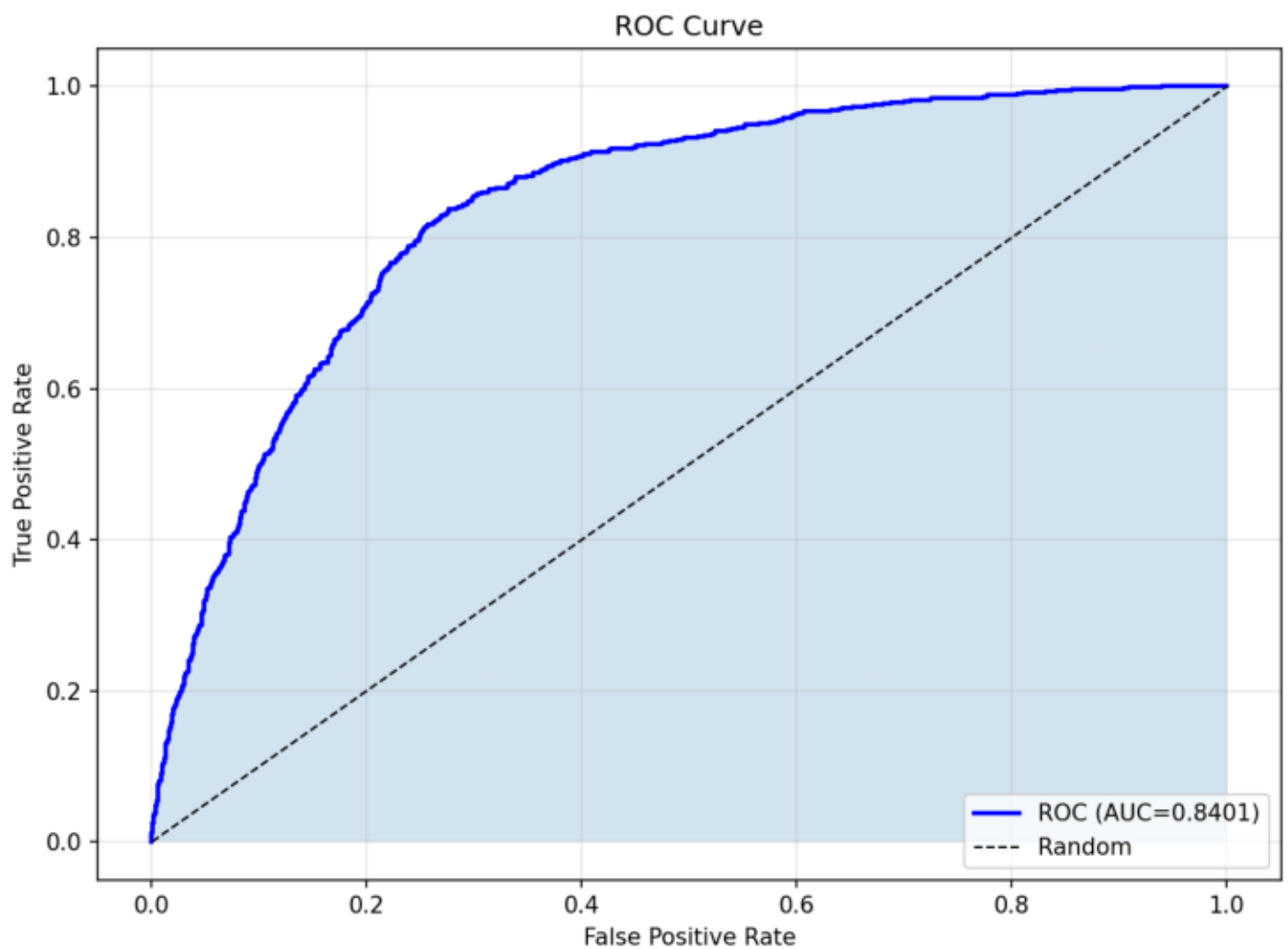
value_date_quarter=2024Q4 - 1) Model Effectiveness (2/3)

Metric	Figure	Explanation
ks_threshold	0.2400	This is the threshold that maximizes KS separation. Next: treat it as a candidate operating point, then compare it with business capacity and policy limits.
confusion_matrix[TN]	1744	True negatives correctly cleared. Next: preserve this base while tuning the alert threshold.
confusion_matrix[FP]	229	False positives create review cost without SAR conversion. Next: reduce this count with threshold tuning or sharper risk features.
confusion_matrix[FN]	373	False negatives are missed SAR cases. Next: if this count is unacceptable, lower the threshold or add features that capture suspicious behavior earlier.
confusion_matrix[TP]	432	True positives are correctly escalated cases. Next: assess whether this volume is sufficient relative to expected SAR capture goals.
precision_at_k[10]	1.0000	Shows how accurate the top-10 highest-risk queue is. Next: align K with investigator capacity and confirm that top alerts stay high quality.
precision_at_k[50]	0.8200	Shows how accurate the top-50 highest-risk queue is. Next: align K with investigator capacity and confirm that top alerts stay high quality.
precision_at_k[100]	0.8000	Shows how accurate the top-100 highest-risk queue is. Next: align K with investigator capacity and confirm that top alerts stay high quality.

value_date_quarter=2024Q4 - 1) Model Effectiveness (3/3)

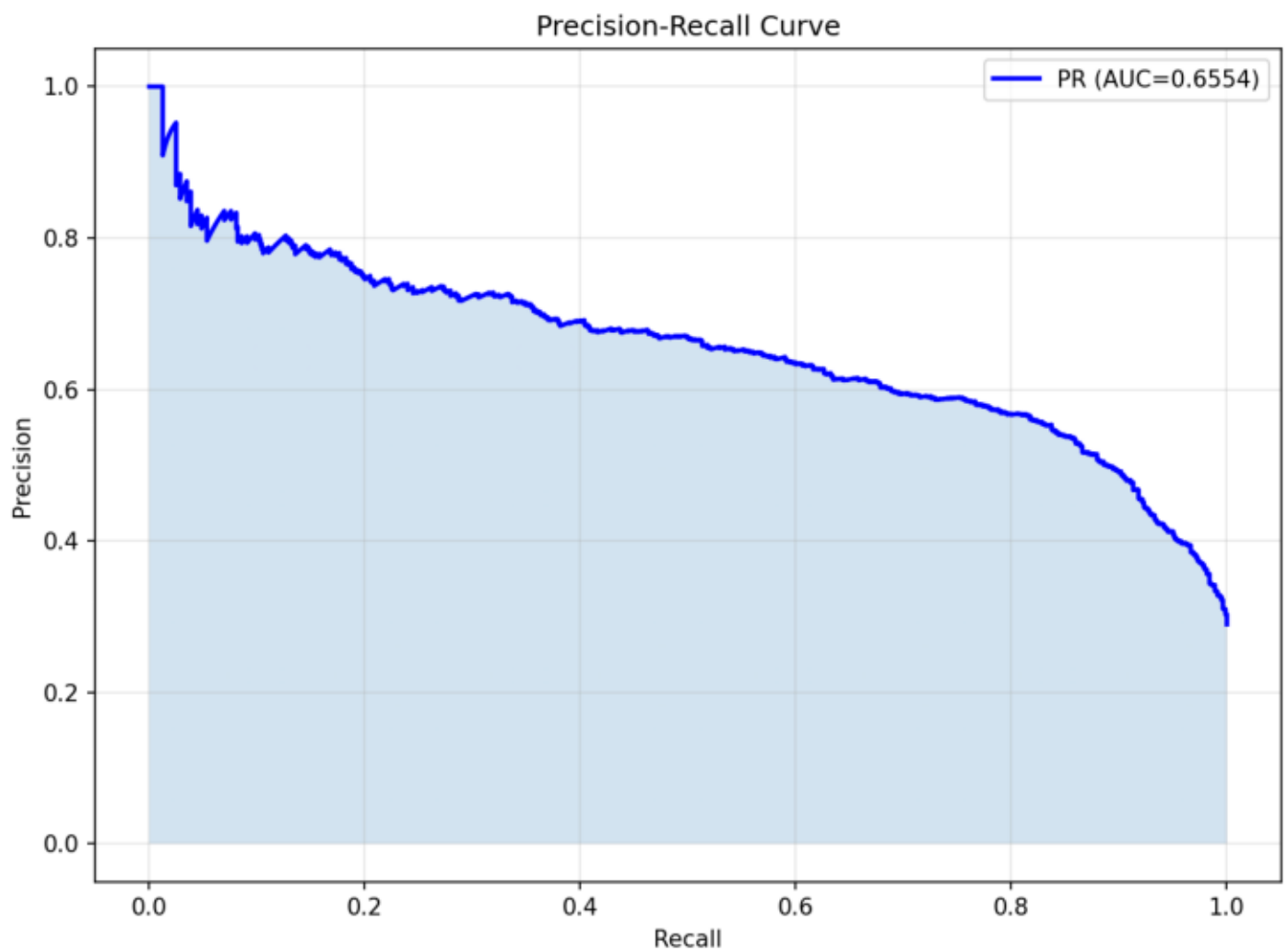
Metric	Figure	Explanation
precision_at_k[200]	0.7600	Shows how accurate the top-200 highest-risk queue is. Next: align K with investigator capacity and confirm that top alerts stay high quality.
precision_at_k[500]	0.6780	Shows how accurate the top-500 highest-risk queue is. Next: align K with investigator capacity and confirm that top alerts stay high quality.
recall_at_k[10]	0.0124	Shows how much of the positive population is captured within the top-10 queue. Next: increase K if coverage is too low, or improve ranking quality if capacity cannot expand.
recall_at_k[50]	0.0509	Shows how much of the positive population is captured within the top-50 queue. Next: increase K if coverage is too low, or improve ranking quality if capacity cannot expand.
recall_at_k[100]	0.0994	Shows how much of the positive population is captured within the top-100 queue. Next: increase K if coverage is too low, or improve ranking quality if capacity cannot expand.
recall_at_k[200]	0.1888	Shows how much of the positive population is captured within the top-200 queue. Next: increase K if coverage is too low, or improve ranking quality if capacity cannot expand.
recall_at_k[500]	0.4211	Shows how much of the positive population is captured within the top-500 queue. Next: increase K if coverage is too low, or improve ranking quality if capacity cannot expand.

value_date_quarter=2024Q4 - 1) Model Effectiveness



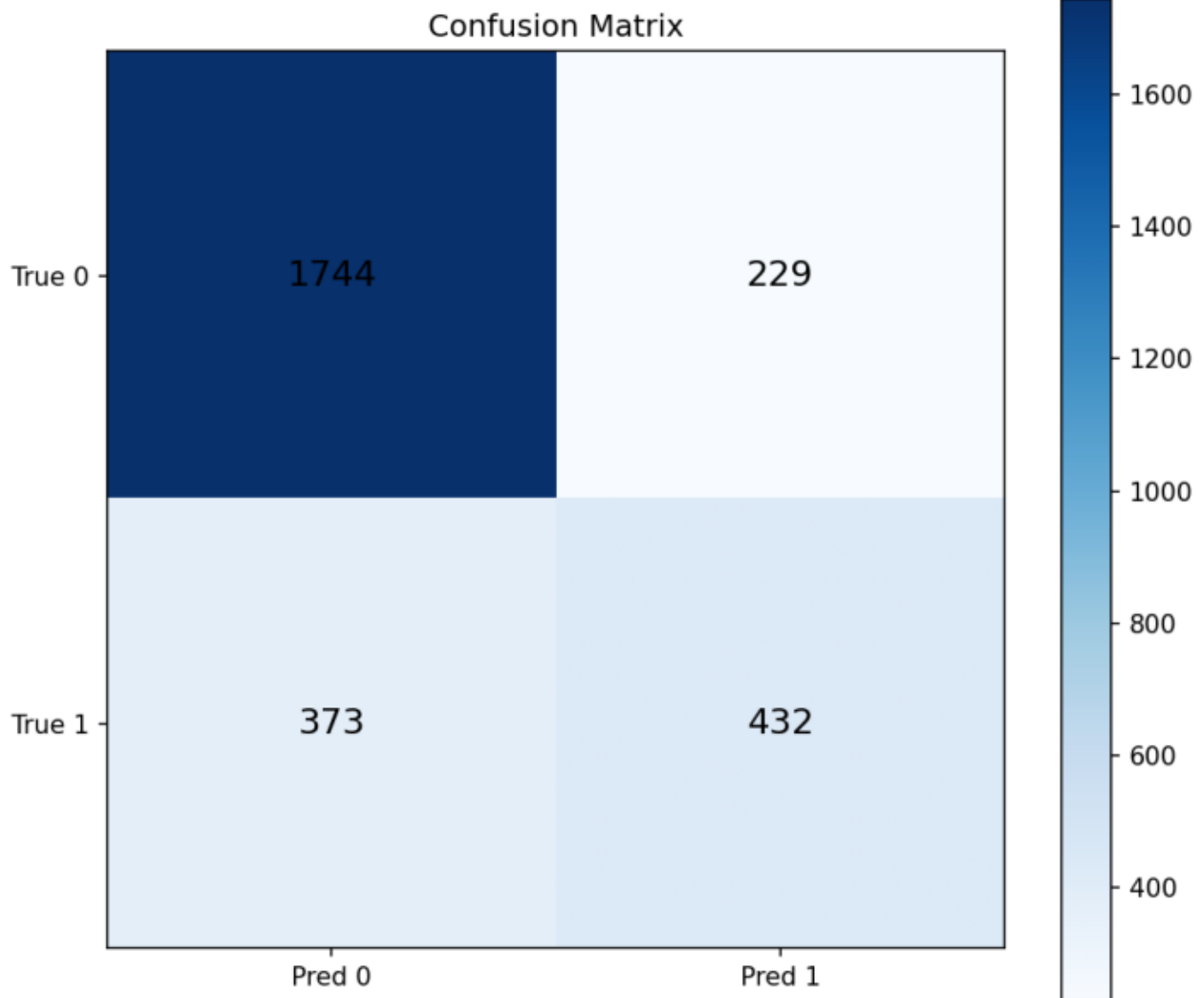
roc_curve: AUC-ROC=0.8401; curve above diagonal indicates strong separation.

value_date_quarter=2024Q4 - 1) Model Effectiveness



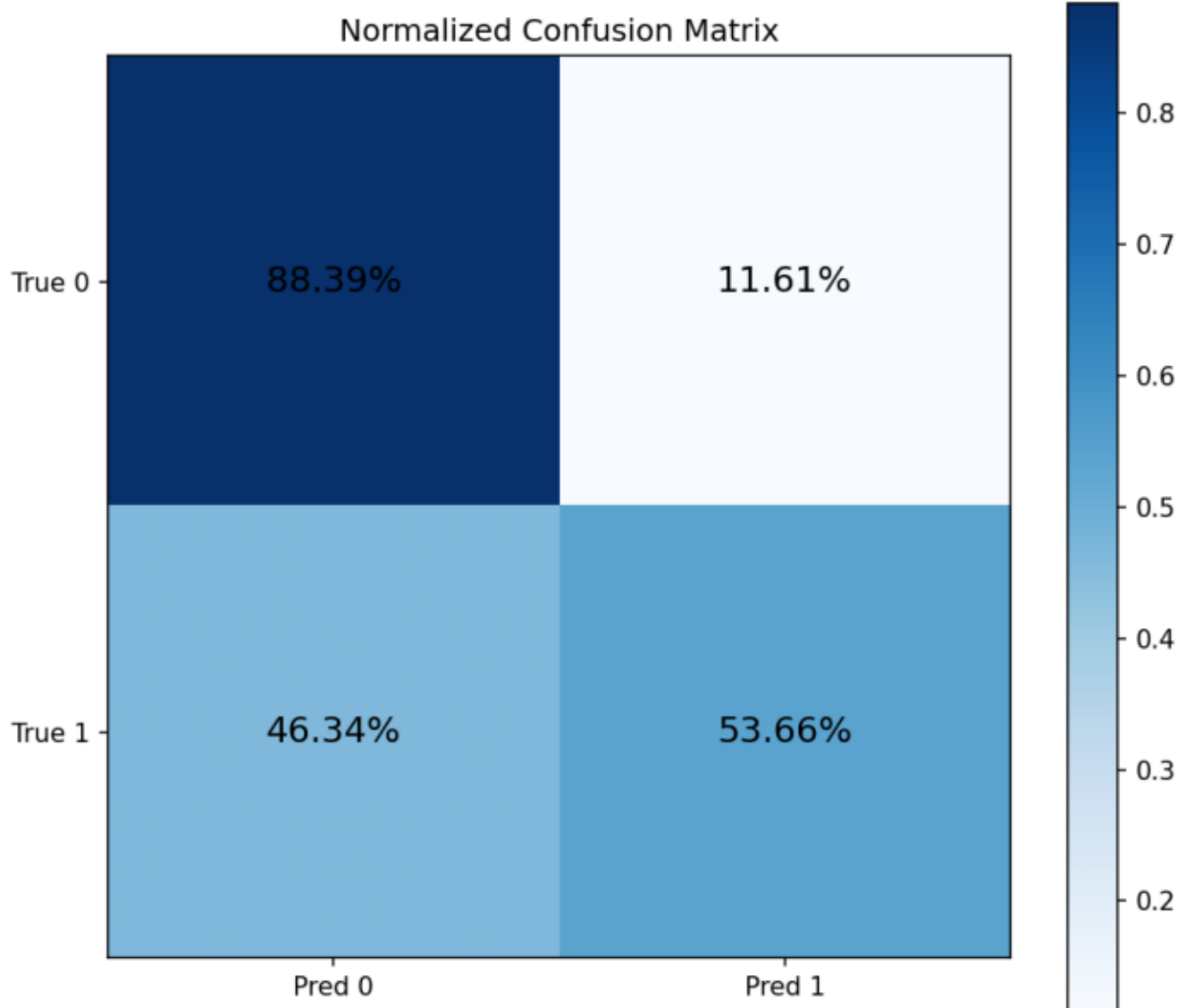
pr_curve: AUC-PR=0.6554 vs baseline 28.98%; higher curve implies better precision at recall.

value_date_quarter=2024Q4 - 1) Model Effectiveness



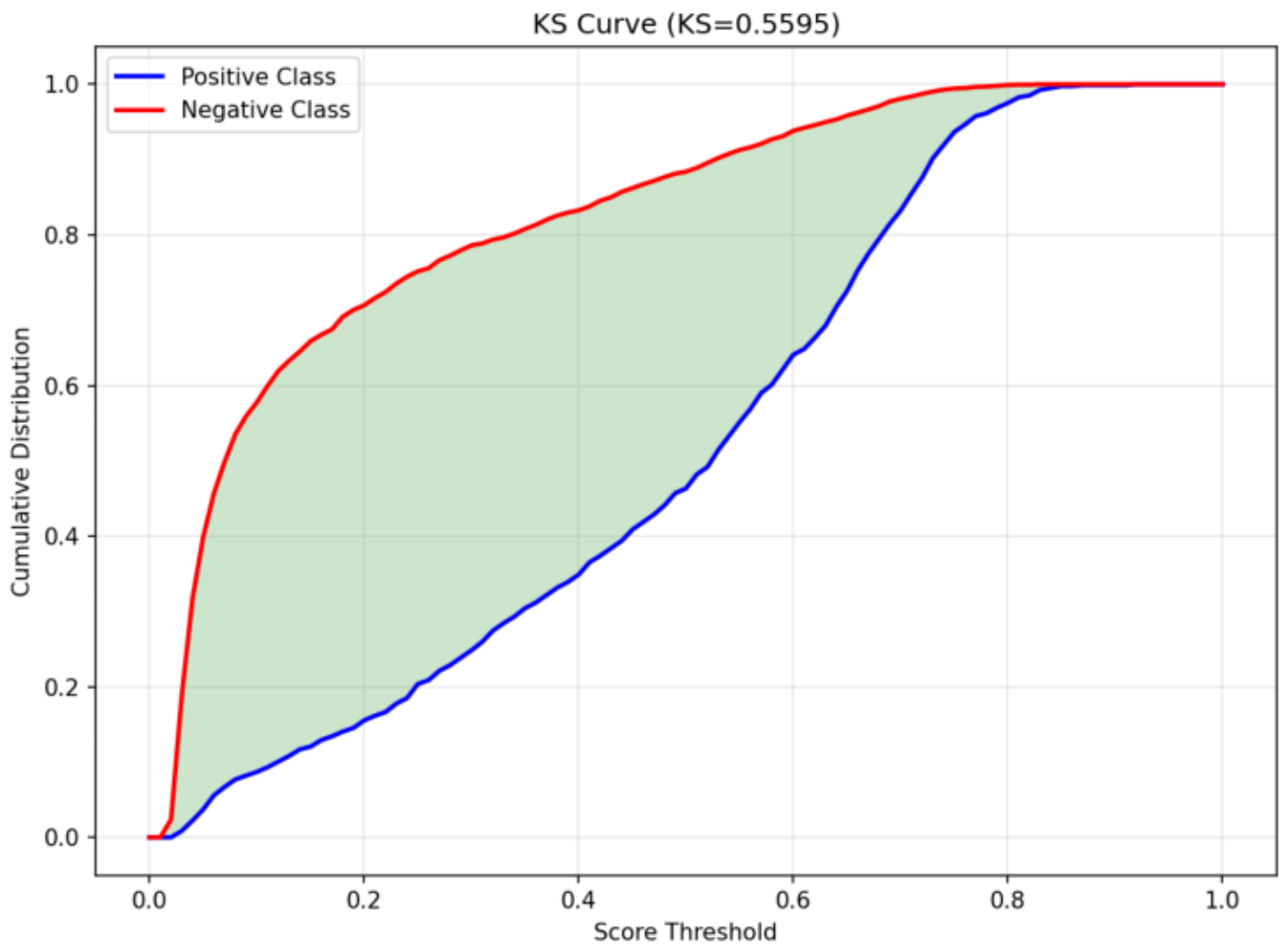
confusion_matrix: At threshold 0.50: TP=432, FP=229, FN=373, TN=1744.

value_date_quarter=2024Q4 - 1) Model Effectiveness



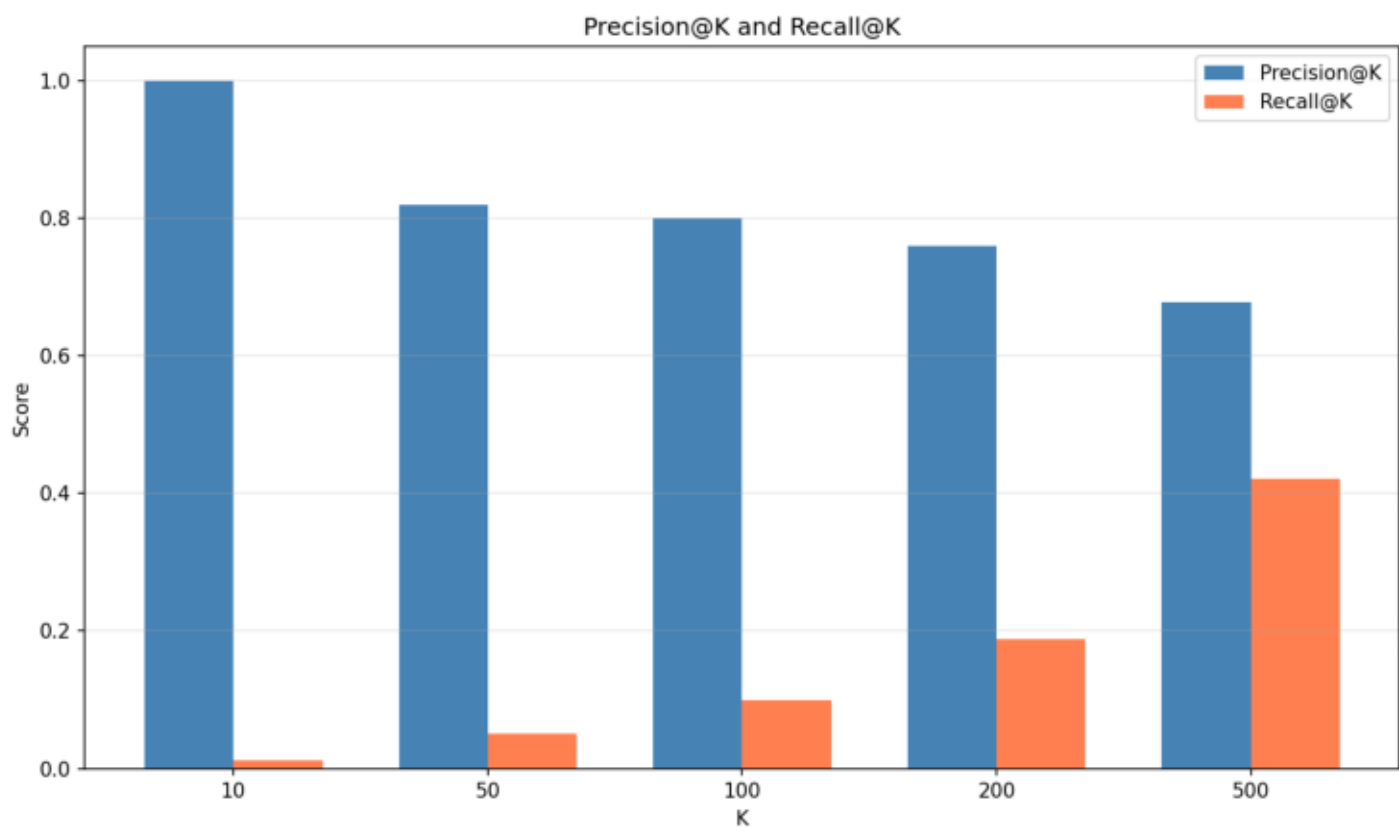
confusion_matrix_norm: Normalized rates: TPR=53.66%, TNR=88.39%.

value_date_quarter=2024Q4 - 1) Model Effectiveness



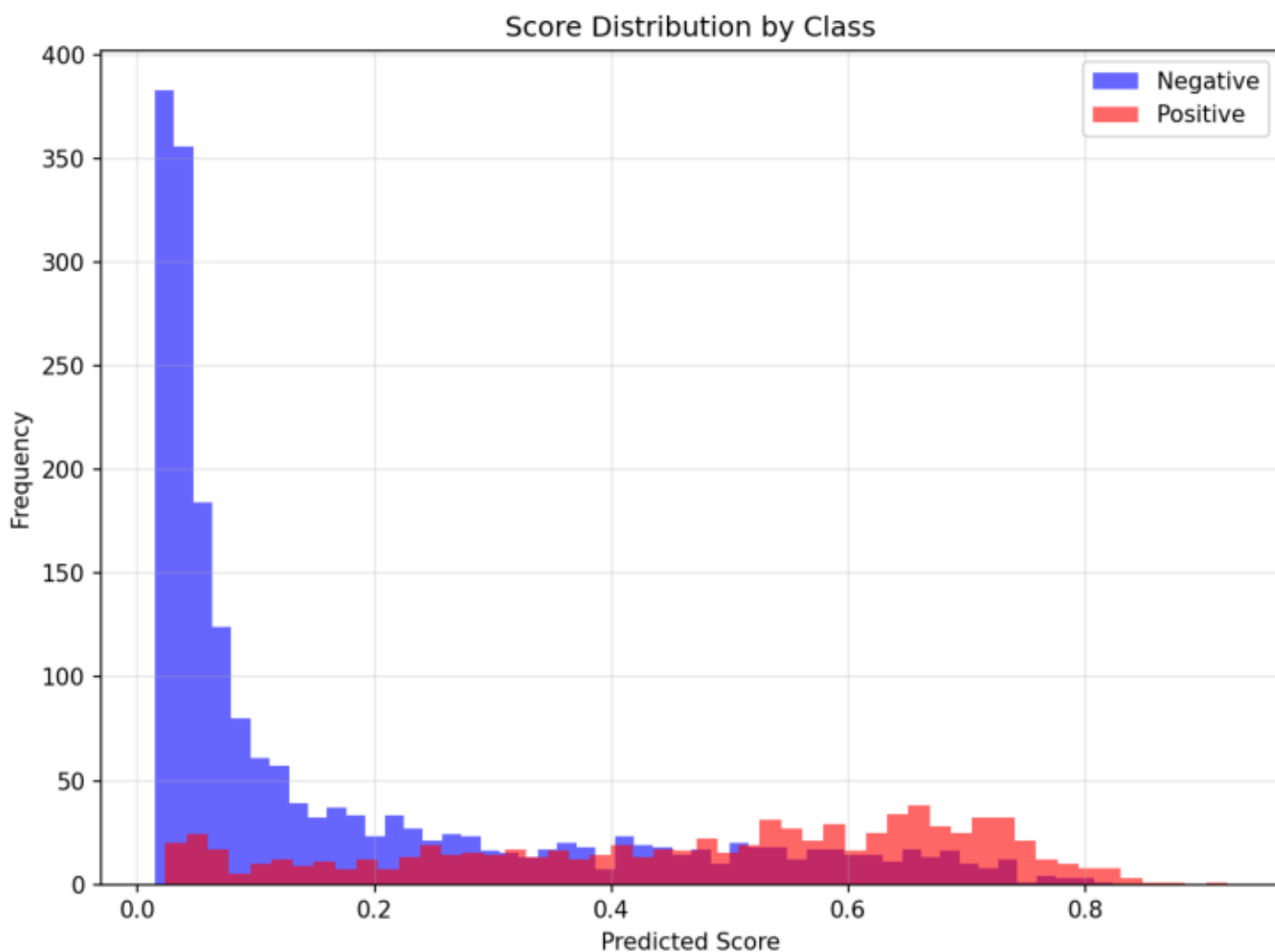
ks_curve: Maximum separation KS=0.5595 at threshold 0.24.

value_date_quarter=2024Q4 - 1) Model Effectiveness



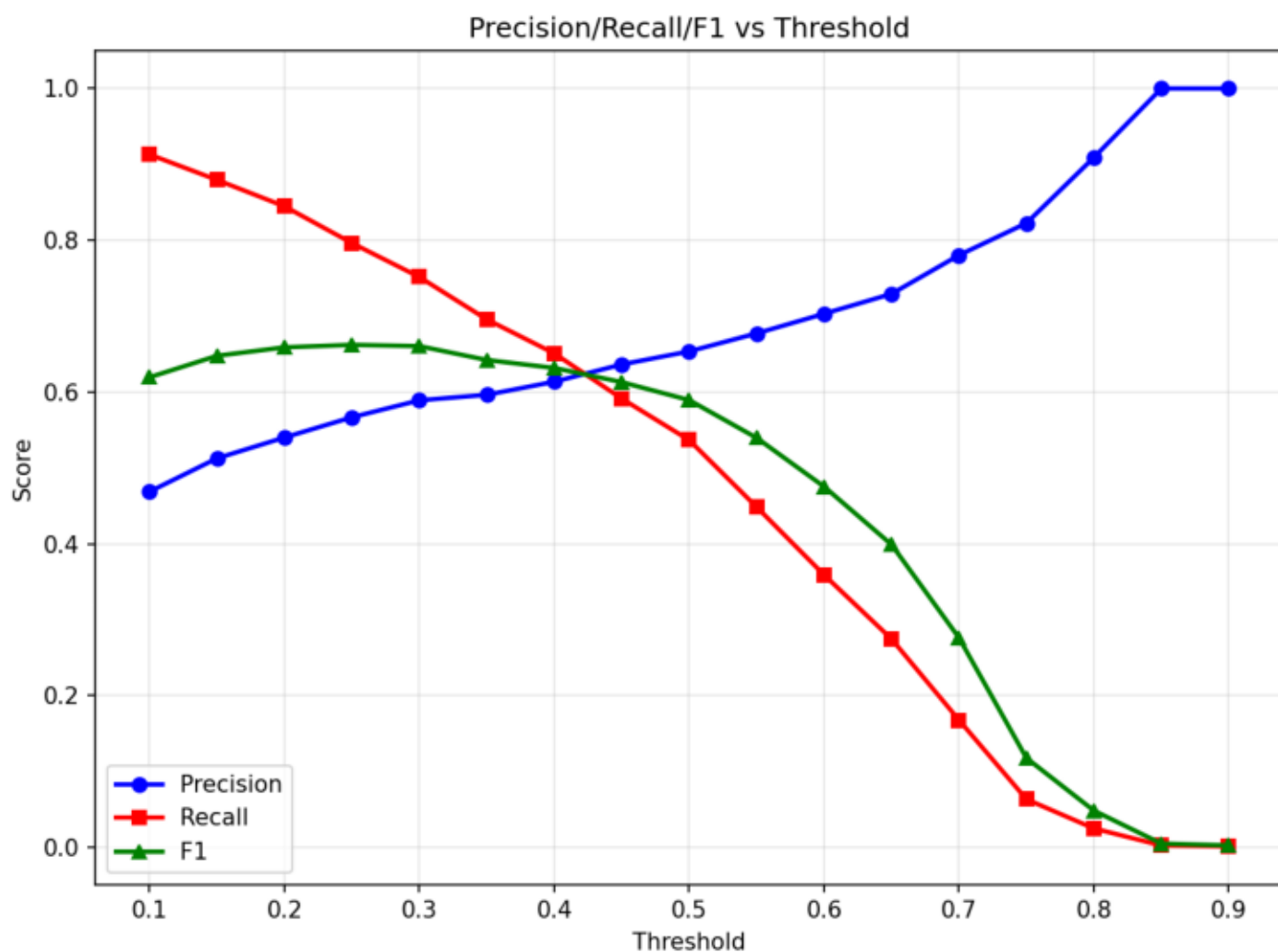
precision_recall_at_k: Bars show precision/recall as you expand the review queue.

value_date_quarter=2024Q4 - 1) Model Effectiveness



score_distribution: Mean score: positive=0.474, negative=0.174 (larger gap is better).

value_date_quarter=2024Q4 - 1) Model Effectiveness



threshold_analysis: Best F1 in tested grid is 0.6622 at threshold 0.25.

value_date_quarter=2024Q4 - 2) Model Efficiency (1/4)

Metric	Figure	Explanation
fpr	0.1161	Measures the share of negatives incorrectly flagged; 0.1161 is high. Next: if review demand is too high, raise the threshold or add features that better suppress false alerts.
tn	1744	Counts negatives correctly cleared without manual review. Next: preserve this clearance rate while checking that recall does not fall below policy expectations.
fp	229	Counts non-events that would still be sent to review. Next: use this with staffing capacity to decide whether threshold or feature refinement is required.
threshold	0.5000	This is the active operating cutoff used to flag alerts. Next: tune it jointly with recall, precision, and review capacity rather than in isolation.
fpr_at_thresholds[thresholds.t_0.05]	0.6006	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.05. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.10]	0.4222	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.10. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.

value_date_quarter=2024Q4 - 2) Model Efficiency (2/4)

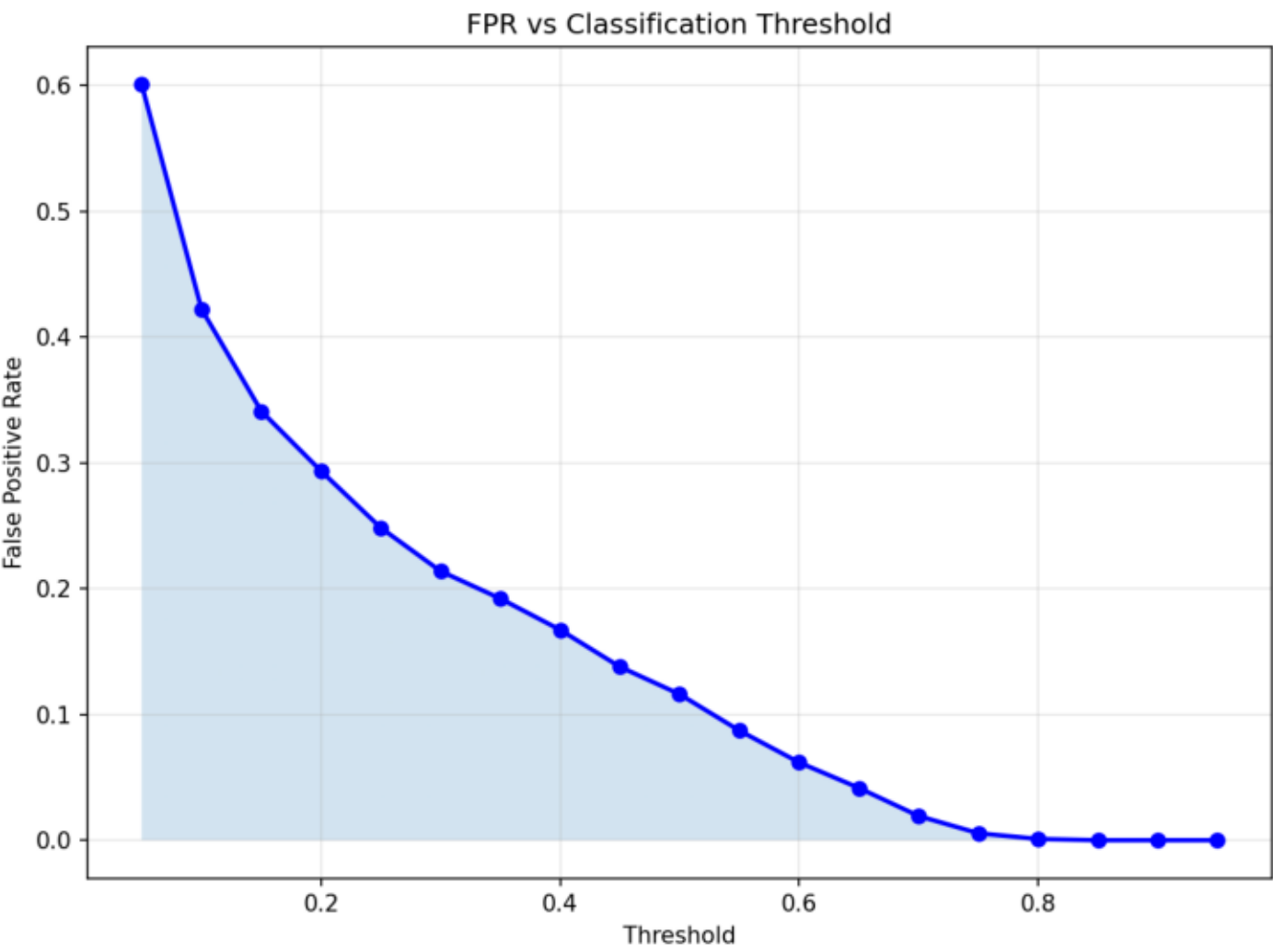
Metric	Figure	Explanation
fpr_at_thresholds[thresholds.t_0.15]	0.3411	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.15. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.20]	0.2935	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.20. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.25]	0.2484	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.25. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.30]	0.2139	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.30. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.35]	0.1921	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.35. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.40]	0.1673	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.40. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.

value_date_quarter=2024Q4 - 2) Model Efficiency (3/4)

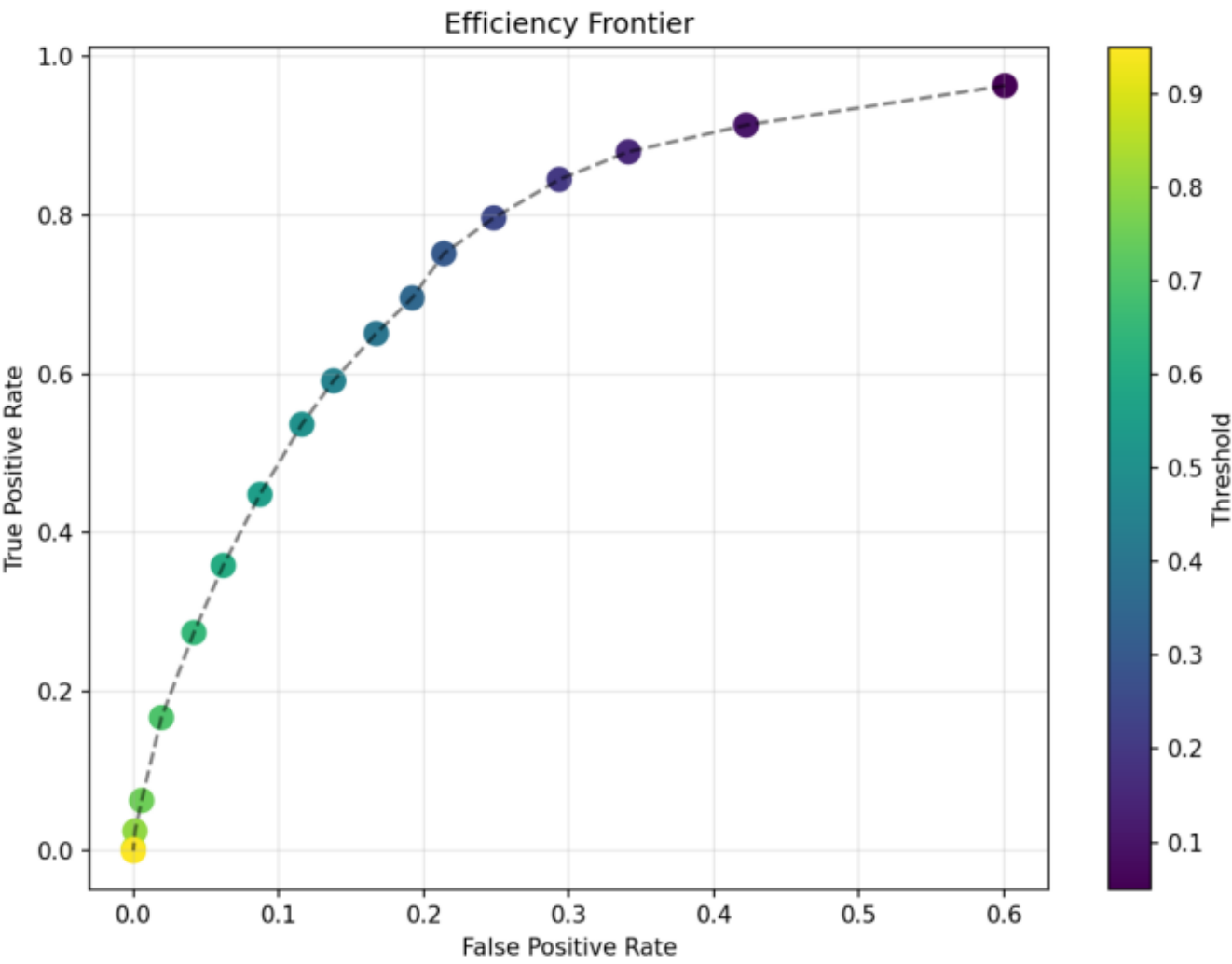
Metric	Figure	Explanation
fpr_at_thresholds[thresholds.t_0.45]	0.1379	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.45. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.50]	0.1161	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.50. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.55]	0.0872	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.55. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.60]	0.0618	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.60. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.65]	0.0416	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.65. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.70]	0.0193	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.70. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.

value_date_quarter=2024Q4 - 2) Model Efficiency (4/4)

Metric	Figure	Explanation
fpr_at_thresholds[thresholds.t_0.75]	0.0056	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.75. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.80]	0.0010	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.80. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.85]	0.0000	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.85. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.90]	0.0000	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.90. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.95]	0.0000	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.95. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.

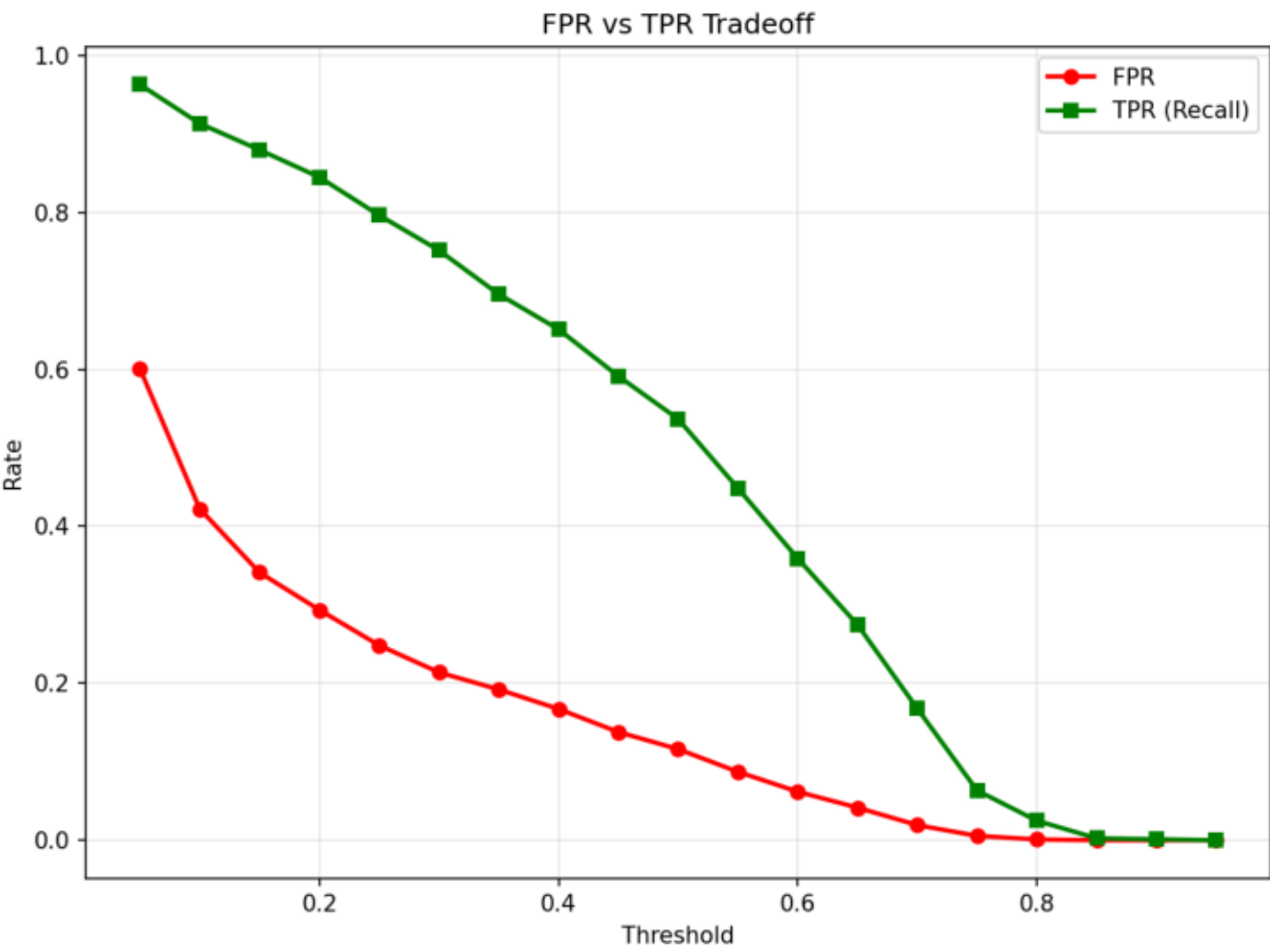


value_date_quarter=2024Q4 - 2) Model Efficiency



efficiency_frontier: Each point shows the FPR/TPR tradeoff; move toward the top-left for better efficiency.

value_date_quarter=2024Q4 - 2) Model Efficiency



fpr_tpr_tradeoff: FPR and TPR curves highlight how recall drops as you reduce false positives.

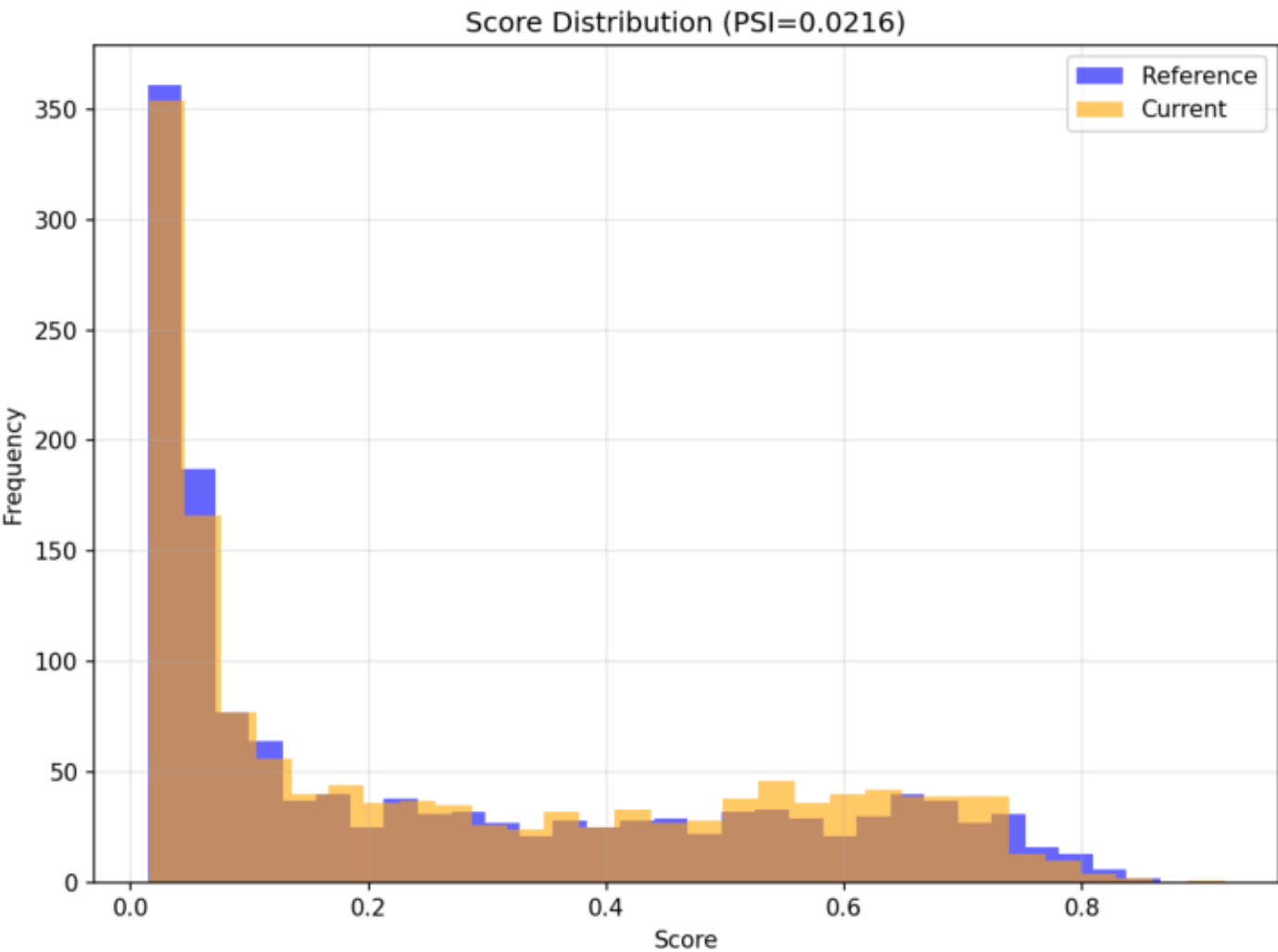
value_date_quarter=2024Q4 - 3) Model Stability (1/2)

Metric	Figure	Explanation
psi	0.0216	Measures score-distribution shift between reference and current populations; 0.0216 is negligible. Next: investigate segment mix, data drift, and recalibration needs when PSI is material.
cv_auc_roc_mean	0.8398	Average AUC-ROC across validation folds. Next: compare it with overall AUC-ROC; a large gap can indicate overfitting or unstable sampling.
cv_auc_roc_std	0.0188	Standard deviation of AUC-ROC across folds. Next: if variability is high, simplify the model or increase training stability before deployment.
cv_auc_pr_mean	0.6578	Average AUC-PR across validation folds. Next: confirm that precision-recall performance remains acceptable under resampling.
cv_auc_pr_std	0.0323	Standard deviation of AUC-PR across folds. Next: high dispersion suggests unstable alert quality and warrants deeper validation.
bootstrap_auc_roc_mean	0.8129	Bootstrap mean AUC-ROC estimates expected ranking performance under repeated sampling. Next: compare it with the point estimate to judge optimism in the current test run.
bootstrap_auc_roc_ci_lower	0.7923	Lower confidence bound for bootstrap AUC-ROC. Next: use it as a conservative performance floor in model-risk discussions.
bootstrap_auc_roc_ci_upper	0.8354	Upper confidence bound for bootstrap AUC-ROC. Next: review the interval width with the lower bound to assess uncertainty.

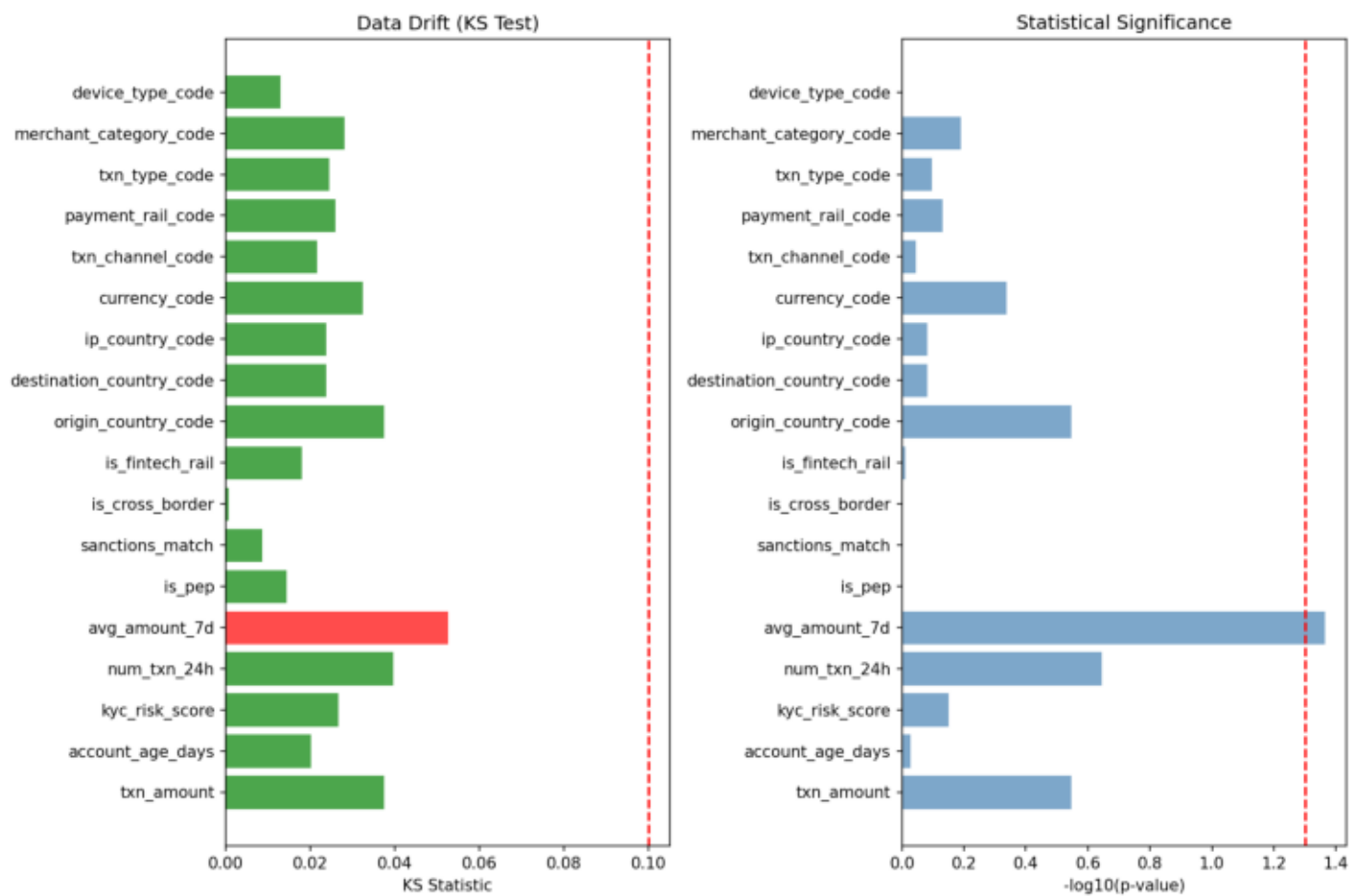
value_date_quarter=2024Q4 - 3) Model Stability (2/2)

Metric	Figure	Explanation
concept_drift_detected	False	No material concept-drift signal was detected in the evaluated slices. Next: retain ongoing monitoring because stable historical results do not guarantee future stability.
concept_drift_score	0.0261	Quantifies the magnitude of performance change across ordered slices. Next: rising scores should trigger deeper review of segment behavior and model refresh timing.

value_date_quarter=2024Q4 - 3) Model Stability

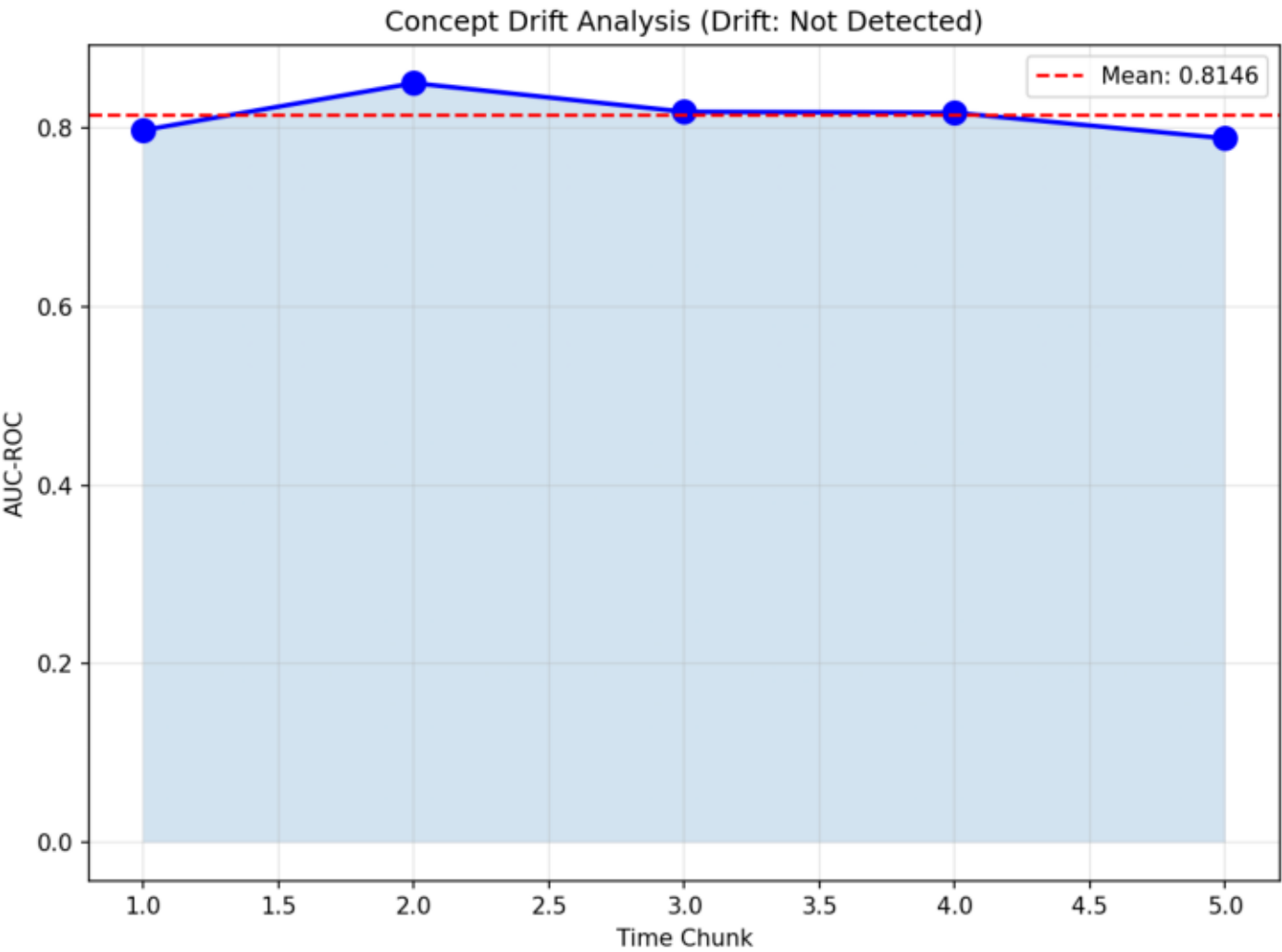


value_date_quarter=2024Q4 - 3) Model Stability



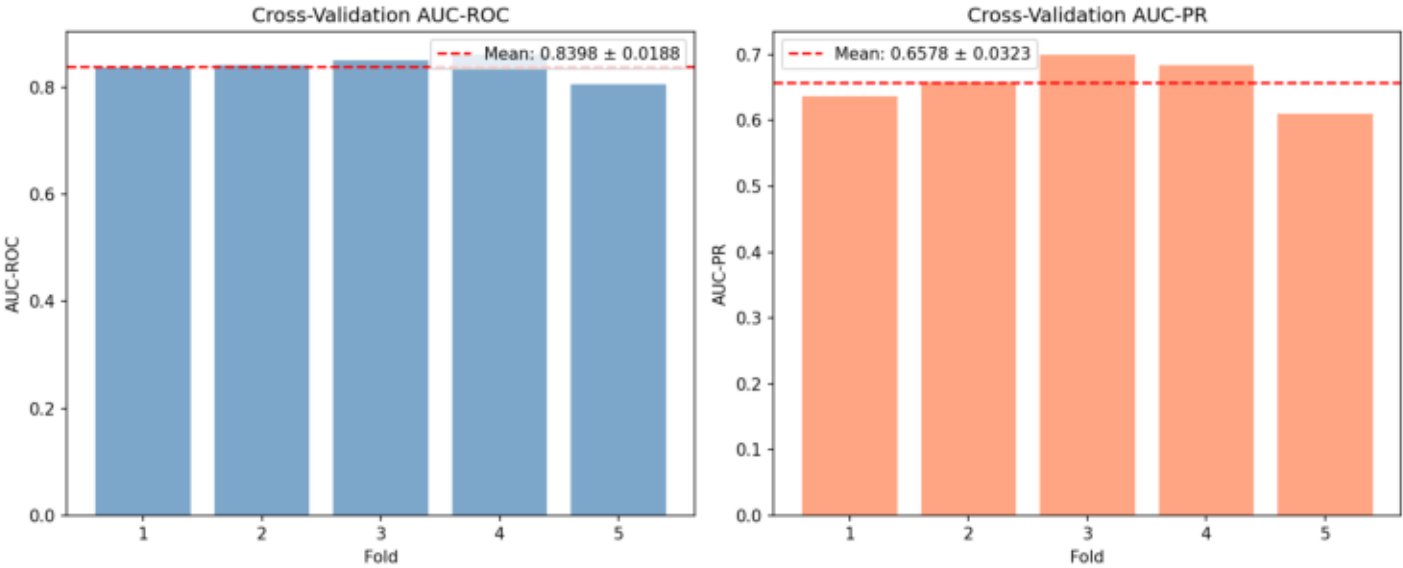
data_drift_heatmap: 1/18 features show drift; red bars highlight flagged features.

value_date_quarter=2024Q4 - 3) Model Stability



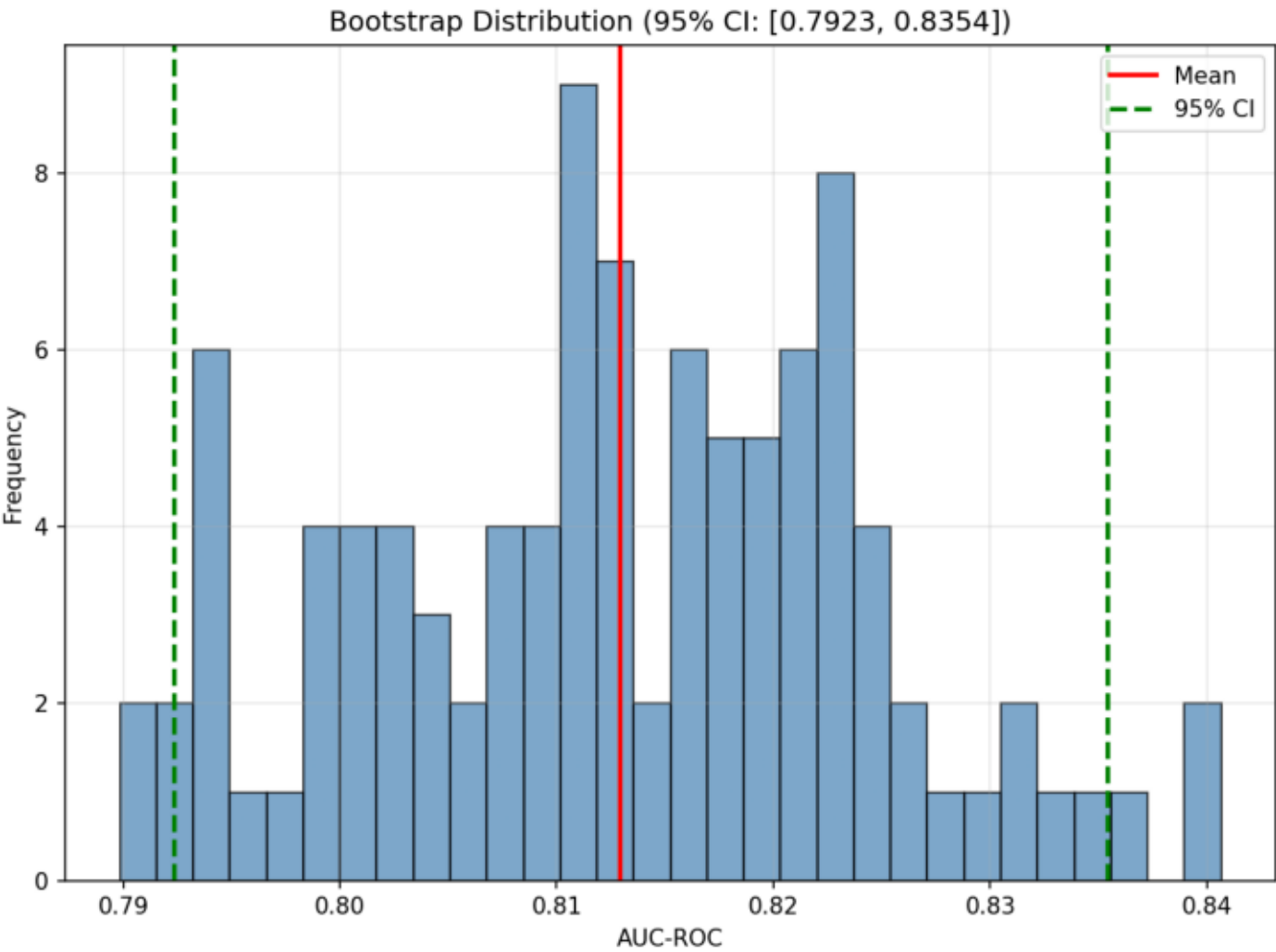
concept_drift: AUC across time chunks shows whether performance is stable over time.

value_date_quarter=2024Q4 - 3) Model Stability



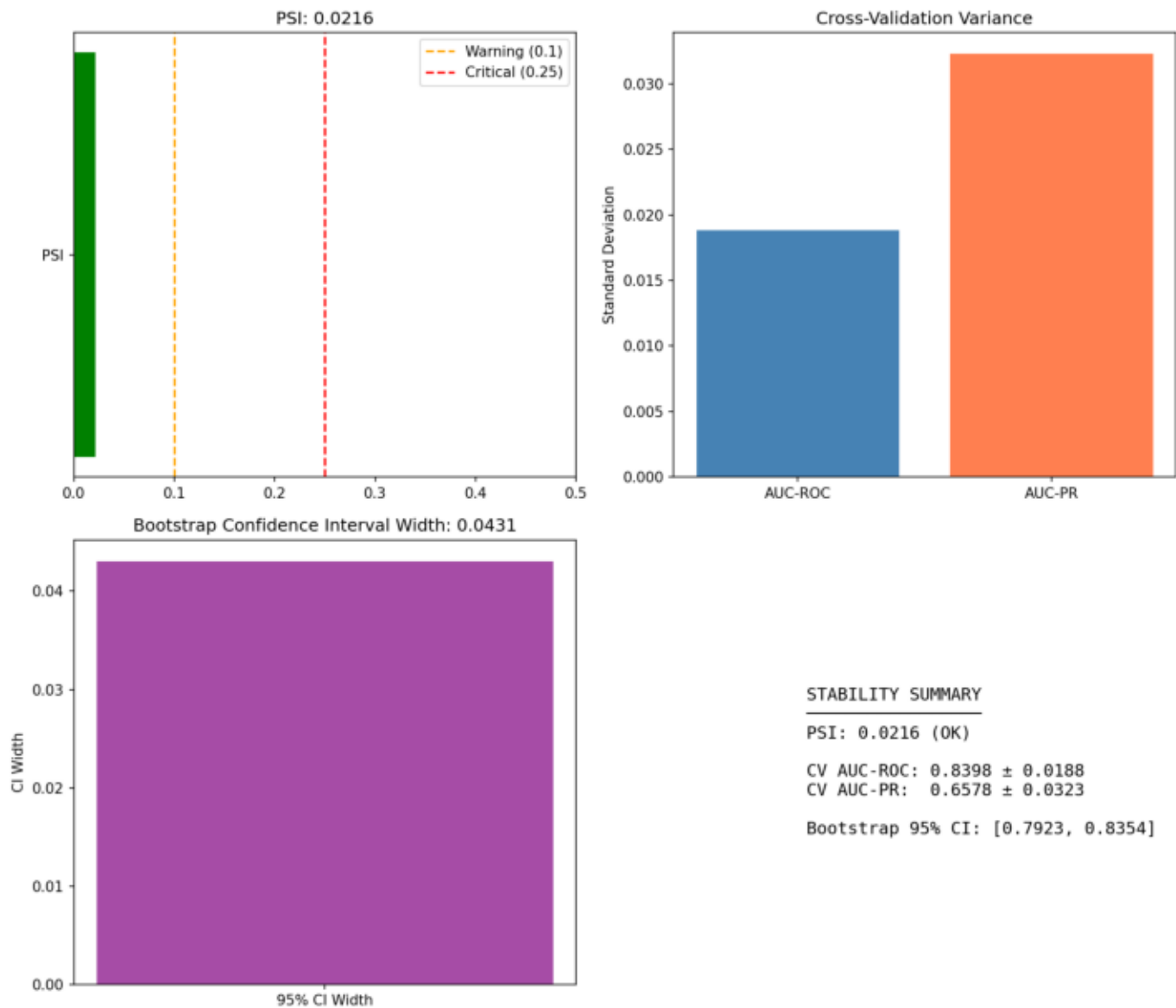
cv_results: Fold scores cluster around ROC 0.8398 and PR 0.6578.

value_date_quarter=2024Q4 - 3) Model Stability



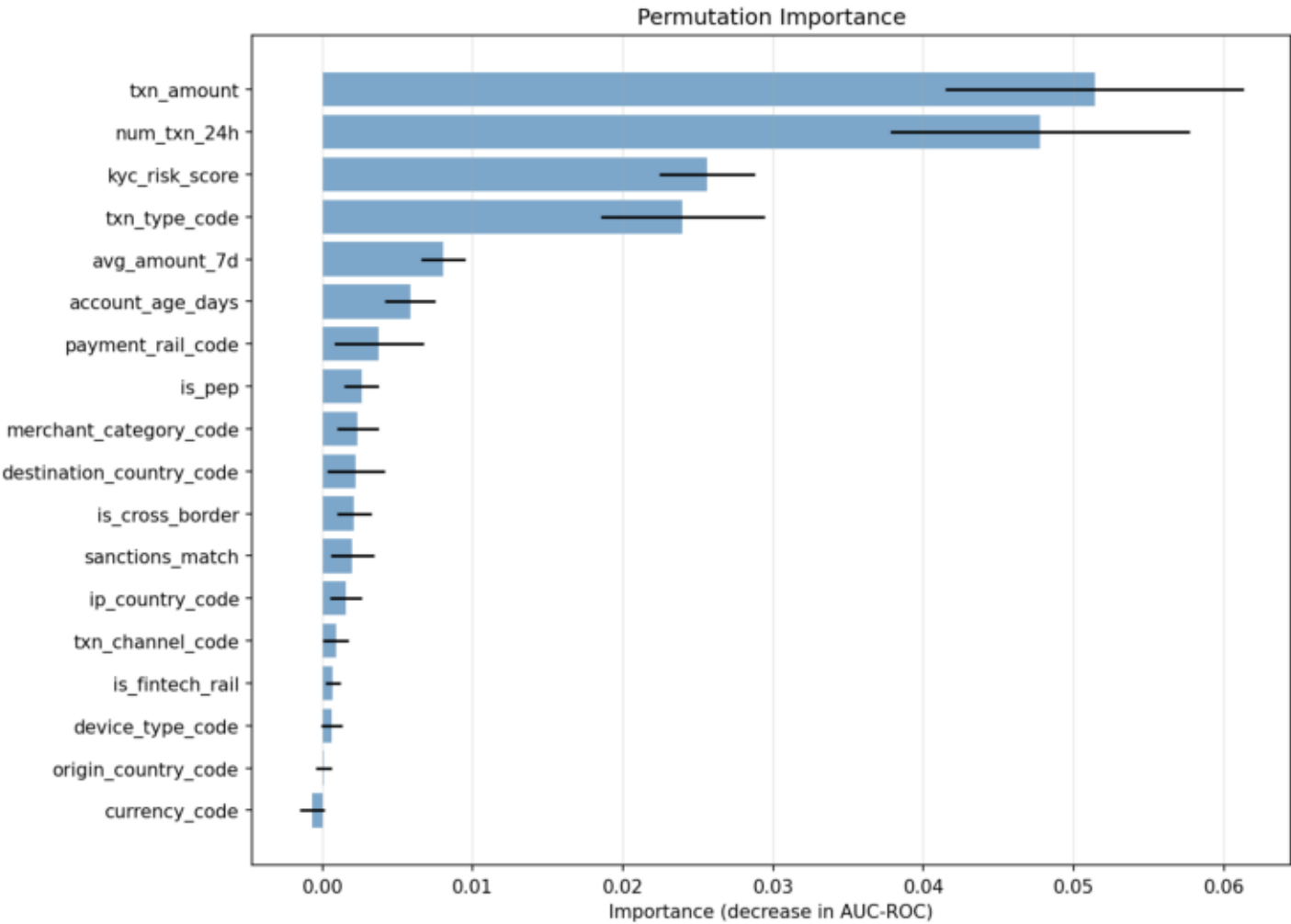
bootstrap_distribution: Bootstrap CI [0.7923, 0.8354] reflects performance stability.

value_date_quarter=2024Q4 - 3) Model Stability



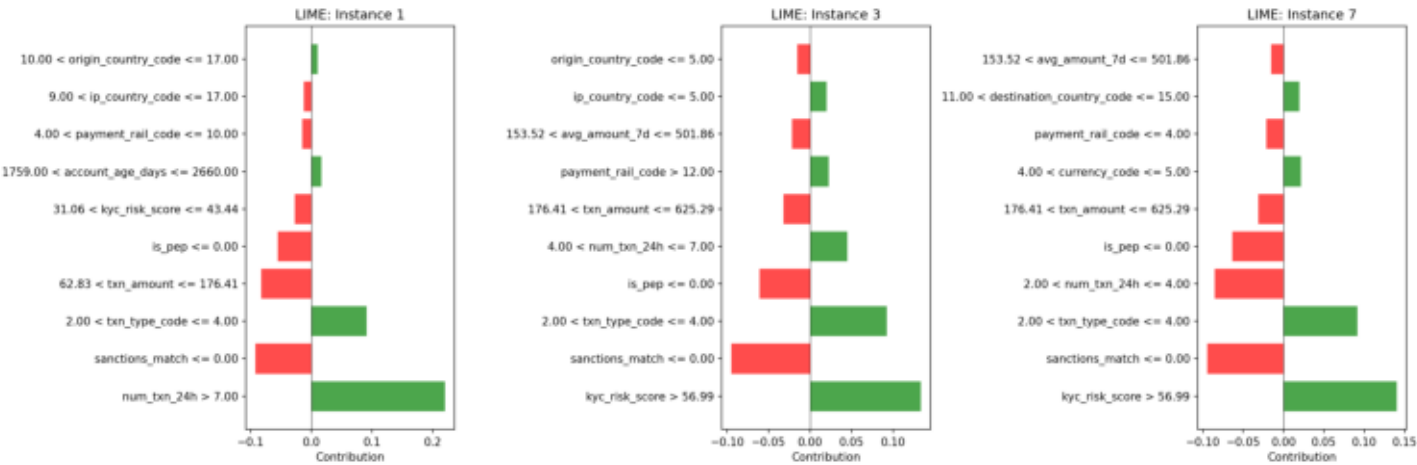
Metric	Figure	Explanation
model_type	tree	Identifies the estimator family so explanation methods can be interpreted in the right context. Next: confirm the chosen explanation methods are appropriate for this model class.
methods_used	permutation, pdp, ice, lime	Lists the interpretability methods produced in this run. Next: compare signals across methods before using them in documentation or model challenge.
shap_status	unavailable	Summarizes interpretability evidence for this run. Next: use these outputs to document model drivers and challenge unexpected dependencies.
shap_reason	No module named 'shap'	Summarizes interpretability evidence for this run. Next: use these outputs to document model drivers and challenge unexpected dependencies.
perm_top_features	txn_amount, num_txn_24h, kyc_risk_score, txn_type_code, avg_amount_7d ...	Shows features with the largest performance drop when permuted, which highlights global importance. Next: validate that the top drivers align with business logic, feature governance, and policy expectations.
lime_instances	3	Counts the locally explained observations reviewed with LIME. Next: examine representative true-positive and false-positive cases to understand local decision logic.

Metric	Figure	Explanation
pdp_features	txn_amount, avg_amount_7d, account_age_days, kyc_risk_score, num_txn_24h ...	Lists features with partial-dependence plots. Next: review whether average model response is monotonic, stable, and operationally sensible.
ice_features	txn_amount, avg_amount_7d, account_age_days, kyc_risk_score	Lists features with individual conditional expectation plots. Next: inspect heterogeneity and interactions that may be hidden in average-only views.



permutation_importance: Bars show how much AUC drops when each feature is permuted.

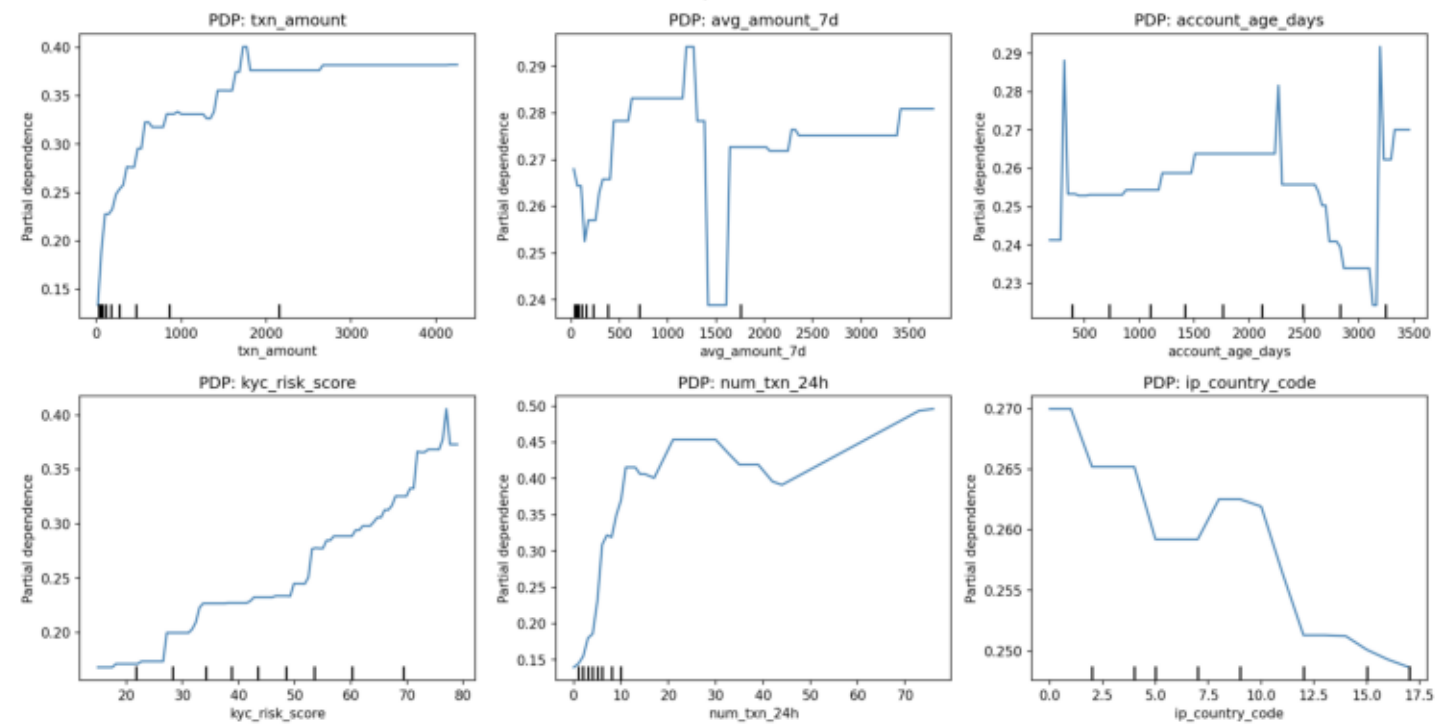
value_date_quarter=2024Q4 - 4) Model Interpretability



lime_explanation: LIME bars show local feature contributions for selected instances.

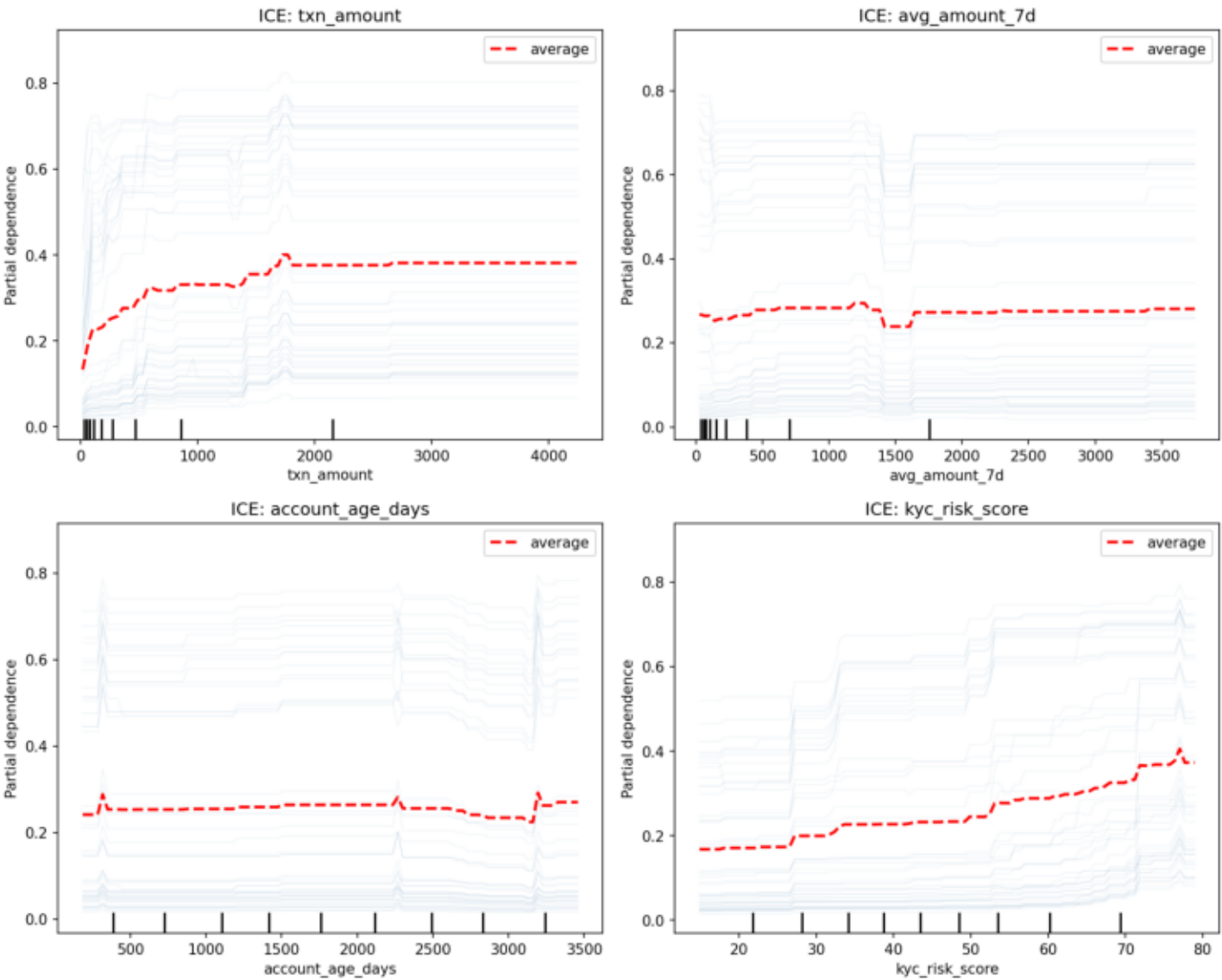
value_date_quarter=2024Q4 - 4) Model Interpretability

Partial Dependence Plots



pdp: PDP curves show average model response as a feature varies.

Individual Conditional Expectation (ICE) Plots



ice: ICE curves show per-instance responses as a feature varies.